SOLUTIONS TO CHARACTER THEORY OF FINITE GROUPS BY I. MARTIN ISAACS

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Abstract. This is a solutions manual to I. Martin Isaacs' book *Character Theory of Finite Groups*. This solutions manual is currently a work-in-progress that I have been incrementally piecing together since 2021, and so far, I only have complete solutions for each of the first five chapters and the eleventh chapter with only partial solutions for the sixth chapter.

Many of the solutions presented are my own work. However, some of the solutions were taken from the Math Stack Exchange or from the proofs of certain other books where these problems appear as theorems. I have tried my best to cite all of these sources when applicable.

I have tried my best to maintain consistent notation with Isaacs. However, there may be certain inconsistencies noticed with my notation for which I would like to apologize in advance, i.e., using $H \leqslant G$ versus $H \subseteq G$ to denote the containment of a subgroup H. I have also tried to consistently include the original "hints" and "notes" that Isaacs includes with his problems. For certain problems, I also have some personal comments, such as when certain recent advances have been made regarding the results presented in a problem, when my solution utilizes a somewhat nonstandard result, or when I believe my proof differs from Isaacs' likely intended solution. These are labeled as "Remarks" and are placed between the problem statement and my proof, underneath any hints and notes if present.

I welcome and encourage any questions, corrections, and comments on this work. I may be reached via email at chris.w.herbig@gmail.com.

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1. Algebras, modules, and representations

Problem 1.1. Let V be an A-module. Show that V is completely reducible iff the intersection of all of the maximal submodules of V is trivial.

Hint: To prove "if," embed V into a sum of irreducible modules. Recall that our definition of "module" requires finite dimensionality.

Note: This problem is ""dual" to Theorem 1.10 which implies that V is completely reducible iff it is the sum of its minimal submodules. Theorem 1.10 is true much more generally than we have proved it. It holds for arbitrary modules over rings. Problem 1.1, however, is not valid in this greater generality. A counterexample is the regular module of the ring of integers $\mathbb{Z}$.

Proof. Let V be completely reducible. Write $V = \oplus_{i=1}^{n} W_i$ where each W_i is an irreducible submodule. Observe that for $1 \leqslant j \leqslant n$, each submodule of the form

$$\bigoplus_{i=1,\, i\neq j}^{n} W_i \tag{1.1}$$

is maximal, and hence,

$$\bigcap_{j=1}^{n} \bigoplus_{i=1,\, i\neq j}^{n} W_i = 0. \tag{1.2}$$

The intersection of all maximal submodules must then be 0 as well.

Conversely, let $\mathfrak{V}$ be the set of all maximal submodules of V, and assume $\bigcap_{W\in\mathfrak{V}} W$ is trivial. The finite dimensionality of V allows us to choose a finite subset $\mathfrak{W}$ of $\mathfrak{V}$ having a trivial intersection. For a $W \in \mathfrak{W}$, we can define an A-module homomorphism from V onto V/W by sending $v \longmapsto v + W$. This extends to an injective A-module homomorphism from V into $\bigoplus_{W\in\mathfrak{W}} V/W$. Note that the map has trivial kernel since the intersection of these maximal submodules is trivial. Since $\oplus_{W\in\mathfrak{W}} V/W$ is a finite direct sum of irreducible A-modules, the sum is completely reducible. So, V can be identified with a submodule of a completely reducible module and is therefore completely reducible itself. □

Problem 1.2. Let $\mathfrak{X}$ and $\mathfrak{N}$ be representations of an F-algebra, A. A nonzero matrix is said to *intertwine* $\mathfrak{X}$ and $\mathfrak{N}$ if $P\mathfrak{X}(a) = \mathfrak{N}(a)P$ for each $a \in A$. Assume $\mathfrak{X}$ and $\mathfrak{N}$ are irreducible.

(a) If P intertwines $\mathfrak{X}$ and $\mathfrak{N}$, then show that P is square and nonsingular.
(b) Assume F is algebraically closed, and P and Q intertwine $\mathfrak{X}$ and $\mathfrak{N}$. Show $P = \lambda Q$ for some $\lambda \in F$.

Proof. Part (a) is essentially an application of Schur's Lemma. Assume that $\mathfrak{X}$ and $\mathfrak{N}$ have degree n and m respectively. By letting V and W denote respectively the

n-dimensional and m-dimensional vector spaces of row vectors with entries in F, we can define A-module actions on these vector spaces by defining $v \cdot a := v\mathfrak{X}(a)$ and $w \cdot a := w\mathfrak{N}(a)$ for $v \in V$ and $w \in W$. Notice from the relation $P\mathfrak{X}(a) = \mathfrak{N}(a)P$ that P is an $m \times n$ matrix, so in particular, P may be viewed as a linear transformation from W to V. The relation $P\mathfrak{X}(a) = \mathfrak{N}(a)P$ also implies that P commutes with the action of A on these modules, so P may in fact be viewed as an A-module homomorphism. By Schur's Lemma and the irreducibility of W and V, P is either an A-module isomorphism or zero. However, P is assumed to be nonzero, so P is an A-module isomorphism and (*a fortiori*) an isomorphism of F-vector spaces. Thus, P is a nonsingular, square matrix.

For part (b), assume P and Q are nonzero matrices such that $P\mathfrak{X}(a) = \mathfrak{N}(a)P$ and $Q\mathfrak{X}(a) = \mathfrak{N}(a)Q$ for each $a \in A$. Then, by part (a), Q is invertible, and we may write

$$Q^{-1}P\mathfrak{X}(a) = \mathfrak{X}(a)Q^{-1}P \tag{1.3}$$

for $a \in A$. Again, let V be the F-vector space consisting of $\dim(\mathfrak{X})$-dimensional row vectors with entries in F. The action of $\mathfrak{X}$ on V gives V an A-module structure. Now, (1.3) implies that $Q^{-1}P$ commutes with the action of A and may therefore be viewed as an A-module endomorphism of V. In other words, we may view $Q^{-1}P$ as an element of $\mathbf{E}_A(V)$. In particular, it follows from Corollary 1.6 that $Q^{-1}P = \lambda I$ for some $\lambda \in F$. We obtain that $P = \lambda Q$, as desired. □

Problem 1.3. Show that an algebra A is semisimple if and only if each A-module V is completely reducible.

Proof. If each A-module is completely reducible, then the regular action of A on itself, A°, also yields a completely reducible A-module, and so A is semisimple.

Conversely, let A be semisimple, and let V be an A-module. Write $A = \oplus_{i=1}^n I_i$ where the I_i are irreducible A-modules (or equivalently, minimal right ideals of A). Choose $v \in V$, and consider the A-module homomorphism from I_i into V given by $x \longmapsto v \cdot x$. Denote the image of this map as $W_{i,v}$. Since I_i is irreducible, its image under the given map is also irreducible. If we can show that V is a sum of the irreducible submodules $W_{i,v}$, then the claim will follow from Theorem 1.10. To see this, write

$$1 = \sum_{i=1}^{n} e_i, \text{ for } e_i \in I_i. \tag{1.4}$$

For arbitrary $v \in V$, we may write $v = v \cdot 1 = \sum_{i=1}^n v \cdot e_i \in \sum_{i=1}^n W_{i,v}$. Since v was arbitrary, it now follows that $V = \sum_{i=1}^n \sum_{v \in V} W_{i,v}$, and the claim now follows from Theorem 1.10. □

Problem 1.4. Let A be an algebra. For an A-module V, let $\operatorname{ann}(V) = \{a \in A \mid Va = 0\}$. Define $J(A) = \bigcap_{M \in \mathcal{M}(A)} \operatorname{ann}(M)$ where $\mathcal{M}(A)$ is a representative set of irreducible A-modules. Show

(a) $\mathrm{ann}(V)$ is an ideal of A for all V.
(b) $VJ(A) \subsetneq V$ for every nonzero A module V.
(c) $J(A)^n = 0$ for some integer n.
(d) If I is a right ideal of A such that $I^m = 0$ for some integer m, then $I \leqslant J(A)$.

Note: The ideal $J(A)$ is called the *Jacobson radical* of A; $\mathrm{ann}(V)$ is the annihilator of V and a right ideal I with $I^m = 0$ is said to be *nilpotent*.

Proof. For (a), to see that $\mathrm{ann}(V)$ is a right ideal, choose $a \in \mathrm{ann}(V)$ and $r \in A$. We see that $V \cdot (ar) = (V \cdot a) \cdot r = 0 \cdot r = 0$. Thus $ar \in \mathrm{ann}(V)$, and $\mathrm{ann}(V)$ is a right ideal. To see that $\mathrm{ann}(V)$ is a left ideal, notice that $V \cdot r \subseteq V$. So, $V \cdot (ra) \subseteq V \cdot a = 0$, and it follows that $ra \in \mathrm{ann}(V)$ and that $\mathrm{ann}(V)$ is a left ideal.

For part (b), let W be a maximal submodule of V. V/W is irreducible, so $J(A)$ annihilates V/W. Equivalently, $VJ(A) \subseteq W \subsetneq V$, as desired.

For part (c), observe that $J(A)^i$ is an A-module, and apply part (b) to obtain $J(A)^{i+1} \subsetneq J(A)^i$ if $J(a)^i$ is nonzero. Since $J(A)$ is finite dimensional, $J(A)^n = 0$ for some $n \leqslant \dim J(A)$.

For part (d), let $I \leqslant A$ with $I^m = 0$, and assume that $I \nsubseteq J(A)$. There then exists some irreducible module M such that $MI \neq \{0\}$. Since MI is a nonzero submodule of M, we have that $MI = M$. It follows that $MI^m = M$, but $I^m = 0$, so this is a contradiction. □

Problem 1.5. Prove that the following are equivalent for an algebra A:

(a) $J(A) = 0$.
(b) A has no nonzero nilpotent right ideals.
(c) A has no nonzero nilpotent two-sided ideals.
(d) A is semisimple.

Hint: If V is irreducible, then $\mathrm{ann}(V)$ is an intersection of maximal right ideals of A.

Proof. (a) $\implies$ (b) follows from part (d) of Problem 1.4. (b) $\implies$ (c) is obvious since any two-sided ideal is a right ideal. (c) $\implies$ (a) holds by part (b) of Problem 1.4 since $J(A)$ is a nilpotent two-sided ideal of A.

For (d) $\implies$ (a), we assume that A is semisimple and write $A = \oplus_{i=1}^{n} W_i$. Now, we rewrite the multiplicative identity 1 using this decomposition:

$$1 = \sum_{i=1}^{n} e_i \text{ for } e_i \in W_i. \tag{1.5}$$

Now, each e_i lies in an irreducible A-module, so $J(A)e_i = 0$ for each i, implying that $J(A) \cdot 1 = \sum_{i=1}^{n} J(A)e_i = 0$, forcing $J(A)$ to be zero.

For (a) $\implies$ (d), we use the hint provided. Choose an irreducible A-module M, and let $m \in M$. We define an A-module homomorphism by setting $a \longmapsto m \cdot a$. As M

is irreducible and the homomorphism is clearly nonzero, the map is surjective, and thus, the kernel is a maximal right ideal of A containing all elements $a \in A$ for which $m \cdot a = 0$. As such, we call the kernel of this map $\operatorname{ann}(m)$. Now, observe that

$$\operatorname{ann}(M) = \bigcap_{m \in M} \operatorname{ann}(m). \tag{1.6}$$

Finally, let $\mathcal{M}(A)$ be a representative set of irreducible A-modules. By the definition of $J(A)$ and the above equation, we have

$$J(A) = \bigcap_{M \in \mathcal{M}(A)} \bigcap m \in M \operatorname{ann}(m). \tag{1.7}$$

Now, $J(A) = 0$ by assumption. Since each $\operatorname{ann}(m)$ above is a maximal right ideal of A (and therefore a maximal A-submodule of A), it now follows from Problem 1.1 that A is completely reducible, i.e., semisimple. □

Problem 1.6. Let V be a completely reducible A-module. Show that A_V is semisimple.

Note: A consequence of Problem 1.6 is that the hypothesis that A is semisimple in the Double Centralizer Theorem 1.16 may be dropped since this theorem is really about A_M for an irreducible module M.

Proof. We shall appeal to Problem 1.5. Notice that we may view V and each of its A-submodules as A_V-modules. Also notice that any irreducible A-submodule of V remains irreducible when viewed as an A_V-submodule. In particular, since V is completely reducible as an A-module, it will remain completely reducible when viewed as an A_V-module. Fix $c \in J(A_V)$, and write $V = \bigoplus_{i=1}^{n} W_i$ where the W_i are irreducible A_V-submodules. Notice that $c \in \operatorname{ann}_{A_V}(W_i)$ for each i, and therefore, $c \in \operatorname{ann}_{A_V}(V)$. However, by definition, A_V is identified with a subalgebra of $\operatorname{End}(V)$, and therefore, c is the zero element of $\operatorname{End}(V)$. Since c was chosen arbitrarily from $J(A_V)$, we have that $J(A_V) = 0$, which implies A_V is semisimple. □

Problem 1.7. Let V be an irreducible A-module. Show that $|\mathcal{M}(A_V)| = 1$.

Proof. Assume for a contradiction that A_V possesses an irreducible module W which is not isomorphic to V. By Problem 1.6, A_V is semisimple, and so, Theorem 1.15 is applicable. We see that A_V possesses an ideal $W(A_V)$ contained in $\operatorname{ann}_{A_V}(V)$, and $W(A_V)$ is nonzero by Lemma 1.14. It now suffices to show that $\operatorname{ann}_{A_V}(V) = 0$. If c is an element of A_V which annihilates V, then since A_V is identified with a subalgebra of $\operatorname{End}(V)$, we would see that this element is necessarily identified with the zero transformation on V, forcing $c = 0$. Thus, $\operatorname{ann}_{A_V}(V) = 0$, and it follows that $W(A_V) = 0$ as well. □

Problem 1.8. Let $H \leqslant G$ be groups, and let F be a field with characteristic prime to $[G : H]$. Let V be an $F[G]$-module with submodule W. Suppose there exists an

$F[H]$-module $U_0 \leqslant V$ such that $V = W \oplus U_0$. Show there exists and $F[G]$-submodule $U \leqslant V$ such that $V = W \oplus U$.

Note: This generalization of Maschke's Theorem 1.9 is due to D. G. Higman.

Proof. The natural projection π onto W with respect to U_0 is an $F[H]$-module homomorphism, but not necessarily an $F[G]$-module homomorphism. Now choose a left transversal T of G with respect to H, and define

$$\vartheta(v) = \frac{1}{[G:H]} \sum_{t \in T} \pi(vt)t^{-1}. \tag{1.8}$$

It is clear that ϑ is $F[H]$-linear. We claim that ϑ is invariant under choice of a right transversal T, for if $t = t'h$ for some $h \in H$, then $\pi(vt)t^{-1} = \pi(vt'h)h^{-}1(t')^{-1} = \pi(vt')(t')^{-1}$.

We now show that ϑ is an $F[G]$-module homomorphism in a similar manner to the proof of Maschke's Theorem:

$$\vartheta(vg) = \frac{1}{[G:H]} \sum_{t \in T} \pi(vgt)t^{-1} = \frac{1}{[G:H]} \sum_{t \in T} \pi(vgt)(gt)^{-1}g. \tag{1.9}$$

If T is a left transversal, then so is $gT = \{gt \mid t \in T\}$ since left multiplication by g merely permutes cosets. We have established that ϑ is invariant under any choice of a transversal, so

$$\vartheta(vg) = \frac{1}{[G:H]} \sum_{t \in T} \pi(vgt)(gt)^{-1}g = \vartheta(v)g. \tag{1.10}$$

The proof concludes with deducing that $\ker \vartheta$ is an $F[G]$-submodule with complement W. □

Problem 1.9. Let F be a field of characteristic $p > 0$. Suppose $p \,\big|\, |G|$, and show that $F[G]$ is not semisimple.

Hint: $(\sum_{g \in G} g)^2 = 0$.

Proof. We appeal to Problem 1.5 by showing that $F[G]$ has a nonzero nilpotent ideal. Set $a = \sum_{g \in G} g$. By Cayley's Theorem, the maps from G to itself given by $x \longmapsto xg$ and $x \longmapsto gx$ are bijections for fixed g. For any $h \in G$, we therefore have $ha = ah = a$. In particular, we conclude that the one-dimensional subspace of $F[G]$ generated by a is an ideal of $F[G]$. This ideal is also nilpotent since $a^2 = |G|a = 0$, where the last equality holds since we assume $\mathrm{char}(F) \mid |G|$. The claim now follows from Problem 1.5. □

Problem 1.10. Let M be an A-module. Show that M is completely reducible if and only if $MJ(A) = 0$.

Hint: $A/J(A)$ is semisimple.

Proof. Assume M is completely reducible, and write $M = \oplus_{i=1}^{n} W_i$ for irreducible modules W_i. We have that $MJ(A) = \oplus_{i=1}^{n} [W_i J(A)]$. Since each W_i is irreducible, each $W_i J(A)$ is zero by the definition of $J(A)$, and $MJ(A) = 0$ consequently.

Conversely, let $MJ(A) = 0$. With the provided hint in mind, observe that M is an $A/J(A)$-module which is semisimple since $A/J(A)$ is semisimple (which is seen by showing that $J(A/J(A))$ is trivial). If we write $M = \oplus_{i=1}^{n} W_i$ for irreducible $A/J(A)$-submodules W_i, then the action of $A/J(A)$ extends to an action of A on M. Here, one sees that each W_i remains irreducible as an A-module. Hence, V is expressible as a direct sum of irreducible A-modules and is therefore completely reducible. $\square$

2. Group representations and characters

Problem 2.1. Prove the following.

(a) Let $\mathfrak{X}$ be an irreducible F-represenation of G over an arbitrary field. Show that $\sum_{g\in G}\mathfrak{X}(g)=0$ unless $\mathfrak{X}$ is the principal representation.
(b) Let $H\leqslant G$ and $g\in G$ be chosen so that all elements of Hg are in the same conjugacy class of G. Let χ be a $\mathbb{C}$-character such that $[\chi_H,1_H]=0$. Show $\chi(g)=0$.

Hint: (b) Compute the trace of $\sum_{h\in H}\mathfrak{X}(hg)$, where $\mathfrak{X}$ affords χ.

Proof. For part (a), we take a module theoretic approach. We first observe that the set
$I=\{\lambda\cdot\sum_{g\in G}g\mid\lambda\in F\}$ is a one-dimensional subalgebra of $F[G]$ since

$$\left(\sum_{g\in G}g\right)^2=|G|\sum_{g\in G}g. \tag{2.1}$$

Now, let V be the set consisting of $\dim(\mathfrak{X})$-row vectors with entries in F. We may view V as an $F[G]$-module under the action defined by $v\cdot g:=v\mathfrak{X}(g)$. Then, since V is irreducible, $VI=0$ or $VI=V$ since VI is necessarily a submodule of V. If $VI=0$, then right multiplication by $\sum_{g\in G}g$ is the linear transformation sending each $v\in V$ to zero. The matrix of this linear transformation (under the standard basis) is the zero matrix, so $\sum_{g\in G}\mathfrak{X}(g)=0$. Otherwise, if $VI=V$, then for each $v\in V$, we may write $v=w\cdot(\sum_{g\in G}g)$ for some $w\in V$. Now, for any $h\in G$, we have that

$$v\cdot h=w\cdot\left(\sum_{g\in G}g\right)h=w\cdot\left(\sum_{g\in G}g\right)=v. \tag{2.2}$$

In particular, each $h\in G$ acts as the identity transformation on V. Thus, $\mathfrak{X}(h)$ is the identity matrix for each $h\in G$. However, since $\mathfrak{X}$ is assumed to be irreducible, $\mathfrak{X}$ is necessarily the principal representation. This proves part (a).

For part (b), let $\mathfrak{X}$ afford the character χ. Our assumptions imply that $\mathfrak{X}_H$ decomposes as follows:

$$\begin{bmatrix}\mathfrak{N}_1 & & & \\ & \mathfrak{N}_2 & & \\ & & \ddots & \\ & & & \mathfrak{N}_n\end{bmatrix} \tag{2.3}$$

for irreducible, nonprincipal representations $\mathfrak{N}_i$ of H. Thus, by part (a), the sums $\sum_{h\in H}\mathfrak{N}_i(h)$ are zero for each i, and so, $\sum_{h\in H}\mathfrak{X}(h)=0$ as well. We prove the statement by computing $\sum_{h\in H}\mathfrak{X}(hg)$ in two ways. On one hand,

$$\sum_{h\in H}\mathfrak{X}(hg)=\left[\sum_{h\in H}\mathfrak{X}(h)\right]\cdot\mathfrak{X}(g)=0\cdot\mathfrak{X}(g)=0 \tag{2.4}$$

Also, since we are assuming any two elements of gH are conjugates, and since traces are invariant under conjugation by matrices,

$$\operatorname{tr}\left[\sum_{h\in H}\mathfrak{X}(hg)\right]=\sum_{h\in H}\operatorname{tr}\mathfrak{X}(hg)=\sum_{h\in H}\operatorname{tr}\mathfrak{X}(g)=|H|\chi(g) \tag{2.5}$$

Comnbining the above two equations, we have $|H|\chi(g)=0$, and so $\chi(g)=0$. □

Problem 2.2. Prove the following.

(1) Let χ be a character of G. Show that χ is afforded by a representation $\mathfrak{X}$ such that the entries of $\mathfrak{X}(g)$ for all g lie in some finite extension of the field $\mathbb{Q}$.
(2) Let $\epsilon=e^{2\pi i/n}$ where $n=|G|$. Let σ be a field automorphism of $\mathbb{Q}(\epsilon)$ and define χ^σ by $\chi^\sigma(g)=\sigma[\chi(g)]$. Show that χ^σ is a character and χ^σ is irreducible if and only if χ is irreducible.

Proof. The discussion on page 22 on $\bar{\mathbb{Q}}$-characters of a group G tells us that the set of irreducible $\bar{\mathbb{Q}}$-characters is exactly the same as the set of irreducible $\mathbb{C}$-characters of G. This implies that there is a $\bar{\mathbb{Q}}$-representation affording any $\mathbb{C}$-character. Since G is finite, the set of all entries of all matrices of this $\bar{\mathbb{Q}}$-representation generate a finite extension of $\mathbb{Q}$. (Recall that any extension of a field by finitely many algebraic elements is again a finite extension). This shows part (a).

Let $\mathfrak{X}$ be a K-representation affording χ for some finite extension K of $\mathbb{Q}$ containing $\mathbb{Q}(\epsilon)$. For each element of the Galois group of $\mathbb{Q}(\epsilon)$, we wish to choose a field automorphism defined on the field K which restricts to this automorphism. Let $\sigma\in\operatorname{Gal}(\mathbb{Q}(\epsilon))$, and let K' be the normal closure of K in $\mathbb{Q}$. A result from Galois theory states that $K'/\mathbb{Q}$ is a finite extension since $K/\mathbb{Q}$ is a finite extension, so we lose no generality in assuming that $\mathfrak{X}$ is a K'-representation. Another result from Galois theory states that the restriction map from $\operatorname{Gal}(K')$ to $\operatorname{Gal}(\mathbb{Q}(\epsilon))$ is surjective since both field extensions are Galois, so we may choose a $\tau\in\operatorname{Gal}(K')$ such that $\tau|_{\mathbb{Q}(\epsilon)}=\sigma$.

Now, define the action of τ on $\mathrm{GL}_n(K')$ by applying τ to each individual entry. It is routine to verify that this map is an isomorphism from $\mathrm{GL}_n e(K')$ onto itself, so $\tau\circ\mathfrak{X}$ is a K'-representation. Since $\tau|_{\mathbb{Q}(\epsilon)}=\sigma$ and since the values of χ lie in $\mathbb{Q}(\epsilon)$, we have that $\operatorname{tr}(\tau\circ\mathfrak{X})=\sigma(\operatorname{tr}\mathfrak{X})=\chi^\sigma$. It follows that χ^σ is a character of G under the representation $\tau\circ\mathfrak{X}$.

To see that χ^σ is irreducible if and only if χ is, we invoke the orthogonality relations. Observe that

$$[\chi^\sigma,\chi^\sigma]=\frac{1}{|G|}\sum_{g\in G}\chi^\sigma(g)\chi^\sigma(g^{-1})=\frac{1}{|G|}\sum_{g\in G}(\chi(g)\chi(g^{-1}))^\sigma \tag{2.6}$$

$$= \left(\frac{1}{|G|}\sum_{g\in G}\chi(g)\chi(g^{-1})\right)^{\sigma} = [\chi,\chi]^{\sigma}.$$

Since the inner product of any two characters is an integer, we have that $[\chi,\chi]^{\sigma} = [\chi,\chi]$, and so, $[\chi^{\sigma},\chi^{\sigma}] = [\chi,\chi]$. Now, for any character ψ of G, ψ is irreducible iff $[\psi,\psi] = 1$ by Corollary 2.17. It now follows that χ is irreducible iff χ^{σ} is irreducible. □

Problem 2.3. Let χ be a character of G. Define $\det\chi : G \to \mathbb{C}^{\times}$ as follows. Choose a $\mathfrak{X}$ affording χ, and set

$$(\det\chi)(g) = \det\mathfrak{X}(g)$$

Show that $\det\chi$ is a uniquely defined linear character of G.

Proof. The determinant is a group homomorphism from $\mathrm{GL}_n(\mathbb{C})$ into $\mathbb{C}^{\times}$. So, if $\mathfrak{X}$ is a representation affording χ, then $\det\circ\mathfrak{X}$ is a homomorphism from G into $\mathbb{C}^{\times}$ making $\det\chi$ a linear character of G.

To prove the second claim, we demonstrate that any representation similar to $\mathfrak{X}$ yields the same determinant character. In particular, for an invertible matrix P, we have

$$\det(P^{-1}\mathfrak{X}(g)P) = \det(P)^{-1}\det(\mathfrak{X}(g))\det(P) = \det(\mathfrak{X}(g)) = \det\chi(g) \tag{2.7}$$

for each $g \in G$, and from this, it follows that $\det\chi$ is invariant under any choice of $\mathfrak{X}$ affording χ. □

Problem 2.4. Prove the following.

(a). Let G be a nonabelian group of order eight. Show that G has a unique nonlinear irreducible character χ. Show $\chi(1) = 2$, $\chi(z) = -2$, and $\chi(y) = 0$ for $z \in G' - \{1\}$ and $y \in G - G'$.
(b). If $G \cong D_8$, show that $\det\chi \neq 1_G$.
(c). If $G \cong Q_8$, show that $\det\chi = 1_G$.

Hint: Show that $\ker(\det\chi)$ contains all elements of order 4. Use Lemma 2.15.

Note: Although D_8 and Q_8 have identical character tables, the map $\det : \mathrm{Irr}(G) \to \mathrm{Irr}(G)$ is not the same for both groups.

Proof. D_8 and Q_8 are the only nonabelian groups of order 8 up to isomorphism. For the moment, letting G be either of these these groups, one sees that G' $(= \mathbf{Z}(G))$ has order 2 and that G/G' is isomorphic to the Klein-four group. From the latter fact, we see that G has exactly four linear characters by Corollary 2.23(b). Denote these characters as χ_i for $i = 1,2,3,4$ with $\chi_1 = 1_G$. Combining this information with the

formula $|G| = \sum_{\chi \in \mathrm{Irr}(G)} \chi(1)^2$, we obtain that

$$8 = |G| = \sum_{i=1}^{k} \chi_i(1)^2 = 1 + 1 + 1 + 1 + \sum_{i=5}^{k} \chi_i(1)^2, \tag{2.8}$$

where k is the number of irreducible characters of G. Notice that for $i \geqslant 5$, each $\chi_i(1) \geqslant 2$ since all four linear characters have been accounted for already. Thus, one sees that the rightmost sum above consists only of a single term whose value is four. Thus, $\chi_5(1) = 2$.

To calculate the values of χ_5, we begin with the element $z \in G' - 1$. Notice, that since $z \in G'$, z lies in the kernel of each of the linear characters. So, by using the regular character of G, we have

$$0 = \rho_G(z) = \sum_{i=1}^{5} \chi(1)\chi(z) = 1 + 1 + 1 + 1 + 2\chi_5(z). \tag{2.9}$$

Solving for $\chi_5(z)$ yields $\chi_5(z) = -2$, as desired.

Any nonprincipal irreducible character of the Klein-four group takes the value 1 on the identity and on exactly one other nonidentity element. It also takes the value -1 on the remaining two elements. Thus, since we may identify irreducible characters of G/G' with irreducible characters of G in a natural way (see Lemma 2.22(b)), it now follows that for any element of $G - G'$, two of the four nonprincipal linear characters will take the value -1 on this element, and the remaining nonprincipal linear character will take the value 1 on this element. Now, evaluating the regular character of G at any $y \in G - G'$, we obtain

$$0 = \rho_G(y) = \sum_{i=1}^{5} \chi(1)\chi(y) = 1 + 1 - 1 - 1 + 2\chi_5(y). \tag{2.10}$$

Again, solving for $\chi_5(y)$ gives $\chi_5(y) = 0$, as desired. This completes (a).

Before proving that (b) or (c), we first prove that $\ker \det \chi_5$ contains all elements of order four in G. We see that χ_5 is faithful from part (a), so if $\mathfrak{X}$ affords χ_5, then $\mathfrak{X}$ is injective. If $o(g) = 4$, the injectivity of $\mathfrak{X}$ implies $\mathfrak{X}(g)$ has order four, so $\mathfrak{X}(g)$ has at least one of its two eigenvalues equal to a primitive fourth root of unity, say $\pm i$. If $g \in G - G'$, then $\chi_5(g) = 0$. This implies that the other eigenvalue of $\mathfrak{X}(g)$ is $\mp i$. Now, the determinant of $\mathfrak{X}(g)$ is $\pm i \cdot \mp i = 1$.[1] This proves our claim.

Since each element of $Q_8 - Q_8'$ has order four and since each linear character of Q_8 contains Q_8' in its kernel, this implies that $\det \chi_5$ is the principal character of G when $G = Q_8$.

[1]Notice here that we are using a well-known fact from linear algebra, which states that the trace and determinant of a matrix is obtained respectively by taking the sum and product of the eigenvalues of the matrix. This can be reasoned easily with Jordan canonical form.

On the other hand, there exists $g \in D_8 - D_8'$ having order 2. Again, since $\mathfrak{X}$ is faithful, this implies that at least one of the eigenvalues of $\mathfrak{X}(g)$ is a primitive second root of unity, namely -1. Since $\chi_5(g) = 0$, this implies that the other eigenvalue must be 1. Thus, the determinant of $\mathfrak{X}(g)$ is $-1 \cdot 1 = -1$. Hence the determinant of χ_5 in this case is not the principal character. □

Problem 2.5. Find a real representation of D_8 which affords the irreducible character χ of degree two. Show that such a real representation does not exist for Q_8.

Proof. For D_8, we use the presentation

$$D_8 = \langle a, b \mid a^4 = b^2 = 1, ba = a^3b \rangle \tag{2.11}$$

A real representation $\mathfrak{X}$ is obtained from imagining the action of D_8 on four points arranged in a square and centered at the origin in $\mathbb{R}^2$. In particular, D_8 acts naturally on the set $\{(0,1),(1,0),(0,-1),(-1,0)\} \subset \mathbb{R}^2$. Here, a rotates these points around the origin 90°, and b reflects the points over the y-axis. Matrix representations for these actions are given below:

$$\mathfrak{X}(a) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ and } \mathfrak{X}(b) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{2.12}$$

One verifies that the character of the representation described above is the character χ_5 in Problem 2.4.

For Q_8, we use the presentation $\langle a, b, \bar{e} \mid \bar{e}^2 = 1, a^2 = b^2 = \bar{e}, ba = a^3b \rangle$. Now, assume for a contradiction that $\mathfrak{X}$ is a real representation of Q_8 affording the its unique irreducible character of degree 2. We have established in Problem 2.4 that $\mathfrak{X}(a)$ has eigenvalues i and $-i$. Thus, since $\mathfrak{X}(a)$ is diagonalizable (having finite order), this matrix is similar to any other 2×2 matrix having these two eigenvalues. As such, we may replace $\mathfrak{X}$ with a similar representation $\mathfrak{X}'$ such that

$$\mathfrak{X}'(a) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{2.13}$$

which is seen to have eigenvalues i and $-i$ as well. Now, since $\mathfrak{X}(a)$ and $\mathfrak{X}'(a)$ are similar, real matrices, they are similar via a conjugation from a real matrix (see Lemma 1 below for a proof of this fact). Thus, $\mathfrak{X}'$ is real since $\mathfrak{X}$ is real. It is therefore of no loss to consider $\mathfrak{X}'$ instead of $\mathfrak{X}$.

Now, observe that $\mathfrak{X}'(b)$ takes the form

$$\mathfrak{X}'(b) = \begin{bmatrix} x & z \\ y & -x \end{bmatrix} \tag{2.14}$$

for $x, y, z \in \mathbb{R}$. Notice the the two entries on the main diagonal are necessarily negatives of each other since $\operatorname{tr} \mathfrak{X}'$ takes the value zero on elements of order 4 in Q_8 by Problem 2.4.

Now, the relation $ba = a^3b$ implies that

$$\begin{bmatrix} x & z \\ y & -x \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x & z \\ y & -x \end{bmatrix}, \tag{2.15}$$

which yields

$$\begin{bmatrix} z & -x \\ -x & -y \end{bmatrix} = \begin{bmatrix} y & -x \\ -x & -z \end{bmatrix}, \tag{2.16}$$

which implies $y = z$. Also, since the determinant character of this representation is principal by Problem 2.4, we have that

$$\det \mathfrak{X}'(b) = -x^2 - yz = 1. \tag{2.17}$$

Using the fact that $y = z$, we obtain that $x^2 + y^2 = -1$, which has no real solutions. Thus, $\mathfrak{X}'(a)$ and $\mathfrak{X}'(b)$ cannot be real simultaneously, and the result now holds. □

Sublemma. Let A and B be matrices in $\mathbb{R}$ which are similar via a complex matrix P. There then exists a real matrix Q such that $A = QBQ^{-1}$.

Proof. We may write P into its real and imaginary part: $P = X + iY$. Now, $AX + iAY = XB + iBY$. Since $\{1, i\}$ is a basis of $\mathbb{C}$ over $\mathbb{R}$, we obtain that $AX = XB$ and $AY = YB$. (Note that although Y and X are real, they aren't necessarily invertible, so we are not finished yet.) In particular, we may write $A(X + tY) = (X + tY)B$ for $t \in \mathbb{C}$. It now suffices to show there exists a real number t for which $\det(X + tY) \neq 0$. However, this determinant is a polynomial in t, and this polynomial is nonzero since evaluating the polynomial at i gives a nonzero value. Thus, this polynomial has finitely many zeros, and there then exists a real value of t for which $\det(X + tY) \neq 0$. Now, $X + tY$ is real and conjugates B to A. □

Problem 2.6. Let χ and ψ be characters of G. Define $\chi\psi : G \to \mathbb{C}$ by $(\chi\psi)(g) = \chi(g)\psi(g)$.

(a). If $\psi(1) = 1$, show $\chi\psi$ is a character.
(b). If $\psi(1) = 1$, show that $\chi\psi \in \mathrm{Irr}(G)$ if and only if $\chi \in \mathrm{Irr}(G)$.
(c). If $\psi = \bar{\chi}$ and $\chi(1) > 1$, show that $\chi\psi \notin \mathrm{Irr}(G)$.

Note: In Chapter 4, we will show that $\chi\psi$ is always a character.

Proof. For (a), let $\psi(1) = 1$ and let χ be afforded by $\mathfrak{X}$. Since ψ is a group homomorphism of G into $\mathbb{C}$, and scalars commute with matrices, we can define a group homomorphism from G into $\mathrm{GL}_{\chi(1)}(\mathbb{C})$ by mapping $g \longmapsto \psi(g)\mathfrak{X}(g)$. This representation affords $\chi\psi$.

For part (b), assume that $\chi, \psi \in \mathrm{Irr}(G)$ with $\psi(1) = 1$. Notice that since $\psi(g)$ is a root of unity for all $g \in G$, we have $\psi(g)\overline{\psi(g)} = 1$ for all $g \in G$. By part (a), $\psi\chi$ is

a character, so we need only prove that $[\psi\chi, \psi\chi] = 1$ by Corollary 2.17. Using the orthogonality relations, we obtain:

$$[\psi\chi, \psi\chi] = \frac{1}{|G|}\sum_{g\in G}\psi(g)\chi(g)\overline{\psi(g)\chi(g)} = \frac{1}{|G|}\sum_{g\in G}\psi(g)\overline{\psi(g)}\chi(g)\overline{\chi(g)} \tag{2.18}$$

$$= \frac{1}{|G|}\sum_{g\in G}\psi(g)\overline{\psi(g)}\chi(g)\overline{\chi(g)} = \frac{1}{|G|}\sum_{g\in G}\chi(g)\overline{\chi(g)} = [\chi, \chi] = 1.$$

This proves the reverse direction of (b). For the forward direction, if we assume that $\psi\chi$ is irreducible, then multiplying $\psi\chi$ by $\overline{\psi}$ yields χ, which is irreducible by the forward direction. This proves the forward direction of (b).

We prove part (c) now. Let $\chi \in \mathrm{Irr}(G)$ be nonlinear, and set $\psi = \overline{\chi}$. We apply the orthogonality relations. Since χ is a character, $[\chi, \chi] \geqslant 1$. Applying this, we obtain:

$$1 \leqslant [\chi, \chi] = \frac{1}{|G|}\sum_{g\in G}\chi(g)\bar{\chi}(g) = [\chi\psi, 1_G]. \tag{2.19}$$

Thus, 1_G is a constituent of $\chi\psi$. Thus, $\chi\psi$ is irreducible iff $\chi\psi = 1_G$. This implies though that $\chi(1) = \psi(1) = 1$, contrary to our assumptions. □

Problem 2.7. Let G be an abelian group and write $\hat{G} = \mathrm{Irr}(G)$.

(a). Show that $\hat{G}$ is an abelian group under the multiplication defined in Problem 2.6.
(b). If $H \leqslant G$, let $H^{\perp} = \{\lambda \in \hat{G} \mid H \subseteq \ker\lambda\}$. Show that $\perp$ is a bijection from the subgroups of G onto the set of subgroups of $\hat{G}$.
(c). Prove that $G \cong \hat{G}$.

Hints: There is a natural isomorphism of G onto $\hat{\hat{G}}$. Use this for (b). For (c), use the fundamental theorem of abelian groups.

Remark: The goal of this problem is to incrementally work towards a proof of the fact that an abelian group is isomorphic to its character group. It is interesting that these two groups are not canonically isomorphic, i.e., one cannot construct an explicit isomorphism between these two groups in full generality. Interesting examples of uncanonical isomorphisms are somewhat rare in mathematics, and in some sense, they deserve to be celebrated. However, this will require us to supply an indirect (if not slightly awkward) proof.

Proof. Under this operation, the principal character 1_G is the identity. Closure under the operation follows from Problem 2.4(a). Associativity and commutativity are clear. If $\lambda \in \hat{G}$, then $\bar{\lambda}$ is the inverse.

For (b), there are numerous details to be verified which are routine computations and are thus better left to the reader, so we merely sketch the proof here. Given

$H \leqslant G$, it is clear that $H^\perp$ is a subgroup of $\hat{G}$. We first demonstrate that there is a natural isomorphism between G and $\hat{\hat{G}}$. For $g \in G$, we set $\hat{g}(\lambda) = \lambda(g)$. The function $\hat{g}$ clearly defines an irreducible character of G, and one easily verifies that $\hat{g} = \hat{h}$ iff $g = h$, and so $\hat{\hat{G}}$ is exactly the set $\{\hat{g} \mid g \in G\}$. One then easily verifies that the map $g \longmapsto \hat{g}$ is an isomorphism from G to $\hat{\hat{G}}$. One then sees that $H^{\perp\perp} = \{\hat{g} \in \hat{\hat{G}} \mid g \in H\}$, and so the map $H \longmapsto H^{\perp\perp}$ is a bijection of the subgroups of G onto the subgroups of $\hat{\hat{G}}$. In particular, since $\perp\perp$ is a bijection of subgroups, it follows that $\perp$ injectively maps the subgroups of G into the subgroups of $\hat{G}$ and that $\perp$ surjectively maps the subgroups of $\hat{G}$ onto the subgroups of $\hat{\hat{G}}$. However, setting $K = \hat{G}$, by the same logic, $\perp\perp$ is a bijection from the subgroups of K to those of $\hat{\hat{K}}$, implying that $\perp$ is an injection (and therefore also a bijection) from $\hat{G}$ to $\hat{\hat{G}}$. So, $\perp$ therefore has an inverse $\circ : \hat{\hat{G}} \to \hat{G}$. From this, it follows that $\perp = \perp\perp \circ$ is a bijection (being a composition of bijections) from the subgroups of G onto the subgroups of $\hat{G}$, as desired.

Part (c) is proven by first showing that $G \cong \hat{G}$ when G is cyclic of prime power order. Indeed, one sees that the map $\perp$ reverses containment, i.e., if $H \leqslant K \leqslant G$, then $K^\perp \leqslant H^\perp$, and since the subgroup lattice of a finite group is a chain iff it is cyclic of prime power order and since $\perp$ is a bijective map of subgroups, the claim follows. Since any finite abelian group is expressible as a direct product of cyclic groups of prime power order, we may inductively assume that (c) holds for the abelian groups G_1 and G_2, that is, $G_1 \cong \hat{G}_1$ and $G_2 \cong \hat{G}_2$. Now, one verifies that $\hat{G}_1 \times \hat{G}_2 \cong \widehat{G_1 \times G_2}$ via the map $(\lambda_1, \lambda_2) \longmapsto \lambda_1 \times \lambda_2$ where $(\lambda_1 \times \lambda_2)(x, y) = \lambda_1(x)\lambda_2(y)$. Thus, $G_1 \times G_2 \cong \widehat{G_1 \times G_2}$, proving part (c). □

Problem 2.8. Let χ be a faithful character of G. Show that $H \leqslant G$ is abelian if and only if every irreducible constituent of χ_H is linear.

Proof. If H is abelian then every irreducible character of H is linear, so therefore, the restriction of χ to H will only have linear irreducible constituents.

Conversely, assume that each irreducible constituent of χ_H is linear, and χ is faithful. We appeal to Corollary 2.23(a) by showing that $H' = 1$ which is equivalent ot H being abelian. Now, since χ is faithful, its restriction to H is faithful as well. Write $\chi_H = \sum_i n_i \psi_i$ for linear characters ψ_i of H. The intersection of the kernels of each ψ_i is trivial since χ_H has trivial kernel, so the intesection of the kernels of all linear characters of H is trivial. By Corollary 2.23(a), $H' = 1$, and H is abelian. □

Problem 2.9. Prove the following statements.

(a). Let χ be a character of an abelian group A. Show that

$$\sum_{x \in A} |\chi(x)|^2 \geqslant |A|\chi(1)$$

(b). (b) Let $A \leqslant G$ with A abelian and $[G : A] = n$. Show that $\chi(1) \leqslant n$ for all $\chi \in \mathrm{Irr}(G)$.

Proof. Write $\chi = \sum_i n_i\psi_i$ for nonzero n_i and irreducible ψ_i. We have

$$|\chi(x)|^2 = \chi(x)\bar{\chi}(x) = \sum_i n_i^2\psi_i(x)\bar{\psi}_i(x) + \sum_{i\neq j} n_i n_j \psi_i(x)\bar{\psi}_j(x)$$

If we sum the reightmost expression over $x \in A$, the sum on the right is zero by the first Orthogonality Relations, so we have that

$$\frac{1}{|A|}\sum_{x\in A}|\chi(x)|^2 = \frac{1}{|A|}\sum_{x\in A}\sum_i n_i^2\psi_i(x)\bar{\psi}_i(x) \geqslant \frac{1}{|A|}\sum_{x\in A}\sum_i n_i\psi_i(x)\bar{\psi}_i(x) = \chi(1)$$

Multiplying the leftmost and rightmost expressions by $|A|$ yields the desired result for part (a).

For part (b), we have the following inequality:

$$|A|\chi(1) \leqslant \sum_{x\in A}|\chi(x)|^2 \leqslant \sum_{x\in G}|\chi(x)|^2 = |G|[\chi,\chi] = |G|$$

Thus, $\chi(1) \leqslant [G : A]$. □

Problem 2.10. Suppose $G = \cup_{i=1}^n A_i$ where each A_i is an abelian subgroup of G and $A_i \cap A_j = 1$ for $i \neq j$.

(a). Let $\chi \in \mathrm{Irr}(G)$. Show that if $\chi(1) > 1$, then $\chi(1) \geqslant |G|/(n-1)$.

(b). If G is nonabelian, then $|A_i| \leqslant n-1$ for each i, and $n - 1 \geqslant \sqrt{|G|}$.

Hints: For (a), bound $\sum_{g\in G}|\chi(g)|^2$ using Problem 2.9(a). For (b), use Problem 2.9(b).

Proof. For part (a), by our assumptions we have $|G| = 1 + \sum_{i=1}^n |A_i - \{1\}|$. So, using Problem 2.9(a), we have

$$\chi(1)[|G| + (n-1)] = \chi(1)\sum_{i=1}^n |A_i| \leqslant \sum_{i=1}^n\sum_{x\in A}|\chi(x)|^2$$

$$= \sum_{g\in G}|\chi(g)|^2 + (n-1)\chi(1)^2 = |G| + (n-1)\chi(1)^2.$$

By rearranging the leftmost and rightmost sides of the inequality above, we obtain

$$|G|(\chi(1)-1)/(n-1) \leqslant \chi(1)(\chi(1)-1)$$

Since $\chi(1) > 1$, we divide both sides by $\chi(1) - 1$, and we have the desired result for part (a).

For part (b), we use the inequality from part (a), and divide both sides by $|A_i|$:

$$\frac{\chi(1)}{|A_i|} \geqslant \frac{[G : A_i]}{(n-1)}.$$

By Problem 2.9(b), we have that $|G : A_i| \geqslant \chi(1)$ for each i from which, we obtain:

$$\frac{\chi(1)}{|A_i|} \geqslant \frac{[G : A_i]}{n-1} \geqslant \frac{\chi(1)}{n-1}$$

We obtain $|A_i| \leqslant n-1$ by considering the right and left sides in the inequality above. Using this inequality, we now write

$$|G| + (n-1) = \sum_{i=1}^{n} |A_i| \leqslant n(n-1) = (n-1)^2 + (n-1),$$

from which, we obtain $|G| \leqslant (n-1)^2$. □

Problem 2.11. Let $g \in G$. Show that g is conjugate to g^{-1} if and only if $\chi(g)$ is real for all characters χ of G.

Note: An element of a group which is conjugate to its inverse is called a *real* element. If G has any real elements other than 1, then G must necessarily have even order.

Proof. First assume $\chi(g)$ is real for each χ of G. We use the Second Orthogonality Relation:

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(g)\overline{\chi(g^{-1})} = \sum_{\chi \in \mathrm{Irr}(G)} \chi(g)^2 \tag{2.20}$$

This expression either equals $|C_G(g)|$ or equals zero depending on whether g is conjugate to its inverse or not. Since the principal character is included in the above sum and since each $\chi(g)$ is real, this expression is strictly greater than zero. Therefore, the sum must equal $|C_G(g)|$, and we conclude g is conjugate to g^{-1}.

Conversely, let g be conjugate to g^{-1}. We have that

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(g)\overline{\chi(g^{-1})} = \sum_{\chi \in \mathrm{Irr}(G)} \chi(g)^2 = |C_G(g)| \tag{2.21}$$

Also, since g is conjugate to itself, we have

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(g)\overline{\chi(g)} = \sum_{\chi \in \mathrm{Irr}(G)} |\chi(g)|^2 = |C_G(g)| \tag{2.22}$$

Combining these two equations, we have that $\sum_{\chi \in \mathrm{Irr}(G)} \chi(g)^2 = \sum_{\chi \in \mathrm{Irr}(G)} |\chi(g)|^2$. Now, observe that $\Re(\chi(g)^2) \leqslant |\chi(g)|^2$ with equality holding iff $\chi(g)^2$ is real. □

Problem 2.12. Let $|G| = n$, and let $g \in G$. Show that $\chi(g)$ is rational for every character χ of G iff g is conjugate to g^m for every m such that $\gcd(n, m) = 1$.

Hints: Let ϵ be a primitive nth root of 1 in $\mathbb{C}$ and let $E = \mathbb{Q}(\epsilon)$. Let $\mathscr{G}$ be the Galois group of E over $\mathbb{Q}$. Given $\gcd(m, n) = 1$, show that there exists $\sigma \in \mathscr{G}$ with $\chi(g^m) = \chi(g)^\sigma$ for all $g \in G$ and all characters χ. Conversely, for every $\sigma \in \mathscr{G}$, there is an m such that this formula holds.

Proof. First, notice that it clearly suffices to prove the problem under the assumption that χ is irreducible.

Let G be a group, and let E denote the field generated by a primitive $|G|$-th root of unity, and consider $\sigma \in \mathrm{Gal}(E/\mathbb{Q})$. The action of σ is determined by where it sends a primitive $|G|$-th root of unity ϵ, and σ must send a primitive n-th root of unity to another primitive nth root of unity for any n. There thus exists an integer m with $\gcd(m, |G|) = 1$ such that $\epsilon^\sigma = \epsilon^m$ for all roots of unity lying in E. Conversely, if m is an integer such that $\gcd(m, |G|) = 1$, then one routinely verifies that the map $\sigma : E \to E$ defined by $\epsilon \longmapsto \epsilon^m$ defines a Galois automorphism of E. Thus, the group $\mathrm{Gal}(E/\mathbb{Q})$ is naturally isomorphic to $(\mathbb{Z}/|G|\mathbb{Z})^\times$.

Let $\chi \in \mathrm{Irr}(G)$, and let $\mathfrak{X}$ be a representation affording χ. Given $g \in G$, $\chi(g)$ is the sum of eigenvalues of $\mathfrak{X}(g)$, and the eigenvalues of $\mathfrak{X}(g)$ are $|G|$-th roots of unity. Thus, if $\mathfrak{X}(g)$ has (possibly indistinct) eigenvalues $\{\epsilon_i\}$, then $\chi(g)^\sigma = \sum_i \epsilon_i^m = \chi(g^m)$ (where the last equality is seen by considering the diagonalization of $\mathfrak{X}(g)$).

Now, assume that $g \in G$ is conjugate to every element of the form g^m for $\gcd(m, |G|)$. By the above paragraph, this implies that $\chi(g)$ is fixed by every Galois automorphism of E, which happens only if $\chi(g)$ is rational. Since χ was arbitrary, this implies that the "if" direction.

Conversely, if $\chi(g)$ is rational for each $\chi \in \mathrm{Irr}(G)$, then this implies that $\chi(g)^\sigma = \chi(g)$ for each $\sigma \in \mathrm{Gal}(E)$. In particular, by the first paragraph above, we have that $\chi(g^m) = \chi(g)$ for each m such that $\gcd(m, |G|) = 1$. In particular, $\chi(g^m)\overline{\chi(g)}$ is a nonnegative rational number for each such m. By applying the Second Orthogonality relations, we obtain that

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(g^m)\overline{\chi(g)} = \sum_{\chi \in \mathrm{Irr}(G)} \chi(g)^2 > 0,$$

where m satisfies $\gcd(m, |G|) = 1$ and where the strict inequality holds since the term corresponding to $\chi = 1_G$ is of course positive. □

Problem 2.13. Let $|G'| = p$ a prime. Assume that $G' \subseteq \mathbf{Z}(G)$. Show that $\chi(1)^2 = |G : Z(G)|$ for every nonlinear $\chi \in \mathrm{Irr}(G)$.

Proof. Since $G' \subseteq \mathbf{Z}(G) \subseteq \mathbf{Z}(\chi)$, we know that $G/\mathbf{Z}(\chi)$ is abelian. By Theorem 2.31, we know that $[G : \mathbf{Z}(\chi)] = \chi(1)^2$, so it now suffices to show that $\mathbf{Z}(G) = \mathbf{Z}(\chi)$. We begin by showing that $\ker\chi \cap G' = 1$. Since $|G'| = p$, either $\ker\chi \cap G' = 1$ or

$G' \subseteq \ker\chi$. In the latter case, χ defines a character on the abelian group G/G' which is an abelian group, so χ is a linear character contrary to our assumption. We can now assume that $\ker\chi \cap G' = 1$.

We now show that $\mathbf{Z}(\chi) = \mathbf{Z}(G)$. Let $g \in G$ and $z \in \mathbf{Z}(\chi)$. We of course have that $[g,z] \in G'$. Also, since $\mathbf{Z}(\chi)/\ker\chi = \mathbf{Z}(G/\ker\chi)$, we have that $[g,z] \in \ker\chi$, so $[g,z] \in G' \cap \ker\chi = 1$ for each $g \in G$, and $z \in \mathbf{Z}(\chi)$. Thus, each $z \in \mathbf{Z}(\chi)$ commutes with every $g \in G$. $\mathbf{Z}(\chi) \subseteq Z(G)$. The other direction of containment holds from Corollary 2.28. It now follows that $\chi(1)^2 = [G : \mathbf{Z}(G)]$. □

Problem 2.14. Let $H \subseteq G' \cap Z(G)$ be cyclic of order n and let m be the maximum order of elements of G/H. Assume that n is a prime power and show that $|G| \geqslant n^2 m$.

Hints: Choose $\chi \in \mathrm{Irr}(G)$ with $H \cap \ker\chi = 1$. Let $\lambda = \det\chi$ (as in Problem 2.3). We have $\chi_H = \chi(1)\mu$ with $\mu \in \mathrm{Irr}(H)$. Using λ and μ, show that $\chi(1) \geqslant n$. Finish the proof using Problem 2.9(b).

Note: The assumption on n can be removed using Corollary 5.4.

Proof. Since H is cyclic of order $n = p^a$, it has a unique subgroup of order p. Since $\bigcap\{\ker\chi \mid \chi \in \mathrm{Irr}(g)\} = 1$, there is a $\chi \in \mathrm{Irr}(G)$ such that $\ker\chi$ intersects this subgroup of order p trivially. Therefore, a $\chi \in \mathrm{Irr}(G)$ can be chosen so that $\ker\chi \cap H = 1$. Since $H \subseteq Z(G)$, we have that $\chi_H = \chi(1)\mu$ for some $\mu \in \mathrm{Irr}(H)$. Further, since $H \subseteq G'$, we have that $\mu^{\chi(1)} = \det\chi = 1$.

Now, since χ is faithful on H, μ is then a faithful linear character. Now, letting $H = \langle h \rangle$, then $\mu(h)$ is a primitive n-th root of unity. Since $\mu(h)^{\chi(1)} = 1$, this implies that $\chi(1) \geqslant n$.

Let m be the maximum order of any element of G/H, and let $o(kH) = m$. Notice that $K = \langle h, k \rangle$ is an abelian subgroup of G having order nm. By Problem 2.9(b) and the inequality from the previous paragraph, we have $n \leqslant \chi(1) \leqslant [G : K]$. We conclude that $|G| = |G : K|nm \leqslant n^2 m$, as desired. □

Problem 2.15. Let $\chi \in \mathrm{Irr}(G)$ be faithful, and suppose $H \leqslant G$ and $\chi_H \in \mathrm{Irr}(H)$. Show that $C_G(H) = \mathbf{Z}(G)$.

Proof. The relation $\mathbf{Z}(G) \subseteq C_G(H)$ is obvious so we prove the other direction of containment. Let $g \in C_G(H)$. We appeal to Lemma 2.25. Let $\mathfrak{X}$ afford the irreducible, faithful character χ of G. Since $\mathfrak{X}_H$ is irreducible and $\mathfrak{X}(g)$ commutes with every $\mathfrak{X}(h)$ for $h \in H$, we have that $\mathfrak{X}(g) = \epsilon I$ for some $\epsilon \in \mathbb{C}$ by Lemma 2.25. Thus, by Lemma 2.27(a), $g \in \mathbf{Z}(\chi)$. Since χ is faithful and irreducible, $\mathbf{Z}(\chi) = \mathbf{Z}(G)$, so $g \in \mathbf{Z}(G)$. □

Problem 2.16. Let $H \leqslant G$. Let χ be a character of G which vanishes on $G - H$. Assume either $H = 1$ or G is abelian. Show that $|G : H|$ divides $\chi(1)$.

Hint: Let λ be an irreducible constituent of χ_H. Under either hypothesis, find $\mu \in \mathrm{Irr}(G)$ with $\mu_H = \lambda$. Compute $[\chi, \mu]$ and conclude $|G : H| \mid [\chi_H, \lambda]$.

Proof. If $H = 1$, then χ notice $\chi = a\rho_G$ for some $a \in \mathbb{C}$. We claim that a is a nonnegative integer. Indeed, we have

$$[\chi, 1_G] = a[\rho_G, 1_G] = a.$$

Since χ is a character, a must be a nonnegative integer. Thus, $\chi(1) = a|G|$, and the claim follows in this case.

Let G be abelian, and write $\chi = \sum_{\mu \in \mathrm{Irr}(G)} n_\mu \mu$ for nonnegative integers n_μ. We also write $\chi_H = \sum_{\lambda \in \mathrm{Irr}(H)} m_\lambda \lambda$. Let $(\mu)_H = \lambda$ for some $\mu \in \mathrm{Irr}(G)$. Since χ vanishes on $G - H$, we have by the First Orthogonality Relations that

$$m_\lambda |H| = |H|[\chi_H, \lambda]_H = |G|[\chi, \mu]_G = n_\mu |G| \tag{2.23}$$

from which it follows that m_λ is a multiple of $|G : H|$. Since χ_H is the restriction of the character χ, any irreducible constituent λ of χ_H is the restriction of some irreducible constituent μ of χ. Thus, by the above equation, $m_\lambda = m'_\lambda |G : H|$ for some integer m'_λ, where λ is any irreducible constituent of χ_H. From this, we obtain

$$\chi_H(1) = \sum_\lambda m_\lambda \lambda(1) = |G : H| \sum_\lambda m'_\lambda,$$

and it follows that $|G : H|$ divides $\chi(1)$. □

Problem 2.17. Let $A < G$ be abelian, and assume there exists $\chi \in \mathrm{Irr}(G)$ with $\chi(1) = [G : A]$. Show that G has a nontrivial normal abelian subgroup.

Hint: Show that χ vanishes on $G - A$.

Proof. From Problem 2.9(a), we have that

$$|A|\chi(1) \leqslant \sum_{x \in A} |\chi(x)|^2 \leqslant \sum_{x \in G} |\chi(x)|^2 = |G|[\chi, \chi] = |G|$$

If $\chi(1) = [G : A]$, then the inequalities above become equalities. We deduce that χ vanishes on $G - A$. Now, the conjugacy classes on which χ does not vanish generate a subgroup of A which is necessarily normal in G. This normal subgroup is of course nontrivial, or otherwise χ would be the regular character of G, which cannot be irreducible unless $G = 1$. □

Problem 2.18. Let $A \lhd G$, and suppose $C_G(a) = A$ for each $a \in A - \{1\}$. Assume that G/A is abelian. Show that G has exactly $(|A| - 1)/[G : A]$ nonlinear irreducible characters, and that these all have degree equal to $[G : A]$ and vanish on $G - A$. Assume that $A < G$.

Note: The group G of Problem 2.18 is a special case of a Frobenius group. The character theory of such groups will be discussed more fully later. Observe that although the hypotheses of 2.18 are very special, this situation does arise frequently. Some examples are A_4 and nonabelian groups of order pq where p and q are primes with $p/(q - 1)$.

Proof. Taking the advice of the hint and setting $k = |\operatorname{Irr}(G)|$, we show by counting conjugacy classes that

$$k \leqslant 1 + (|A| - 1)/|G : A| + (|G| - |A|)/|A| \tag{2.24}$$

and then, using characters, show that

$$k \geqslant |G : A| + (|G| - |G : A|)/|G : A|^2 \tag{2.25}$$

Both the right sides of (2.24) and (2.25) are easily verified to be the same expression, so this will show that $k = (|A| - 1)/|G : A| + |G : A|$. Then, by deducing that $A = G'$, we will have shown that the number of nonlinear irreducible characters is $(|A| - 1)/|G : A|$ as desired.

We begin by proving (2.24). Since each nonidentity element of A centralizes A, we deduce that the size of conjugacy classes to which such an element belong is $|G : A|$. The union of these conjugacy classes is exactly A since A is normal. Therefore, the number of conjugacy classes contained in A is precisely $1 + (|A| - 1)/|G : A|$. We claim that the number of conjugacy classes of $G - A$ is less than $(|G| - |A|)/|A|$. If $g \in G - A$, then $|\mathrm{Cl}(g)| \geqslant |A|$. This is because if $a, b \in A - \{1\}$, then $a^{-1}ga = b^{-1}gb$ implies $a = b$ since this relation forces $C_G(ab^{-1}) \neq A$. Thus, each conjugacy class contained in $G - A$ has size greater than $|A|$. It now follows that the number of conjugacy classes contained in $G - A$ is less than $(|G| - |A|)/|A|$, and we have shown (2.24).

To prove (2.25), we see that since G/A is abelian, G has at least $|G : A|$ linear characters. We choose a set X of exactly $|G : A|$ linear characters from $\operatorname{Irr}(G)$. By Problem 2.9(b), we have that $\chi(1) \leqslant |G : A|$ for each $\chi \in \operatorname{Irr}(G)$, so

$$|G| - |G : A| = \sum_{\chi \in \operatorname{Irr}(G) - X} \chi(1)^2 \leqslant |G : A|^2(|\operatorname{Irr}(G)| - |X|).$$

So,

$$|G : A| + (|G| - |G : A|)/|G : A|^2 \leqslant |\operatorname{Irr}(G)| = k,$$

which proves (2.25).

Combining the results of (2.24) and (2.25), we have shown that $k = (|A| - 1)/|G : A| + |G : A|$. We also observe that equality holds in (7) only when each $\chi(1) = |G : A|$ for each $\chi \in \operatorname{Irr}(G) - X$. Thus, each nonlinear irreducible character has degree $|G : A|$, and the only linear characters of G are the $|G : A|$ linear characters in X. Since the number of linear characters is equal to $|G : G'|$ and $G' \leqslant A$, it follows that $A = G'$. The fact that each irreducible nonlinear character vanishes on $G - A$ was established in the proof to Problem 2.17. □

Problem 2.19. Let $E = \langle x_1, x_2, x_3, x_4 \rangle$ be an elementary abelian group of order 16. Let $P = \langle y \rangle$ be a cyclic group of order 3. Let P act on E in the following way:

$$x_1^y = x_2; \quad x_2^y = x_1x_2; \quad x_3^y = x_4; \quad x_4^y = x_3x_4 \tag{2.26}$$

Let $G = E \rtimes P$ under the above action. Show that G is a group with a trivial center that has no faithful irreducible character.

Hint: The smallest possible degree of a faithful character of E is 4.

Proof. We claim that $G' = E$. We see that $E \subseteq G'$ since $[x_1, y] = x_1x_2$, $[x_2, y] = x_1$, $[x_3, y] = x_3x_4$, and $[x_4, y] = x_3$. These four commutators generate E. Now, E is a normal subgroup of index three in G, so $G' \subseteq E$ since G' is the smallest subgroup of G such that G/G' is abelian. It follows that G has exactly three irreducible linear characters by Corollary 2.23 since $[G : G'] = 48/16 = 3$. Since G is not cyclic, none of these linear characters can be faithful, since there is no injective homomorphism of a noncyclic group G into a cyclic group such as the multiplicative group of $\mathbb{C}$. We now turn to showing that G has no faithful nonlinear irreducible characters.

We also claim that for each $x \in E - \{1\}$, $|C_G(x)| = 16$. Clearly, $E \subseteq C_G(x)$ since E is abelian. Now, y does not commute with any nonidentity element of E, so $C_G(x) \neq G$. Since $[G : E]$ is prime, it follows $C_G(x) = E$, and each centralizer of a nonidentity element of E has order 16. Now, each nonidentity element therefore belongs to a conjugacy class of size three.

We claim that $G - E$ is a union of two conjugacy classes of size 16. By the Sylow Theorems, we deduce that G has exactly 16 Sylow 3-subgroups, and each subgroup is conjugate to one another. Deduce that y is not conjugate to y^{-1}, and it follows that $G - E$ splits into two conjugacy classes of size 16 each.

Combining the results of the two above paragraphs, we have that there are eight conjugacy classes of G with the following sizes: $1, 3, 3, 3, 3, 3, 16, 16$. In particular, G has trivial center since $\{1\}$ is the only conjugacy class of size 1.

Now, notice by the second paragraph that the hypothesis of Problem 2.18 is satisfied, so that any nonlinear character of G has degree 3. Clearly, the restriction of any such character to E (an elementary abelian 2-group of order 16) decomposes as a sum of three linear characters. We may view E as an $\mathbb{F}_2$ vector space of dimension 4, and the kernel of any linear character of E is a dimension-3 subspace. One verifies that the intersection of three dimension three subspaces in $\mathbb{F}_2^4$ must have dimension at least 1. Thus, the kernel of any nonlinear character of G contains a subgroup of order 2 in E and is therefore not faithful. As for the linear characters, their kernel is $G' = E$ and are therefore not faithful either. Thus G possesses no faithful irreducible characters. □

Problem 2.20. Let $\mathfrak{X}$ and $\mathfrak{N}$ be irreducible $\mathbb{C}$-representations of G and define $a_{ij}(g)$ and $b_{ij}(g)$ by $\mathfrak{X}(g) = (a_{ij}(g))$ and $\mathfrak{N}(g) = (b_{ij}(g))$. Write

$$S_{pqrs} = \sum_{g\in G} a_{pq}(g) b_{rs}(g^{-1})$$

Show that $S_{pqrs} = 0$ if $\mathfrak{X}$ and $\mathfrak{N}$ are not similar. If $\mathfrak{X} = \mathfrak{N}$, show that $S_{pqrs} = 0$ unless $p = s$ and $q = r$ in which case $S_{pqqp} = |G|/\deg \mathfrak{X}$.

Hint: Let $P_{qr} = \sum_{g\in G} \mathfrak{X} E_{qr} \mathfrak{N}(g^{-1})$ where E_{qr} is the $(\deg \mathfrak{X}) \times (\deg \mathfrak{N})$ matrix with all entries zero except the (q, r) entry which equals one. Note that $\mathfrak{X} P_{qr} = P_{qr} \mathfrak{X}$. Use Schur's Lemma. For the last statement use Lemma 2.25 and compute $\operatorname{tr}(P_{qr})$.

Note: The results of this problem are called the *Schur relations*. They can be used to give another proof of the orthogonality relations.

Proof. We define $P_{qr} = \sum_{g\in G} \mathfrak{X}(g) E_{qr} \mathfrak{N}(g^{-1})$ where E_{qr} is a $\dim \mathfrak{X} \times \dim \mathfrak{N}$ matrix of all zeroes except for a one in the qr entry. The key observation is that S_{pqrs} is the ps entry of P_{qr}. Now, for $h \in G$, we have that $h \sum_{g\in G} g = \sum_{g\in G} g$, so it follows that

$$\mathfrak{X}(h) \sum_{g\in G} \mathfrak{X}(g) E_{qr} \mathfrak{N}(g^{-1}) = \sum_{g\in G} \mathfrak{X}(hg) E_{qr} \mathfrak{N}(g^{-1}) = \sum_{g\in G} \mathfrak{X}(g) E_{qr} \mathfrak{N}(g^{-1})$$

We can factor out a $\mathfrak{N}(h)$ to the right:

$$\sum_{g\in G} \mathfrak{X}(g) E_{qr} \mathfrak{N}(g^{-1}) = \sum_{g\in G} \mathfrak{X}(g) E_{qr} \mathfrak{N}(g^{-1}h) = \sum_{g\in G} \mathfrak{X}(g) E_{qr} \mathfrak{N}(g^{-1}) \mathfrak{N}(h)$$

Thus, P_{qr} interwines $\mathfrak{X}$ and $\mathfrak{N}$ in the language of Problem 1.2. We have proven in Problem 1.2 that any nonzero matrix which intertwines two representations is square and nonsingular. That is, $\mathfrak{X}$ is similar to $\mathfrak{N}$ if P_{qr} is nonzero. If $\mathfrak{X}$ and $\mathfrak{N}$ are not similar, then $P_{qr} = 0$. Therefore $S_{pqrs} = 0$.

Now, if $\mathfrak{X} = \mathfrak{N}$, then we deduce that P_{qr} commutes with each $\mathfrak{X}(g)$. By Lemma 2.25, P_{qr} is a scalar matrix. We calculate the trace of P_{qr}:

$$\operatorname{tr} P_{qr} = \sum_{g\in G} \operatorname{tr}\left[\mathfrak{X}(g) E_{qr} \mathfrak{X}(g^{-1})\right] = \sum_{g\in G} \operatorname{tr} E_{qr}$$

If $q \neq r$, then $\operatorname{tr} P_{qr} = 0$, and therefore, since P_{qr} is a scalar matrix, P_{qr} is the zero matrix, so $S_{pqrs} = 0$. Otherwise, $\operatorname{tr} P_{qr} = |G|$ if $q = r$. Again, since P_{qr} is scalar, $S_{pqrs} = |G|/\dim \mathfrak{X}$ when $q = r$ and $p = s$. If $p \neq s$, then S_{pqrs} is taken from off the main diagonal, and is zero.

The remark below the problem states that these relations can be used to give an alternative proof to the First Orthogonality Relations. This is seen by observing that for $\chi \in \operatorname{Irr}(G)$ afforded by $\mathfrak{X}$, we have

$$\sum_{g\in G} \chi(g)\chi(g^{-1}) = \sum_{i=1}^{\deg \mathfrak{X}} \sum_{j=1}^{\deg \mathfrak{X}} S_{iijj} = \sum_{i=1}^{\deg \mathfrak{X}} S_{iiii} = \sum_{i=1}^{\deg \mathfrak{X}} |G|/\deg \mathfrak{X} = \deg \mathfrak{X} \cdot \frac{|G|}{\deg \mathfrak{X}} = |G|$$

and if χ and ψ are distinct irreducible characters afforded by $\mathfrak{X}$ and $\mathfrak{N}$ respectively, then the Schur Relations imply

$$\sum_{g\in G} \chi(g)\psi(g^{-1}) = \sum_{i=1}^{\deg \mathfrak{X}} \sum_{j=1}^{\deg \mathfrak{N}} S_{iijj} = \sum_{i=1}^{\deg \mathfrak{X}} \sum_{j=1}^{\deg \mathfrak{N}} 0 = 0$$

which proves the First Orthogonality Relations. □

3. Characters and integrality

Problem 3.1. Let α be an algebraic integer, and suppose that $f(\alpha) = 0$, where $f(x) \in \mathbb{Q}[x]$ is irreducible and monic. Show that $f(x) \in \mathbb{Z}[x]$.

Proof. We claim first that $f(x)$ is a polynomial of minimal degree for which α is a root. If $g(x) \in \mathbb{Q}[x]$ is a monic polynomial of minimal degree for which α is a root, then since $\mathbb{Q}[x]$ is a principal ideal domain, we have that $g(x)$ divides $f(x)$, but since $f(x)$ is irreducible and monic, we have that $f(x) = g(x)$.

Let $h(x)$ denote a monic polynomial of minimal degree in $\mathbb{Z}[x]$ for which α is a root (whose existence is guaranteed by the definition of an algebraic integer). Now, $f(x)$ divides $h(x)$ since $f(x)$ was established to be minimal. If $f(x)$ is a proper divisor of $h(x)$, then $h(x)$ is reducible in $\mathbb{Q}[x]$. However, by Gauss' Lemma, any polynomial which is reducible over $\mathbb{Q}[x]$ is reducible over $\mathbb{Z}[x]$, meaning that $h(x)$ factorizes as $h(x) = p(x)q(x)$ where $p(x)$ and $q(x)$ are nonconstant polynomials in $\mathbb{Z}[x]$. Clearly, α is a root of either $p(x)$ or $q(x)$, but this contradicts the assumed minimal degree of $h(x)$. □

Problem 3.2. Let G be a group, $g \in G$, and let χ be a character of G. Suppose that $|\chi(g)| = 1$. Show that $\chi(g)$ is a root of unity.

Hint: Let $E \subseteq \mathbb{C}$ be the splitting field for $x^n - 1$ over $\mathbb{Q}$. For integral $\alpha \in E$, with $|\alpha| = 1$, let $f_\alpha \in \mathbb{Z}[x]$ be the polynomial of Problem 3.1. Show that only finitely many polynomials can arise this way. Do this by bounding the degree and the coefficients of f_α.

Remark: In fact, given any $\alpha \in \mathbb{C}$ expressible as a sum of roots of unity, we have $|\alpha| = 1$ iff α is a root of unity. This no longer holds if α is merely assumed to be an algebraic integer, that is, there exist algebraic integers having absolute value equal to one despite not being roots of unity.

Proof. Set $\alpha := \chi(g)$ for a $g \in G$, and suppose that $|\alpha| = 1$ so that α belongs to some cyclotomic field extension of $\mathbb{Q}$, say $E := \mathbb{Q}(\epsilon)/\mathbb{Q}$ for some primitive n-th root of unity ϵ. The Galois group $\mathcal{G} := \mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q})$ is abelian of order $\varphi(n)$ (where φ is the totient function) and therefore commutes with complex conjugation restricted to E. For $\sigma \in \mathcal{G}$, we have that

$$(|\alpha|^2)^\sigma = (\alpha\overline{\alpha})^\sigma = \alpha^\sigma(\overline{\alpha})^\sigma = \alpha^\sigma\overline{\alpha^\sigma} = |\alpha^\sigma|^2 \tag{3.1}$$

We deduce from (3.1) that $|\alpha^\sigma| = 1$ for each $\sigma \in \mathcal{G}$.[2]

If α is not a root of unity, then the sequence $\{\alpha, \alpha^2, \alpha^3, \ldots\}$ is a sequence with infinitely many distinct values. Given a minimal polynomial $f_\ell(x) \in \mathbb{Z}[x]$ for any

[2]We remark that the fact that $\mathrm{Gal}(E/\mathbb{Q})$ is abelian was fundamental to this argument. There are indeed algebraic integers with absolute value 1 having Galois conjugates whose absolute values may not equal to 1.

α^ℓ, we can immediately deduce that the degree of $f_\ell(x)$ is bounded by $|E:\mathbb{Q}|$ since $\alpha^\ell \in E$ for each $\ell \in \mathbb{Z}^+$. Thus, if we also show that the coefficients of each $f_\ell(x)$ lie in a set X of a fixed cardinality, then we will have shown that that there are only finitely many minimal polynomials for which the α^ℓ can satisfy. Since any polynomial has finitely many roots, it will follow that the set $\{\alpha, \alpha^2, \alpha^3, \ldots\}$ is finite.

Let $\beta = \alpha^\ell$ for some integer ℓ. Let $f_\ell(x)$ be the minimal polynomial of β, so that

$$f_\ell(x) = \prod_{i=1}^{m}(x - \beta_i)$$

where we take $\beta := \beta_1$. Each β_i is a Galois conjugate of β, so $|\beta_i| = 1$ for each i by (3.1). Now, $f_\ell(x)$ may be expanded out of its product form into $f_\ell(x) = x^n + a_{n-1}x^{n-1} + \ldots a_1 x + a_0$. By Problem 3.1, we have that each $a_i \in \mathbb{Z}$.

Now, we write down a formula for the a_i in terms of the β_i. We have that

$$a_{m-k} = (-1)^k \sum_{\substack{A \subseteq \{\beta_i\} \\ |A|=k}} \prod_{\gamma \in A} \gamma$$

for $k < m$. By the triangle inequality, we have that

$$|a_{m-k}| = \left|(-1)^k \sum_{\substack{A \subseteq \{\beta_i\} \\ |A|=k}} \prod_{\gamma \in A} \gamma\right| \leqslant \sum_{\substack{A \subseteq \{\beta_i\} \\ |A|=k}} \left|\prod_{\gamma \in A} \gamma\right| = \binom{m}{k} \leqslant m! \leqslant |E:\mathbb{Q}|!.$$

Notice that this bound of $|a_i| \leqslant |E:\mathbb{Q}|$ does not depend on ℓ. In particular, any coefficient of any minimal polynomial of α^ℓ for any ℓ lies in the set of integers whose absolute value is at most $|E:\mathbb{Q}|!$, as desired. □

Problem 3.3. Show that no simple group can have an irreducible character of degree 2.

Hint: Problem 2.3 is relevant.

Proof. If G is a simple abelian group, then all irreducible characters of G are linear, so let G be a nonabelian simple group with $\chi \in \mathrm{Irr}(G)$ having degree 2. Now, by Theorem 3.11, we have that $\chi(1) \mid |G|$, so $|G|$ is even. Let $h \in G$ have order two. Letting $\mathfrak{X}$ be a representation affording χ, consider the eigenvalues of $\mathfrak{X}(h)$, which must each be second roots of unity. If the eigenvalues of $\mathfrak{X}(h)$ are either both 1 or both -1, then this respectively implies that $\ker\chi \neq 1$ or $\mathbf{Z}(\chi) \neq 1$, a contradiction to the assumption of simplicity of G. Thus, $\mathfrak{X}(h)$ has two distinct eigenvalues, namely 1 and -1. However, $\det\chi(h) = \det\mathfrak{X}(h) = (1)(-1) = -1$ (since the determinant of a matrix equals the product of its eigenvalues). Thus, $\det\chi$ is a nontrivial linear character, implying that G' is a proper subgroup of G, again contradicting the simplicity of G. □

Problem 3.4. Let G be a simple group, and supposed $\chi \in \mathrm{Irr}(G)$ with $\chi(1) = p$ for some prime p. Show that a Sylow p-subgroup of G has order p.

Hint: If the Sylow p-subgroup P is nonabelian, then $\mathbf{Z}(P) \subseteq \mathbf{Z}(\chi)$.

Proof. First note that G has a nontrivial Sylow p-subgroup by Theorem 3.11. Since irreducible characters of abelian groups are linear, we can assume G is nonabelian. We can also assume that χ is a faithful character since otherwise, $\ker\chi$ would be a nontrivial normal subgroup of G. If G has an abelian Sylow p-subgroup, then Theorem 3.13 applies which states that $p \mid |G|$, but $p^2 \nmid |G|$, so G has a Sylow p-subgroup of order p.

Now, assume that $P \in \mathrm{Syl}_p(G)$ is nonabelian. Since is faithful, Problem 2.8 implies that at least one irreducible constituent of χ_P is nonlinear. By Theorem 3.11, the degree of such a nonlinear character must divide $|P|$ while also being less than or equal to p. It follows that χ_P has exactly one irreducible constituent of degree p, so χ_P is irreducible. It follows from Lemma 2.27(f) that $\mathbf{Z}(P) = \mathbf{Z}(\chi_P) \leqslant \mathbf{Z}(\chi)$. Since centers of p-groups are nontrivial, this implies $\mathbf{Z}(\chi)$ is a nontrivial normal subgroup of G, a contradiction. □

Problem 3.5. Suppose $A \subseteq G$ is abelian and $|G : A|$ is a prime power. Show that $G' < G$.

Proof. Let $a \in A - \{1\}$. Since A is abelian, we have $A \subseteq C_G(a)$, so a belongs to a conjugacy class of prime power order. Theorem 3.9 implies then that G has a nontrivial, proper normal subgroup.

Assume the statement of the problem is not true, and select a counterexample G to the statement with minimal order. We choose a nontrivial, proper normal subgroup N of G and consider the group G/N. We have that AN/N is an abelian subgroup of G/N, and AN/N has prime power index in G/N. Our induction hypithesis implies that G/N has a proper derived subgroup. Of course, if $(G/N)' < (G/N)$, then $G' < G$, which proves the claim. □

Problem 3.6. Let G be a p-group and suppose $\chi \in \mathrm{Irr}(G)$. Show that $\chi(1)^2 \big| |G : Z(\chi)|$.

Proof. By Theorem 3.11, $\chi(1) \mid |G|$, and so, $\chi(1)^2$ is a power of p. Also, by Corollary 2.30, $\chi(1)^2 \leqslant |G : Z(\chi)|$. Thus, $\chi(1)^2$ is a power of p which is less than or equal to $|G : Z(\chi)|$ which is also a power of p. This implies that $\chi(1)^2 \big| |G : Z(\chi)|$. □

Problem 3.7. Let $\chi \in \mathrm{Irr}(G)$ be faithful and suppose $\chi(1) = p^a$ for some prime p. Let $P \in \mathrm{Syl}_p(G)$ and suppose that $C_G(P) \nsubseteq P$. Show that $G' < G$.

Hint: Let $Q \subseteq C_G(P)$ be a p'-subgroup, $Q \neq 1$. Show that $Q \cap \mathbf{Z}(\chi) \neq 1$ and consider $\det\chi$.

Proof. Since $C_G(P)$ is not contained in P, we can choose a p'-subgroup $Q \leqslant C_G(P)$. It suffices to produce a nonidentity element $g \in Q \cap \mathbf{Z}(\chi)$. If such an element can be chosen, then it will follow that $\det \chi(g) \neq 1$. Indeed, in this case, $\chi(g) = \chi(1)\epsilon$ for some primitive n-th root of unity ϵ where $p \nmid n$, and further, since χ is faithful, we see that $\epsilon \neq 1$. One also sees that $\det \chi(g) = \epsilon^{\chi(1)}$ (since $\chi(g)$ is the trace of a scalar matrix). Since $\gcd(n, \chi(1)) = 1$ by assumption, we see that $\epsilon^{\chi(1)}$ is also a primitive n-th root of unity. In particular, $\det \chi$ is a nontrivial linear character, and $G' < G$.

Assume for a contradiction that no nonidentity p'-element of $C_G(P)$ lies in $\mathbf{Z}(\chi)$. Notice that any p'-element $h \in C_G(P)$ lies in a conjugacy class of p'-cardinality since $|h^G| = |G : C_G(h)|$ divides $|G : P|$. Thus, the hypothesis of Theorem 3.8 is satisfied, and we conclude that $\chi(h) = 0$ for all nonidentity p'-elements of $C_G(P)$. In particular, the restriction of χ to any p'-subgroup Q of $C_G(P)$ is an integer multiple of the regular character ρ_Q by Problem 2.16, which implies that $|Q| \mid \chi(1)$, contradicting the assumption that $\chi(1) = p^a$. □

Problem 3.8. Let χ be a (possibly reducible) character of G which is constant on $G - \{1\}$. Show that $\chi = a1_G + b\rho_G$, where $a, b \in \mathbb{Z}$ and ρ is the regular character of G. Show that if $G \neq \ker \chi$, then $\chi(1) \geqslant |G| - 1$.

Hint: First show that $\chi = a1_G + b\rho_G$ for some $a, b \in \mathbb{C}$.

Proof. Let $\chi(g) = \alpha$ for $g \in G - \{1\}$, and let $\chi(1) = n$. We have that

$$\chi = \alpha 1_G + \frac{n - \alpha}{|G|}\rho_G, \quad \alpha \in \mathbb{C} \tag{3.2}$$

We note that α is an algebraic integer by by virtue of being a character value. We now compute the inner product $[\chi, 1_G]$:

$$[\chi, 1_G] = \frac{1}{|G|}\sum_{g \in G} \chi(g) = \frac{1}{|G|}\left[\sum_{g \in G} \alpha\right] + \frac{n - \alpha}{|G|} = \alpha + \frac{n - \alpha}{|G|}$$

Now, since χ and 1_G are characters, their inner product is an integer: $m := [\chi, 1_G]$. We have that

$$m - \alpha = \frac{n - \alpha}{|G|} \tag{3.3}$$

Since m is an integer, and α is an algebraic integer, it follows from the above equation that $(n - \alpha)/|G|$ is an algebraic integer. We now solve the above equation for α:

$$\alpha = \frac{-n + |G|m}{|G| - 1}$$

It follows now that $\alpha \in \mathbb{Q}$, but since α is an algebraic integer, we have that $\alpha \in \mathbb{Z}$. Consequently, $(n-\alpha)/|G|$ is also a rational integer by (3.3). So, by setting $a = \alpha$ and $b = (n-\alpha)/|G|$, (3.2) becomes

$$\chi = a1_G + b\rho_G, \quad a, b \in \mathbb{Z}$$

This verifies the first statement.

Let $\ker\chi \neq G$. This implies that $b \neq 0$. We observe that $b > 0$ since otherwise, any nonprincipal constituent of χ would then have negative multiplicity. We also have that $a \geqslant -b$. This is because for any group G, the principal character has a multiplicity of one as a constituent of the regular character, so if $a < -b$, then 1_G would have negative multiplicity in χ. From this, it follows that the minimum value of $\chi(1)$ occurs when $a = -1$ and $b = 1$. In this case, $\chi(1) = |G| - 1$. □

Problem 3.9. Let $\mathscr{K}_1, \mathscr{K}_2, \ldots, \mathscr{K}_k$ be the conjugacy classes of a group G and let $K_1, \ldots, K_k$ be the corresponding class sums. Choose representatives $g_i \in \mathscr{K}_i$, and let a_{ijv} be the integers defined by

$$K_iK_j = \sum_v a_{ijv}K_v$$

Show that

$$a_{ijv} = \frac{|\mathscr{K}_i||\mathscr{K}_j|}{|G|} \sum_{\chi\in\mathrm{Irr}(G)} \frac{\chi(g_i)\chi(g_j)\overline{\chi(g_v)}}{\chi(1)}. \tag{3.4}$$

Hint: Use the ω_x and the second orthogonality relation.

Notes: This formula shows that the a_{ijv} can be calculated from the character table. Therefore, the character table of G can be used to answer questions such as: Is an element $g \in \mathscr{K}_v$ the product of an element of $\mathscr{K}_i$ with one of $\mathscr{K}_j$?

The fact that the a_{ijv} are nonnegative rational integers imposes another necessary condition on an array of complex numbers that the array be a character table for some group.

Since the K_i are a basis for $\mathbf{Z}(\mathbb{C}[G])$, it follows that the character table of G determines $\mathbf{Z}(\mathbb{C}[G])$ up to algebra isomorphism.

Proof. We have that $\chi(g_i) = \chi(1)\omega_\chi(K_i)/|\mathscr{K}_i|$. Substituting this expression into $\chi(g_i)$ and $\chi(g_j)$ in the right side of (3.4), we obtain

$$\frac{1}{|G|} \sum_{\chi\in\mathrm{Irr}(G)} \omega_\chi(K_i)\omega_\chi(K_j)\chi(1)\overline{\chi(g_v)}.$$

We have that $\omega_\chi(K_i)\omega_\chi(K_j) = \sum_w a_{ijw}\omega_\chi(K_w)$ since ω_χ is an algebra homomorphism, so the above equation becomes

$$\frac{1}{|G|}\sum_{\chi\in\mathrm{Irr}(G)}\chi(1)\overline{\chi(g_v)}\sum_w a_{ijw}\omega_\chi(K_w)$$

$$= \frac{1}{|G|}\sum_{\chi\in\mathrm{Irr}(G)}\overline{\chi(g_v)}\sum_w a_{ijw}\chi(g_w)|\mathscr{K}_w|.$$

By rearranging the order of the sums, we obtain

$$\frac{1}{|G|}\sum_w a_{ijw}|\mathscr{K}_w|\sum_{\chi\in\mathrm{Irr}(G)}\chi(g_w)\overline{\chi(g_v)}.$$

We are able to apply the Second Orthogonality Relations to the innermost sum. The innermost sum is nonzero precisely when $v = w$, so we have

$$\frac{1}{|G|}a_{ijv}|\mathscr{K}_v||C_G(g_v)| = a_{ijv},$$

which proves the desired equality since $|\mathscr{K}_v||C_G(g_v)| = |G|$. □

Problem 3.10. We write $[x, y]$ for the commutator $x^{-1}y^{-1}xy$ of x and y in a group G.

(a) Let $g \in G$ and fix $x \in G$. Show that g is conjugate to $[x, y]$ for some $y \in G$ if and only if

$$\sum_{\chi\in\mathrm{Irr}(G)}\frac{|\chi(x)|^2\overline{\chi(g)}}{\chi(1)} \neq 0 \tag{3.5}$$

(b) Show that $g = [x, y]$ for some $x, y \in G$ if and only if

$$\sum_{\chi\in\mathrm{Irr}(G)}\frac{\chi(g)}{\chi(1)} \neq 0 \tag{3.6}$$

Note: We already knew that the character table of a group determines the commutator subgroup. Problem 3.10 says more than this, since the commutator subgroup always does not consist entirely of commutators.

Remark: It is a well-known fact that the commutator subgroup of a group does not always consist entirely of commutators. However, the smallest example of such a group has order 96. Øystein Ore conjectured in 1951 that every element of a nonabelian simple group is expressible as a commutator, and this was shown to be true by Liebeck, O'Brien, Shalev, and Tiep (*J. Eur. Math. Soc.* **12**(4) (2010), pp. 939–1008). Their proof (of course) relies upon the classification of finite simple groups.

Proof. For part (a), we rewrite (3.5) as:

$$\sum_{\chi\in\mathrm{Irr}(G)} \frac{|\chi(x)|^2\overline{\chi(g)}}{\chi(1)} = \sum_{\chi\in\mathrm{Irr}(G)} \frac{\chi(x)\chi(x^{-1})\overline{\chi(g)}}{\chi(1)}$$

Set $\mathscr{H}_i = \mathrm{Cl}(x)$, $\mathscr{H}_j = \mathrm{Cl}(x^{-1})$ and $\mathscr{H}_v = \mathrm{Cl}(g)$. Using (3.4) from Problem 3.9, we may rewrite the right side of the above equation as

$$\frac{a_{ijv}|G|}{|\mathscr{H}_i||\mathscr{H}_j|} = \sum_{\chi\in\mathrm{Irr}(G)} \frac{\chi(x)\chi(x^{-1})\overline{\chi(g)}}{\chi(1)} \tag{3.7}$$

So, the expression is nonzero if and only if a_{ijv} is nonzero. We see that a_{ijv} is the number of ways of expressing a fixed element of $\mathscr{H}_v$ as a product of an element of $\mathscr{H}_i$ and an element of $\mathscr{H}_j$. We have $a_{ijv} = \big|\{(h,k) \in \mathscr{H}_i \times \mathscr{H}_j \mid g = hk\}\big|$. So, if $a_{ijv} \neq 0$ if and only if $g = (x^{-1})^a x^b$ for some $a, b \in G$. We have now that $g^{a^{-1}} = [x, ab^{-1}]$. Conversely, if there existed some $y \in G$ such that g was conjugate to $[x,y]$, then we may write $g^a = x^{-1}y^{-1}xy$. We may also write $y = ba$ for some $b \in g$, so we have that

$$g = ax^{-1}a^{-1}b^{-1}xb = (x^{-1})^a x^b$$

so, $a_{ijv} \neq 0$. This proves part (a).

For part (b), we first observe that g is a commutator iff g^h is commutator for some $h \in G$. This follows from the following identity: $[x,y]^z = [x^z, y^z]$. From this, it follows that if there exists an $x \in G$ such that (3.5) is nonzero, then g is a commutator. We also deduce that for a fixed g, the value of (3.5) for a given x is constant on the conjugacy classes of G. We also observe from (3.7), that (3.5) is nonnegative for any choice of g and x since a_{ijv} is always nonnegative. Combining these facts, we have that g is a commutator if and only if

$$\sum_w \frac{|\mathrm{Cl}(g_w)|}{|G|} \sum_{\chi\in\mathrm{Irr}(G)} \frac{\chi(g_w)\chi(g_w^{-1})\overline{\chi(g)}}{\chi(1)} \neq 0,$$

where w ranges over the conjugacy classes of G and each g_w is a conjugacy class representative of the w-th conjugacy class. We rearrange the expression to obtain the following expression:

$$\sum_{\chi\in\mathrm{Irr}(G)} \frac{\chi(g^{-1})}{\chi(1)} \cdot \frac{1}{|G|} \sum_w |\mathrm{Cl}(g_w)|\chi(g_w)\chi(g_w^{-1}) = \sum_{\chi\in\mathrm{Irr}(G)} \frac{\chi(g^{-1})}{\chi(1)} \cdot [\chi,\chi]$$

$$= \sum_{\chi\in\mathrm{Irr}(G)} \frac{\chi(g^{-1})}{\chi(1)}.$$

We now have that g is a commutator if and only if $\sum_\chi \chi(g^{-1})/\chi(1) \neq 0$ which is clearly equivalent to the complex conjugate being nonzero:

$$\sum_{\chi \in \mathrm{Irr}(G)} \frac{\chi(g)}{\chi(1)} \neq 0.$$

This proves the desired equivalence in (3.6). □

Problem 3.11. Let $g \in G$ be a commutator. Suppose $\gcd(o(g), m) = 1$. Show that g^m is a commutator.

Hint: See the hint to to Problem 2.12.

Proof. Since g is a commutator, we have from Problem 3.10(b) that

$$\sum_{\chi \in \mathrm{Irr}(G)} \frac{\chi(g)}{\chi(1)} \neq 0$$

Now, set $n = |G|$, and let ϵ be a primitive n-th root of unity. Set $\mathcal{G} = \mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q})$. In Problem 2.12, we have shown that for each integer m such that $\gcd(o(g), m) = 1$, there is $\sigma \in \mathcal{G}$ such that $\sigma(\epsilon) = \epsilon^m$, and further, $\chi(g)^\sigma = \chi(g^m)$ for any $g \in G$. It now follows that

$$\sum_{\chi \in \mathrm{Irr}(G)} \frac{\chi(g^m)}{\chi(1)} = \sigma\left(\sum_{\chi \in \mathrm{Irr}(G)} \frac{\chi(g)}{\chi(1)} \right) \neq \sigma(0) = 0,$$

so it follows from Problem 3.10(b) that g^m is a commutator. □

Problem 3.12. Let $\chi \in \mathrm{Irr}(G)$ and $g, h \in G$. Show

$$\chi(g)\chi(h) = \frac{\chi(1)}{|G|} \sum_{z \in G} \chi(gh^z) \tag{3.8}$$

Hint: Let $\mathfrak{X}$ afford χ and let K_i and K_j be the class sums in $\mathbb{C}[G]$ which contain g and h. Use he fact that $\mathfrak{X}(K_i)\mathfrak{X}(K_j) = \mathfrak{X}(K_iK_j)$.

Proof. We set K_i to be the class sum for $\mathrm{Cl}(g)$ and K_j to be the class sum for $\mathrm{Cl}(h)$. We let $\mathfrak{X}$ afford a character $\chi \in \mathrm{Irr}(G)$. One sees that $\mathfrak{X}(K_i)\mathfrak{X}(K_j) = \mathfrak{X}(K_iK_j)$. We evaluate the trace of both sides. Since $K_i, K_j \in Z(\mathbb{C}[G])$, we have that $\mathfrak{X}(K_i) = \omega_\chi(K_i)I$ and $\mathfrak{X}(K_j) = \omega_\chi(K_j)I$. We have that

$$\mathrm{tr}\left(\mathfrak{X}(K_i)\mathfrak{X}(K_j)\right) = \omega_\chi(K_i)\chi(K_j) = \frac{\chi(g)|\mathrm{Cl}(g)|}{\chi(1)}|\mathrm{Cl}(h)|\chi(h). \tag{3.9}$$

We will now evaluate the trace of $\mathfrak{X}(K_iK_j)$. We have

$$\mathrm{tr}\,\mathfrak{X}(K_iK_j) = \sum_{x \in \mathrm{Cl}(g)} \sum_{y \in \mathrm{Cl}(h)} \chi(xy)$$

$$= \frac{1}{|C_G(h)|} \sum_{x \in \mathrm{Cl}(g)} \sum_{z \in G} \chi(xh^z)$$

Now, we have that $(g^w h^z)^{w^{-1}} = (gh^{zw^{-1}})$, so we have that $\chi(g^w h^z) = \chi(gh^{zw^{-1}})$ for each $w \in G$. From this, we have that

$$\frac{1}{|C_G(h)|} \sum_{x \in \mathrm{Cl}(g)} \sum_{z \in G} \chi(xh^z) = \frac{|\mathrm{Cl}(g)|}{|C_G(h)|} \sum_{z \in G} \chi(xh^z)$$

We have now that

$$\mathrm{tr}\, \mathfrak{X}(K_i K_j) = \frac{|\mathrm{Cl}(g)|}{|C_G(h)|} \sum_{z \in G} \chi(xh^z) \tag{3.10}$$

Combining (3.9) and (3.10), we have that

$$\frac{\chi(g)|\mathrm{Cl}(g)|}{\chi(1)} |\mathrm{Cl}(h)| \chi(h) = \frac{|\mathrm{Cl}(g)|}{|C_G(h)|} \sum_{z \in G} \chi(xh^z)$$

which becomes (3.8) after solving for $\chi(g)\chi(h)$. □

Problem 3.13. (Brauer) Let $K_1, K_2, \ldots, K_k$ be the class sums in $\mathbb{C}[G]$. Suppose there exists a $c \in \mathbb{C}$ such that $\sum K_i = c \prod K_i$. Show that $G = G'$.

Hint: Let $1_G \neq \chi \in \mathrm{Irr}(G)$. Show that $\omega_\chi(K_i) = 0$ for some i.

Proof. We wish to show that G has no nonprincipal linear characters. By Theorem 3.15, it follows that an irreducible character is linear if and only if it is nonzero for all $g \in G$. We let $\chi \in \mathrm{Irr}(G)$, and we proceed by showing that $\chi(g)$ vanishes for some $g \in G$. Now, since ω_χ is an algebra homomorphism, we have that

$$\sum_i \omega_\chi(K_i) = c \prod_i \omega_\chi(K_i).$$

We are able to show that the left hand side of the above equation is zero.

$$\sum_i \omega_\chi(K_i) = \sum_i \frac{\chi(g_i)|\mathrm{Cl}(g_i)|}{\chi(1)} = \frac{|G|}{\chi(1)} [\chi, 1_G].$$

If χ is not the principal character, then $[\chi, 1_G] = 0$, and so

$$\sum_i \omega_\chi(K_i) = 0.$$

We now have that

$$c \prod_i \omega_\chi(K_i) = 0,$$

which implies $\omega_\chi(K_i) = 0$ for some i since the definition of c clearly implies that $c \neq 0$. Thus, there is a $g \in G$ such that $0 = \chi(g)|\mathrm{Cl}(g)|/\chi(1)$, so $\chi(g) = 0$ for at least one $g \in G$ which proves χ is nonlinear. It follows that $G = G'$. □

Problem 3.14. (Brauer) In the notation of the previous problem, show that if $G = G'$, then there exists a $c \in \mathbb{Q}$ such that $\sum K_i = c \prod K_i$.

Hint: To show that two elements $a, b \in \mathbf{Z}(\mathbb{C}[G])$, it suffices to prove that $\omega_\chi(a) = \omega_\chi(b)$ for all $\chi \in \mathrm{Irr}(G)$.

Proof. Let G have k conjugacy classes. Let $\{\chi_1, \ldots, \chi_k\}$ denote the set of irreducible characters of G. We begin by defining a linear isomorphism from $\mathbf{Z}(\mathbb{C}[G])$ onto $\mathbb{C}^k$ given by the map

$$K_i \longmapsto \sum_{j=1}^{k} \omega_{\chi_j}(K_i) e_j \tag{3.11}$$

where the e_i are the standard basis vectors of $\mathbb{C}^k$. It is clear that the map is $\mathbb{C}$-linear from the fact that each ω_χ is $\mathbb{C}$-linear. To see that the map is a bijection, we fix a set $\{g_i\}_{i=1}^k$ of conjugacy class representatives and write the map as a composition of two maps:

$$K_i \longmapsto \sum_{j=1}^{k} \omega_{\chi_j}(K_i) e_j = (e_i \longmapsto e_i/\chi_i(1)) \circ (K_i \longmapsto \sum_j \chi_j(g_i)|\mathrm{Cl}(g_i)|e_j)$$

The map given by $(K_i \longmapsto \sum_j \chi_j(g_i)|\mathrm{Cl}(g_i)|e_j)$ is seen to be a bijection by considering the inner products of the images of any two K_i and K_j and invoking the Second Orthogonality relations to conclude that the inner product is nonzero iff $i = j$. The map $(e_i \longmapsto e_i/\chi_i(1))$ is clearly a bijection since each component is being scaled by some nonzero constant. Now, this proves that the map in (3.11) defines a linear bijection. It now follows that if $a, b \in Z(\mathbb{C}[G])$, then $a = b$ if and only if $\omega_\chi(a) = \omega_\chi(b)$ for each $\chi \in \mathrm{Irr}(G)$.

Now, if $G = G'$, then by Theorem 3.15, we have that each nonprincipal character of G is zero for at least one $g \in G$. So, if $\chi \in \mathrm{Irr}(G)$ is not the principal character, $\prod_i \omega_\chi(K_i) = 0$. We have also shown in the proof of Problem 3.13 that $\sum_i \omega_\chi(K_i) = 0$ for each irreducible nonlinear character χ. We also have that $\sum_i \omega_{1_G}(K_i) = \sum_i 1_G(K_i) = |G|$ and $\prod_i \omega_{1_G}(K_i) = \prod_i |\mathrm{Cl}(g_i)| \in \mathbb{Z}$. So, if we set $m := \prod_i |\mathrm{Cl}(g_i)|$, we have that

$$\omega_\chi(\sum_i K_i) = \frac{|G|}{m} \omega_\chi(\prod_i K_i)$$

for all $\chi \in \mathrm{Irr}(G)$ which proves the statement with $c = |G|/m \in \mathbb{Q}$. □

Problem 3.15. (Thompson) Let E be a Galois extension of $\mathbb{Q}$ with Galois group $\mathcal{G}$. Let $\alpha \in E$ be an algebraic integer with the property that α^σ is real and positive for all $\sigma \in \mathcal{G}$. A theorem of Siegel (*Ann. of Math.* **46** (1945) p. 30, Theorem III) asserts

that if $\alpha \neq 1$, then

$$\frac{1}{|\mathcal{G}|} \sum_{\sigma \in \mathcal{G}} \alpha^{\sigma} \geqslant \frac{3}{2}$$

Use this to show that if $\chi \in \mathrm{Irr}(G)$, then $\chi(x)$ is either zero or a root of unity for more than a third of the elements of $x \in G$.

Hints: Mimic the proof of Theorem 3.15. Use Problem 3.2.

Remarks: This is a rather fascinating problem, and there are many things to say about it. First, the result of Siegel quoted is a rather hard result that holds for all algebraic integers. A somewhat easier and more elementary proof is obtained as Lemma 2 in a paper of Cassels (*J. reine angew. Math.* **1969** (1969), pp. 112 – 131), which merely assumes that α is expressible as a sum of roots of unity.

Also, in many situations, much better results are known to hold. For instance, let G be nilpotent, and let p be the smallest prime divisor of $|G|$. As a consequence of Corollary 6.14, any nonlinear irreducible character vanishes on at least $|G|(1 - 1/p)$ elements of G. Moreover, if G is solvable, Fang Bian and Yong Yang has shown in an arXiv note that any irreducible character value is either zero or a root of unity for at least half of all elements of G. A. R. Miller conjectured that this 1/2-bound should hold for all finite groups (*Enseign. Math.* **70** (2024), 151–164), but this was disproven in a note by Navarro and Sambale.

Further, Stan and Zaharescu have shown that any irreducible character of a group takes values expressible as sums of at most two roots of unity on three-fifths of the group's elements ([*J. reine angew. Math.* **637** (2009), pp. 217–234.).

Proof. Fix $\chi \in \mathrm{Irr}(G)$. We assume for a contradiction that G is a finite group where more than two-thirds of the elements of G have a value at χ which is neither zero nor a root of unity. Denote the set of such group elements as R. Now, let ϵ be a primitive $|G|$-th root of unity, and set $\mathcal{G} = \mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q})$. For any $\sigma \in \mathcal{G}$, notice that since $\chi(g)^{\sigma} = \chi(g^m)$ for some integer m with $\gcd(m, |G|)$, we may conclude that each $\sigma \in \mathcal{G}$ permutes the terms of

$$\sum_{r \in R} |\chi(r)|^2.$$

Thus, by Siegel's result quoted in the problem above, the average value of any term in the above sum is at least 3/2. From this fact, it follows that

$$|G| = \sum_{g \in G} |\chi(g)|^2 \geqslant \sum_{r \in R} |\chi(r)|^2 \geqslant \frac{3}{2}|R| > \frac{3}{2} \cdot \frac{2}{3} \cdot |G| = |G|,$$

an impossibility. □

Problem 3.16. (Burnside) Let $|G|$ be odd and suppose $\chi \in \mathrm{Irr}(G)$ is not principal. Show that $\chi \neq \overline{\chi}$.

Hint: Using orthogonality, show that $\chi \neq 1_G$ and $\chi = \bar{\chi}$, then $\chi(1) = 2\alpha$ for some algebraic integer α.

Proof. A group element $g \in G$ has order two if and only if $g^{-1} = g$. If $|G|$ is odd, then G contains no elements of order two, so no element of G is its own inverse. We can now choose a subset $S \subset G$ with $|S| = (|G| - 1)/2$ such that no element of S is the inverse of any other element of S. We have that $G = \{1\} \sqcup S \sqcup S^{-1}$, where S^{-1} contains the inverses of each element of S. Now, suppose $\chi \in \mathrm{Irr}(G)$ is not principal and that $\chi = \overline{\chi}$. Now, for any $g \in G$, we have that $\chi(g) = \overline{\chi(g)} = \chi(g^{-1})$. We combine these facts in looking at $[\chi, 1_G]$:

$$0 = |G|[\chi, 1_G] = \sum_{g \in G} \chi(g) = \chi(1) + 2\sum_{g \in S} \chi(g)$$

From the equation above, we have that $\chi(1) = 2\alpha$ where $\alpha = -\sum_{g \in S} \chi(g)$. Since α is an algebraic integer, we have that $\chi(1)/2$ is a rational integer, and therefore $\chi(1)$ is even. However, by Theorem 3.11, $\chi(1)$ divides $|G|$, but this is impossible since G has odd order. We have reached a contradiction. □

Problem 3.17. (Burnside) Let $|G|$ be odd and suppose that G has exactly k conjugacy classes. Show that

$$|G| \equiv k \bmod 16$$

Hint: If n is an odd integer, then $n^2 \equiv 1 \bmod 8$.

Proof. One verifies that for odd n, we have $n^2 \equiv 1 \bmod 8$ by checking the odd integers up to 7. If G is a group of odd order, Theorem 3.11 implies that $\chi(1)$ is odd for each $\chi \in \mathrm{Irr}(G)$. We have that $\chi(1)^2 \equiv 1 \bmod 8$, and in particular, we have

$$\chi(1)^2 \equiv 1 \bmod 16$$

or

$$\chi(1)^2 \equiv 9 \bmod 16.$$

Most importantly, we observe that in either case, we have

$$\chi(1)^2 + \overline{\chi(1)}^2 \equiv 2 \bmod 16 \tag{3.12}$$

for each $\chi \in \mathrm{Irr}(G)$.

By Problem 3.16, we have that $\chi \neq \overline{\chi}$ for each nonprincipal character $\chi \in \mathrm{Irr}(G)$. So, we can choose a subset $S \subset \mathrm{Irr}(G)$ such that S does not contain the principal character, and each character in S is not the conjugate of any other character in S. We note that $|S| = (k-1)/2$. We may now write

$$|G| = \sum_{\chi \in \mathrm{Irr}(G)} \chi(1)^2 = 1_G(1)^2 + \sum_{\chi \in S} \left[\chi(1)^2 + \overline{\chi(1)}^2\right]$$

By (3.12), when reducing modulo 16, we obtain

$$|G| \equiv 1 + |S| \cdot 2 \equiv k \bmod 16,$$

as desired. □

4. Products of characters

Problem 4.1. Let χ and ψ be characters of G. Express $\det(\chi\psi)$ in terms of $\det(\chi)$ and $\det(\psi)$.

Hint: It suffices to assume that G is cyclic.

Proof. We claim that the following identity holds:

$$\det(\chi\psi) = \det(\chi)^{\psi(1)} \det(\psi)^{\chi(1)}.$$

By restricting, it is of no loss to assume that G is cyclic, say $G = \langle g \rangle$. Thus, χ and ψ are afforded by diagonal representations $\mathfrak{X}$ and $\mathfrak{N}$. Let V and W be the spaces of row vectors with entries in $\mathbb{C}$ with respective dimensions $\chi(1)$ and $\psi(1)$. Also, let $\{e_i\}$ and $\{f_r\}$ be the sets of standard basis vectors for V and W respectively. The character $\chi\psi$ is afforded by the tensor action of G on $V \otimes W$.

Now, since G is cyclic and since $\mathfrak{X}$ and $\mathfrak{N}$ are diagonal representations, we have that $e_i \cdot g = e_i\mathfrak{X}(g) = a_i e_i$ and $f_r \cdot g = f_r\mathfrak{N}(g) = b_r f_r$ for some $a_i, b_r \in \mathbb{C}$. In particular, we have that $\det \chi(g) = \det \mathfrak{X}(g) = \prod_i a_i$ and $\det \psi(g) = \det \mathfrak{N}(g) = \prod_r b_r$.

We also have that

$$(e_i \otimes f_r) \cdot g = (a_i e_i \otimes b_r f_r) = a_i b_r (e_i \otimes f_r).$$

We conclude that the matrix given by the action of g on $V \otimes W$ with respect to the basis $\{(e_i \otimes f_r)\}$ is a diagonal matrix with the diagonal entries being $\{a_i b_r\}$. The determinant of this matrix (that is, $\det \chi\psi(g)$) is then

$$\prod_{i=1}^{\chi(1)} \prod_{r=1}^{\psi(1)} a_i b_r = \prod_{i=1}^{\chi(1)} a_i^{\psi(1)} \prod_{r=1}^{\psi(1)} b_r^{\chi(1)} = \det \chi(g)^{\psi(1)} \det \psi(g)^{\chi(1)},$$

which verifies the identity.

□

Problem 4.2. (Garrison) Let $\mathscr{K}$ be a class of G not contained in any proper normal subgroup. Let K be the corresponding class sum in $\mathbb{C}[G]$ and let m be the number of distinct values of $\omega_\chi(K)$ for $\chi \in \mathrm{Irr}(G)$. Show that every element of G can be written as a product of fewer than m elements of $\mathscr{K}$.

Hint: Mimic the proof of Theorem 4.3 using the Second Orthogonality Relation.

Proof. Denote the classes of G as $\{\mathscr{K}_v\}_{v=1}^k$ and their corresponding class sums as $\{K_v\}$. Choose $\{g_v\}$ to be a set of class representatives. Denote the m distinct values taken by $\omega_\chi(K)$ as $\{\alpha_i\}_{i=1}^m$ where α_1 is chosen so that $\alpha_1 = \omega_{1_G}(K) = |\mathscr{K}|$. We choose a representative set of irreducible characters $\{\chi_i\}$ where $\omega_{\chi_i}(K) = \alpha_i$ and $\chi_1 = 1_G$.

Now, assume for contradiction that we may choose an $h \in G$ such that h is not expressible as a product of fewer than m elements of $\mathscr{K}$. We set

$$X_i = \{\chi \in \mathrm{Irr}(G) \mid \omega_\chi(K) = \omega_{\chi_i}(K)\},$$

and set

$$\beta_i = \sum_{\chi \in X_i} \overline{\chi(h)}\chi(1). \tag{4.1}$$

Before proceeding, we first establish that $\beta_1 \neq 0$, as was done in the proof of Theorem 4.3. We observe that since $\mathscr{K}$ is not a subset of any proper normal subgroup, $\mathscr{K}$ is not contained in the kernel of any nonprincipal irreducible character of G. As such, 1_G is the only irreducible character such that $\omega_\chi(K) = |\mathscr{K}|$, so we may label $\beta_1 = 1$.

We now prove the below identity where j is an integer with $j < m$:

$$\sum_{i=1}^{m} \big(\omega_{\chi_i}(K)\big)^j \beta_i = 0. \tag{4.2}$$

Substituting the expression in (4.1) into the above, we obtain

$$\sum_{i=1}^{m} \big(\omega_{\chi_i}(K)\big)^j \beta_i = \sum_{i=1}^{m} \big(\omega_{\chi_i}(K)\big)^j \sum_{\chi \in X_i} \overline{\chi(h)}\chi(1) = \sum_{i=1}^{m} \sum_{\chi \in X_i} \big(\omega_{\chi_i}(K)\big)^j \overline{\chi(h)}\chi(1) \tag{4.3}$$

Notice that since the ω_χ's are algebra homomorphisms, we have that

$$\big(\omega_{\chi_i}(K)\big)^j = \big(\omega_\chi(K)\big)^j = \omega_\chi(K^j) = \sum_{v=1}^{k} a_{jv}\omega_\chi(K_v) \tag{4.4}$$

for any $\chi \in X_i$ and some integers a_{jv}. Substituting (4.4) into (4.3) yields

$$\begin{aligned} \sum_{i=1}^{m} \big(\omega_{\chi_i}(K)\big)^j \beta_i &= \sum_{i=1}^{m} \sum_{\chi \in X_i} \sum_{v=1}^{k} a_{jv}\omega_\chi(K_v)\overline{\chi(h)}\chi(1) \\ &= \sum_{v=1}^{k} a_{jv}|K_v| \sum_{i=1}^{m} \sum_{\chi \in X_i} \chi(g_v)\overline{\chi(h)} = \sum_{v=1}^{k} a_{jv}|K_v| \sum_{\chi \in \mathrm{Irr}(G)} \chi(g_v)\overline{\chi(h)}. \end{aligned} \tag{4.5}$$

Now, let L be the sum in $\mathbb{C}[G]$ over the elements of the conjugacy class containing h. We have assumed that h is not expressible as a product of fewer than m elements of $\mathscr{K}$, which is equivalent to the coefficient of L in K^j being zero for all $j < m$. In particular, either $a_{jv} = 0$ or h is not conjugate to g_v for $v = 1, \ldots, k$. Thus, by the Second Orthogonality Relations, the rightmost side of (4.5) is seen to be zero, as desired.

Now, form the $m \times m$ matrix M whose (i,j)-entry is $(\omega_{\chi_i}(K))^{j-1}$, and view $b = (\beta_i)$ as an m-row vector. By (4.2), we have that $bM = 0$. However, M is a Vandermonde matrix having nonzero determinant, and further, $(\beta_i) \neq 0$ since we have established above that $\beta_1 = 1$. We have reached a contradiction. $\square$

Problem 4.3. Let $G = H \times K$. Let $\varphi \in \mathrm{Irr}(G)$ and $\vartheta \in \mathrm{Irr}(K)$ be faithful. Show that $\varphi \times \vartheta$ is faithful if and only if $(|Z(H)|, |Z(K)|) = 1$.

Proof. First, assume that $\gcd(|\mathbf{Z}(H)|, |\mathbf{Z}(K)|) \neq 1$. In particular, there exists a prime p dividing the order of the center of both H and K. Let Z_H and Z_K be central subgroups of H and K respectively having order p. As φ is faithful, there exists $h \in Z_H$ such that $\varphi(h) = \varphi(1)\epsilon$ where ϵ is a primitive pth root of unity. Likewise, there exists $k \in Z_K$ such that $\psi(k) = \psi(1)\bar{\epsilon}$. One sees that $(\varphi \times \psi)(h, k) = \varphi(1)\psi(1)\epsilon\bar{\epsilon} = \varphi(1)\psi(1)$. Thus, $(h, k) \in \ker(\varphi \times \psi)$.

Conversely, assume that $\varphi \times \psi$ is not faithful, i.e, there exists $(h, k) \in H \times K$ with $\varphi(h)\psi(k) = \varphi(1)\psi(1)$. Now, by the triangle inequality, we have that $|\varphi(h)| \leqslant \varphi(1)$ and $|\psi(k)| \leqslant \psi(1)$. This implies that $|\varphi(h)| = \varphi(1)$ and $|\psi(k)| = \psi(1)$. We conclude that there exists roots of unity ϵ and η such that $\varphi(h) = \varphi(1)\epsilon$ and $\psi(k) = \psi(1)\eta$. Now, we have that $(\varphi \times \psi)(h, k) = \varphi(1)\psi(1)\epsilon\eta = \varphi(1)\psi(1)$ This implies that $\epsilon = \bar{\eta}$, which implies that ϵ and η are roots of unity having the same order. Since the order of these roots respectively divide the orders of $\mathbf{Z}(H)$ and $\mathbf{Z}(K)$, we have that $\gcd(|\mathbf{Z}(H)|, |\mathbf{Z}(K)|) \neq 1$, as desired. □

Problem 4.4. Suppose that $G = HK$ where $H \subseteq C_G(K)$.

(a). Let $\chi \in \mathrm{Irr}(G)$. Show that $\chi_H = \vartheta(1)\varphi$ and $\chi_K = \varphi(1)\vartheta$ for $\varphi \in \mathrm{Irr}(H)$ and $\vartheta \in \mathrm{Irr}(K)$.

(b). Let $\varphi \in \mathrm{Irr}(H)$ and $\vartheta \in \mathrm{Irr}(K)$ and suppose that $\varphi_{H\cap K}$ and $\vartheta_{H\cap K}$ have a common constituent. Show that there exists a unique $\chi \in \mathrm{Irr}(G)$ with $\chi_H = \vartheta(1)\varphi$ and $\chi_K = \varphi(1)\vartheta$.

Hint: G is a homomorphic image of $H \times K$.

Proof. For part (a), we observe that G is the homomorphic image of $H \times K$ under the map $\Xi : H \times K \to G$ given by $\Xi(h, k) = h \cdot k$. So, χ defines an irreducible character $\hat{\chi}$ on $H \times K$. This character is of the form $\varphi \times \vartheta$ for some $\varphi \in \mathrm{Irr}(H)$ and $\vartheta \in \mathrm{Irr}(K)$. We have that

$$\chi_H(h) = \hat{\chi}(h, 1) = \varphi(h) \cdot \vartheta(1).$$

Likewise, we have that

$$\chi_K(k) = \varphi(1) \cdot \vartheta(k).$$

as desired.

For part (b), we begin by showing existence. Consider the character $\varphi \times \vartheta \in \mathrm{Irr}(H \times K)$. We wish to show that this character may viewed as an irreducible character of G, which is equivalent to verifying that $\ker \Xi \subseteq \ker(\varphi \times \vartheta)$. First, notice that the centralizer condition above implies $H \cap K \leqslant \mathbf{Z}(G)$, so by Lemma 2.27(c), $\varphi_{H\cap K} = \varphi(1)\lambda_1$ and $\vartheta_{H\cap K} = \vartheta(1)\lambda_2$ for $\lambda_1, \lambda_2 \in \mathrm{Irr}(H \cap K)$. However, by our assumption that $\varphi_{H\cap K}$ and $\vartheta_{H\cap K}$ have a common constituent, it follows that $\lambda_1 = \lambda_2$. Notice now that $\ker \Xi = \{(g, g^{-1}) \mid g \in H \cap K\}$, and for $g \in H \cap K$, we have

$(\varphi\times\vartheta)(g,g^{-1}) = \varphi(g)\bar{\vartheta}(g) = \varphi(1)\vartheta(1)\lambda(g)\bar{\lambda}(g) = \varphi(1)\vartheta(1)$. Thus, $\ker\Xi \subseteq \ker(\varphi\times\vartheta)$, and it follows that we may view $\varphi\times\vartheta$ as a character of G. We will refer to this character in $\mathrm{Irr}(G)$ as χ. For any $h\in H$, we have $\chi(h) = \varphi(h)\times\vartheta(1) = \vartheta(1)\varphi(h)$, and similarly, for any $k\in K$, we have that $\chi(k) = \varphi(1)\times\vartheta(k) = \varphi(1)\vartheta(k)$. Thus, $\chi_H = \vartheta(1)\varphi$ and $\chi_K = \varphi(1)\vartheta$. The existence of χ is now verified.

It remains to show the uniqueness. Let $\chi,\psi\in\mathrm{Irr}(G)$ both satisfy the conditions stated above. By the claim in the hint, we may view χ and ψ as irreducible characters of $G\times H$, and so there exists $\varphi_1,\varphi_2\in\mathrm{Irr}(H)$ and $\vartheta_1,\vartheta_2\in\mathrm{Irr}(K)$ such that $\chi = \varphi_1\times\vartheta_1$ and $\psi = \varphi_2\times\vartheta_2$. By the computations in the first paragraph of the proof of Theorem 4.21, we have that $[\varphi_1\times\vartheta_1,\varphi_2\times\vartheta_2] = [\varphi_1,\varphi_2][\vartheta_1,\vartheta_2]$. Thus, if $\chi\neq\psi$, we have that either $\varphi_1\neq\varphi_2$ or $\vartheta_1\neq\vartheta_2$. We may assume by symmetry that $\varphi_1\neq\varphi_2$. However, by assumption, we have that $\chi_H = \psi_H$, which implies $\vartheta_1(1)\varphi_1 = \vartheta_2(1)\varphi_2$. By the linear independence of the irreducible characters, this relation implies $\varphi_1 = \varphi_2$, and likewise, $\vartheta_1 = \vartheta_2$. Thus, $\chi = \psi$, as desired. □

Problem 4.5. Let $G = H\times A$ where A is abelian and let $n > 0$. Show that $\chi^{(n)}\in\mathrm{Irr}(G)$ for each n such that $\gcd(n,|H|) = 1$.

Proof. We write $\chi = \varphi\times\vartheta$ for $\varphi\in\mathrm{Irr}(H)$ and $\vartheta\in\mathrm{Irr}(A)$ by Theorem 4.21. We observe that $\chi^{(n)} = \varphi^{(n)}\times\vartheta^{(n)}$. We claim that $\varphi^{(n)}$ and $\vartheta^{(n)}$ are irreducible characters of H and A respectively. Since $\gcd(|H|,n) = 1$, there is $\sigma\in\mathrm{Gal}(\mathbb{Q}(\epsilon)/\mathbb{Q})$ where ϵ is a primitive $|H|$-th root of unity and $\varphi^\sigma(g) = \varphi(g^n)$, so $\varphi^{(n)}$ is irreducible by Problem 2.2. Also, notice that $\vartheta^{(n)} = \vartheta^n\in\mathrm{Irr}(A)$. Now, $\chi^{(n)} = \varphi^{(n)}\times\vartheta^{(n)}$ is an irreducible character of G since both factors are irreducible. □

Problem 4.6. Let $n > 0$ and assume $\chi^{(n)}\in\mathrm{Irr}(G)$ for every $\chi\in\mathrm{Irr}(G)$. Show that $G = H\times A$, where A is abelian and $(|H|,n) = 1$.

Hints:

(a). Let $d = \gcd(|G|,n)$. Show that it is of no loss to assume that $\gcd(|G|/d,n) = 1$.

(b). Let $A = \bigcap_{\chi\in\mathrm{Irr}(G)}\ker\chi^{(n)}$. Show that $A = \{g\in G\mid g^n = 1\}$ and $|A| = d$.

(c). Let $H = \bigcap\{\ker\chi\mid\chi\in\mathrm{Irr}(G),\ \chi^{(n)} = 1_G\}$. Show $|G:H| = d$.

Remark: The proof for hint (c) of this problem is adapted from the following answer given on the Math Stack Exchange.

Proof. We first attempt to prove each of the hints.

Hint (a) is seen by observing that replacing n with n^k will not affect our assumptions for $k\in\mathbb{N}$. This is because if $(|H|,n^k) = 1$ for any subgroup of H of G, then we must also have $(|H|,n) = 1$, and if $\chi^{(n)}\in\mathrm{Irr}(G)$ for all $\chi\in\mathrm{Irr}(G)$, then we must also have $\chi^{(n^k)}\in\mathrm{Irr}(G)$ for any k. Now, we may choose a k large enough so that for any $p\mid\gcd(|G|,n^k)$, we have that $|G|_p\leqslant(n^k)_p$. In particular, taking $d_k = \gcd(|G|,n^k)$, the choice of such a k ensures that $\gcd(|G|/d_k,n^k) = 1$. Thus, the claim in hint (a) is

valid, and by replacing n with n^k for a suitable k, we may from now on assume that $\gcd(|G|/d, n) = 1$. Notice also that our choice of k implies that $\gcd(d, |G|/d) = 1$

For hint (b), we observe that if $g^n = 1$, then $\chi^{(n)}(g) = \chi(g^n) = \chi(1)$ for each $\chi \in \mathrm{Irr}(G)$. Conversely, if $g^n \neq 1$, then there exists an irreducible character such that $\chi(g^n) \neq \chi(1)$ by Lemma 2.21, so $g^n \notin \ker\chi$ or equivalently, $g \notin \ker\chi^{(n)}$. We have that $A = \{g \in G \mid g^n = 1\}$ as desired. Since A contains only the elements of G with order dividing d and since d and $|G|/d$ are relatively prime, it follows that $|A| = d$.

For (c), we wish to show that $|A| = |G : H|$. The condition that $\chi^{(n)} = 1_G$ implies that $\chi(1) = \chi^{(n)}(1) = 1$. Thus, $G' \leqslant H$ since H is the intersection of the kernels of some linear characters. However, notice that for any linear character $\lambda \in \mathrm{Irr}(G/H)$, we have that $\lambda^{(j)} = \lambda^j$ for any integer j. The set of all λ having order divisible by n is a subgroup of $\mathrm{Irr}(G/H)$. However, by the definition of H, it follows that this subgroup cannot be proper in $\mathrm{Irr}(G/H)$. Thus, $\{\chi \in \mathrm{Irr}(G) \mid \chi^{(n)} = 1_G\} = \mathrm{Irr}(G/H)$.

Consider the class function $\vartheta_n(h) = |\{g \in G \mid h = g^n\}|$. By the hint in (b), we see that $\vartheta_n(1) = |A|$. On the other hand, $\vartheta_n = \sum_{\chi\in\mathrm{Irr}(G)} \nu_\chi\chi$ where $\nu_n(\chi) = [\chi^{(n)}, 1_G]$. By the previous paragraph, one sees that $\nu_n(\chi) = [\chi^{(n)}, 1_G] = 1$ iff $\chi \in \mathrm{Irr}(G/H)$. Otherwise, $\nu_n(\chi) = 0$. We are now able to conclude that

$$d = |A| = \vartheta_n(1) = \sum_{\lambda\in\mathrm{Irr}(G/H)} \lambda(1) = |\,\mathrm{Irr}(G/H)| = |G : H|,$$

which proves the claim in hint (c).

We are now ready to prove the statement in the problem. Combining information from hints (b) and (c), we have that H and A are normal subgroups of relatively prime order such that $|H||A| = |G|$, so $G \cong H \times A$. Furthermore, A is abelian since $A \cong G/H$, and we have established above that $G' \leqslant H$ since H is defined to be the intersection of kernels of some linear characters. This completes the proof. □

Problem 4.7. Let V be a $\mathbb{C}[G]$-module with basis $\{v_1, v_2, \ldots, v_n\}$ and let p be a prime. Let $W = V \otimes V \otimes \ldots \otimes V$ (p times), and define α on W by defining $(x_1 \otimes \ldots \otimes x_p)^\alpha = x_2 \otimes x_3 \otimes \ldots \otimes x_p \otimes x_1$ for all $x_i \in V$. Let ϵ be a primitive p-th root of unity in $\mathbb{C}$. Let $W_1 = \{w \in W \mid w^\alpha = \epsilon w\}$. Show that W_1 is a $\mathbb{C}[G]$-submodule of W which affords the character $\chi_1 = (\chi^p - \chi^{(p)})/p$ where χ is the character afforded by V. Conclude that $\chi^{(m)}$ is a difference of characters for all $m \geqslant 1$.

Hints: Let $\mathscr{W} = \{v_{i_1} \otimes \ldots \otimes v_{i_p} \mid \text{not all } i_j \text{ are equal}\}$. For $w \in \mathscr{W}$, let

$$\bar{w} = w + \epsilon^{-1}w^\alpha + \epsilon^{-2}w^{\alpha^2} + \ldots + \epsilon^{-(p-1)}w^{\alpha^{p-1}}.$$

Now, $\langle\alpha\rangle$ permutes $\mathscr{W}$ into orbits of size p. Let $\mathscr{W}_0$ be a set of representatives of these orbits. Then, $\{\bar{w} \mid w \in \mathscr{W}_0\}$ is a basis for W_1.

Remark: This is a problem whose proof involves lots of statements which are routine to verify but tedious to typeset. As such, the proof is sketched in as much

detail as is reasonable, and some statements in the following argument are left to the reader to verify.

Proof. We first observe that it is clear that W_1 is a subspace of W and that since α commutes with the action of G, W_1 is also a $\mathbb{C}[G]$-submodule. We wish to verify that the set $B := \{\bar{w} \mid w \in \mathscr{W}_0\}$ given in the hint is a basis for W_1. First, notice that $|B| = (n^p - n)/p$ (where the subtraction by n in the numerator appears since we excluded the tensors of the form $x_i \otimes x_i \otimes \ldots \otimes x_i$ when constructing $\mathscr{W}$). The linear independence of B follows from the fact that the w^{α^i} are each distinct basis elements of W for $w \in \mathscr{W}$ and $i = 0, 1, \ldots, p-1$. We prove that B spans W_1 by a counting argument. Define the $\mathbb{C}$-linear map $M : V \to W_1$ given by $v \longmapsto v + \epsilon^{-1}v^{\alpha} + \ldots + \epsilon^{-(p-1)}v^{\alpha^{p-1}}$. One sees from the definition of α that for any $v \in W_1$, we have $M(v) = pv$. In particular, this implies that $\operatorname{rank}(M) \geqslant |B|$. Consider the set

$$B' = \{w - \epsilon^{-i}w^{\alpha^i} \mid w \in \mathscr{W}_0, i = 1, \ldots, p-1\} \cup \{v_i \otimes v_i \otimes \ldots \otimes v_i \mid i = 1, \ldots, p\}.$$

The set B' is seen to be linearly independent with cardinality $(p-1)(n^p - p)/p + p$, which equals $n^p - (n^p - p)/p$. One computes that $M(v) = 0$ for each v in the span of B', i.e., $\operatorname{nullity}(M) \leqslant B'$. We see now by the rank nullity theorem that

$$\dim(V) = \operatorname{rank}(M) + \operatorname{nullity}(M) \leqslant +|B'| = [(n^p - p)/p] + [n^p - (n^p - p)/p] = \dim(V).$$

Thus, equality holds above, and W_1 is exactly the span of B. In particular, B is a basis for W_1.

Fix $g \in G$. We may choose a basis $\{v_i\}$ of V on which g acts diagonally, i.e., for each i, $v_i g = \lambda_i v_i$ for root of unity $\lambda_i \in \mathbb{C}$. Let $B_V = \{v_{i_1} \otimes \ldots \otimes v_{i_p}\}$, and notice that B_V is a basis for V. Applying g to $v \in B_V$, we see that

$$vg = (v_{i_1} \otimes \ldots \otimes v_{i_p})g = \prod_{i_k} \lambda_{i_k} =: \lambda_v.$$

The trace of the action of g on V is hence given by $\sum_{v \in B_V} \lambda_v$. On the other hand, by Theorem 4.1, the trace of the action of g on V is given by $\chi(g)^p$. Also, for the $F[G]$-submodule of V generated by the "diagonal" tensors, i.e., the tensors of the form $v_i \otimes v_i \otimes \ldots \otimes v_i$, the trace of the action of g is given by $\chi(g^p)$. In particular, since the span of $\mathscr{W}$ is a vector space complement of span of the diagonal tensors, we see that

$$\sum_{w \in \mathscr{W}} \lambda_w = \chi(g)^p - \chi(g^p).$$

Further, since the action of g commutes with α, it follows that $\lambda_w = \lambda_{w^\alpha}$ for each $w \in \mathscr{W}$, and so,

$$\chi(g)^p - \chi(g^p) = p \sum_{w \in \mathscr{W}_0} \lambda_w$$

Now, since the trace of g on W_1 is given by $\chi_1(g) := \sum_{\bar{w}\in B} \lambda_{\bar{w}}$, it merely suffices to show that $\sum_{\bar{w}\in B} \lambda_{\bar{w}} = \sum_{w\in\mathscr{W}_0} \lambda_w$. In particular, we show that $\lambda_w = \lambda_{\bar{w}}$. Indeed, when g is applied to an element $\bar{w} \in B$, we obtain

$$\lambda_{\bar{w}}\bar{w} = \bar{w}g = \sum_{i=0}^{p-1} \epsilon^{-i}(wg)^{\alpha^i} = \sum_{i=0}^{p-1} \epsilon^{-i}(\lambda_w w)^{\alpha^i} = \lambda_w \sum_{i=0}^{p-1} \epsilon^{-i}(w)^{\alpha^i} = \lambda_w \bar{w},$$

as desired. It now follows that

$$\chi_1(g) = (\chi(g)^p - \chi^{(p)}(g))/p$$

for any $g \in G$.

To prove that $\chi^{(m)}$ is a difference of characters for any natural number m, we proceed by induction on m. Assume that $\chi^(\ell)$ is a difference of characters for all $\ell < m$. We write $m = pk$ for a prime p and an integer k. We have that $\chi^m = (\chi^k)^p$. By the induction hypothesis, $\chi^{(k)}$ may be written as $\phi - \psi$ for characters ϕ and ψ of G. Now, $\chi^{(m)} = \phi^{(p)} - \phi^{(p)}$. By the above paragraph, we may write $\phi^{(g)} = \phi^p - \phi_1$ and $\psi^{(p)} = \psi^p - \psi_1$. We now have that

$$\chi^{(m)} = (\phi^p + \psi_1) - (\psi^p + \phi_1),$$

and so $\chi^{(m)}$ is a difference of characters, as desired. □

Problem 4.8. Let G be a p-group with Frattini factor group of order greater than or equal to p^{2a-1}. Show that the number of elements of order p in G is $\equiv -1 \bmod p^a$.

Hint: Mimic the proof of Theorem 4.9. Use the fact that $\nu_p(\chi)$ is an integer (Problem 4.7).

Remark: The proof provided is based upon a proof of the same result given in Theorem 9.5 of *Character Theory of Finite Groups* by Bertram Huppert.

Proof. In the notation introduced on page 49, we wish to consider

$$\vartheta_p(1) = \sum_{\chi\in\mathrm{Irr}(G)} \nu_p(\chi)\chi(1)$$

since $\vartheta(1) - 1$ is exactly the number of elements of G having order p. By the previous problem, each $\nu_p(\chi)$ is an integer. We remark that the Frattini factor group $A := G/\Phi(G)$ is an elementary abelian p-group.[3] In particular,

$$A = \bigcap \{\ker\lambda \mid \lambda(1) = 1, \lambda(x^p) = 1 \text{ for each } x \in G\}.$$

So, A acts on the irreducible $\mathrm{Irr}(G)$ in the same way as G/G' does. $\mathrm{Irr}(G)$ is partitioned into orbits under this action which we denote $\{O_i\}$. We choose a representative

[3]See Theorem 5.1.3 in *Finite Groups* by Daniel Gorenstein, for instance.

set $\{\chi_i\}$ for each orbit. We also recall that the action of A on $\mathrm{Irr}(G)$ fixes both $\nu_p(\chi)$ and $\chi(1)$ for each $\chi \in \mathrm{Irr}(G)$. We will show that

$$\vartheta_p(1) = \sum_{\chi \in \mathrm{Irr}(G)} \nu_p(\chi)\chi(1) \equiv \sum_i \nu_p(\chi)|O_i|\chi_i(1) \equiv 0 \bmod p^a$$

by showing that $p^a \,\big|\, |O_i|\chi_i(1)$ for each i, and this will imply the desired congruence above.

We introduce some notation. We define

$$A_\chi = \{\lambda \in \mathrm{Irr}(A) \mid \chi\lambda = \chi\} \text{ and } R_\chi = \{a \in A \mid \lambda(a) = 1 \text{ for each } \lambda \in A_\chi\}$$

First, observe that if $\lambda \in A_\chi$, then χ vanishes off the kernel of λ, and consequently, if $g \in aR_\chi$ for some $a \notin R_\chi$, then $\chi(g) = 0$. Also, observe that A_χ and R_χ are subgroups of $\hat{A}$ and A respectively. In the notation of Problem 2.7, $R_\chi^\perp = A_\chi$. Specifically, one verifies that for any subgroup H of an abelian group K, we have $|K : H| = |H^\perp|$. Using this fact and the Orbit-Stabilizer Theorem, we deduce that $|R_{\chi_i}| = |O_i|$. We claim that $\chi_i(1)^2 \geqslant |G : R_{\chi_i}|$. This is seen by observing that for any $\chi \in \mathrm{Irr}(G)$, we have

$$\chi(1)^2 \geqslant \frac{1}{|R_\chi|} \sum_{h \in R_\chi} |\chi(h)|^2 = \frac{1}{|R_\chi|} \sum_{g \in G} |\chi(g)|^2 = |G : R_\chi|[\chi, \chi] = |G : R_\chi|.$$

Setting $|O_i| = |R_\chi| = p^b$ and multiplying both sides of the above equation by $|O_i|^2$ yields

$$|O_i|^2\chi_i^2(1) \geqslant |A| \cdot p^b$$

for some $b \geqslant 0$. Using our assumption on the order of A, we have that

$$|O_i|^2\chi_i^2(1) \geqslant p^{2a-1+b},$$

so

$$p^{2a-1+b} \,\big|\, |O_i|^2\chi_i^2(1).$$

Since $|O_i|$ and $\chi_i(1)$ are powers of p, it follows that $|O_i|\chi_i(1)$ divides p^a. □

Problem 4.9. If $\chi \in \mathrm{Irr}(G)$, then $|\nu_2(\chi)| \leqslant 1$. Show that no absolute bound exists on $\nu_n(\chi)$ for $n > 1$.

Hint: There exist groups of prime exponent $p \neq 2$ and of exponent 4 with irreducible characters of arbitrarily large degree.

Proof. Let A be the cyclic group of order $n > 2$. Assume first that $p \mid n$ where p is an odd prime, and let m be any positive integer. Let $U(3, p)$ denote the group of upper unitriangular 3×3 matrices with entries in the field of size p. This group has order p^3 and exponent p.[4] One also sees that this group possesses a nonlinear

[4]The fact that $U(3, p)$ has exponent p for $p > 2$ can be proven in an elementary way. Choose $A \in U(3, p)$, and write $A = I + N$ where I is the identity matrix. Observe that $N^p = 0$, and apply the the binomial theorem to conclude that $A^p = (I + N)^p = I$.

irreducible character χ. Let G denote the m-fold product of $U(3,p)$, and let ψ be the m-fold direct product of χ viewed as an irreducible character of G (in the sense of Theorem 4.21). Observe that since G has exponent p and since $p \mid n$, we have $\psi^{(n)}(g) = \psi(g^n) = \psi(1) = \chi(1)^m$. In particular, $\psi^{(m)} = \chi(1)^m 1_G$, and so $\nu_n(\psi) = [\psi^{(n)}, 1_G] = \psi(1) = \chi(1)^m$. Since χ was chosen to be nonlinear and since G is a p-group, we have that $\nu_n(\psi) \geqslant p^m$. However, we can choose m to be arbitrarily large, so a group G and $\chi \in \mathrm{Irr}(G)$ can be chosen so that $\nu_n(\psi)$ is arbitrarily large in the case where n is divisible by an odd prime.

Assume now that n is not divisible by any odd prime so that $n = 2^a$ for some integer $a > 1$. Consider the quaternion group Q_8, and let $\chi \in \mathrm{Irr}(Q_8)$ be the unique irreducible character of degree 2. the group Q_8 has exponent dividing 4. Let m be a positive integer, and again let G denote the m-fold direct product of Q_8. Also, let ψ denote the m-fold direct product of χ which we view as an irreducible character of G in the sense of Theorem 4.21. Observe that $\psi^{(n)}(g) = \psi(g^n) = \psi(1) = \chi(1)^m = 2^m$ for all $g \in G$. (where $\psi(g^n) = 1$ since $4 \mid n$ and G has exponent 4.) We now conclude that $\nu_n(\psi) = [\psi^{(n)}, 1_G] = 2^m$. Since we can choose m to be arbitrarily large, the claim in the problem now follows. □

Problem 4.10. Let V, W, V_1 and W_1 be $F[G]$-modules for an arbitrary field. Fix a particular F-basis in each module and construct $V \otimes W$ and $V_1 \otimes W_1$ as was done for $\mathbb{C}[G]$. Suppose that $\alpha : V \to V_1$ and $\beta : W \to W_1$ are module homomorphisms. Show that there exists a module homomorphism $\gamma : V \otimes W \to V_1 \otimes W_1$ such that $\gamma(v \otimes w) = \alpha(v) \otimes \beta(w)$. Conclude that $V \otimes W$ is determined up to module isomorphism independently of the choice of bases.

Proof. Let $\{v_i\}$ and $\{w_j\}$ be bases for V and W respectively. The desired homomorphism γ may be constructed by defining $\gamma(v_i \otimes w_j) = \alpha(v_i) \otimes \beta(w_j)$ and extending F-linearly.

Given arbitrary $v \in V$ and $w \in W$, we wish to show $\gamma(v \otimes w) = \alpha(v) \otimes \beta(w)$. Writing $v = \sum_i a_i v_i$ and $w = \sum_j b_j w_j$, we may write

$$v \otimes w \sum_{i,j} a_i b_j (v_i \otimes w_j).$$

We have that

$$\alpha(v) \otimes \beta(w) = \Big(\alpha\big(\sum_i v_i\big) \otimes \beta\big(\sum_j b_j w_j\big)\Big) = \Big(\alpha\big(\sum_i v_i\big) \otimes \sum_j b_j \beta(w_j)\Big)$$

$$= \sum_{i,j} a_i b_j (\alpha(v_i) \otimes \beta(w_j)) = \sum_{i,j} a_i b_j \gamma(v_i \otimes w_j) = \gamma(v \otimes w),$$

as desired.

We now show that the map γ is an $F[G]$-module homomorphism. To this end, it suffices to show that $\gamma((v_i \otimes w_j) \cdot g) = \gamma(v_i \otimes w_j) \cdot g$ for each $g \in G$. Now, $\gamma((v_i \otimes w_j) \cdot g) =$

$\gamma(v_i g \otimes w_i g)$, and by the previous paragraph $\gamma(v_i g \otimes w_j g) = (\alpha(v_i g) \otimes \beta(w_j g))$. Now, since α and β are $F[G]$-module homomorphisms, we have that

$$(\alpha(v_i g) \otimes \beta(w_j g)) = (\alpha(v_i)g \otimes \beta(w_j)g) = (\alpha(v_i) \otimes \beta(w_j))g = \gamma(v_i \otimes w_j) \cdot g,$$

as desired.

Let $(V \otimes W)_1$ and $(V \otimes W)_2$ be tensor product constructed from different choices of bases for V and W. Here, we take $\alpha : V \to V$ and $\beta : W \to W$ to be the identity maps. By our work above, we may construct an $F[G]$-module homomorphism $(\alpha \otimes \beta) : (V \otimes W)_1 \to (V \otimes W)_2$. It is clear that if $(\alpha \otimes \beta)$ defines an isomorphism since α and β are isomorphisms. □

Problem 4.11. Let G be simple and let $S \in \mathrm{Syl}_2(G)$ be elementary abelian group of order q. Suppose that $S = C_G(x)$ for all $1 \neq x \in S$. Show that $\nu_2(\chi) = 1$ for all $\chi \in \mathrm{Irr}(G)$ and that $\chi(x) = \chi(1) - q$ for $\chi \neq 1_G$ and $x \in S$ with $x \neq 1$.

Hint: First show that G contains exactly $|G|/q$ involutions and that the remaining elements of G have odd order. This is done without characters using the fact that if x, y are nonconjugate involutions, then $|\mathbf{Z}(\langle x, y\rangle)|$ is even.

Note: This problem is continued as Problem 5.20 where it is proved that $|G| = q(q-1)(q+1)$.

Proof. We shall assume that G possesses $|G|/q$ involutions as stated in the hint, and we further prove that the set of involutions in G forms a single conjugacy class. The proof of these facts are relegated to a sublemma below. Also, notice that the claim that $\nu_2(\chi) = 1$ is obvious when $\chi = 1_G$, so we shall assume throughout that $\chi \neq 1_G$.

We first claim that every element of G either has odd order or an involution. Assume $g \in G$ has even order. We may factorize g like so: $g = xy = yx$, where x is a 2-element and y is a $2'$-element. Since x is a 2-element, it lies in a Sylow 2-subgroup and is therefore an involution. However, the involution x centralizes the $2'$-element y, but since $C_G(x) = S$ by assumption, it follows that $y = 1$. This proves the claim.

We now claim that the map $g \longmapsto g^2$ permutes the elements of odd order in G. Indeed, write $|G| = qm$ so that m is odd and every element of odd order in g has order dividing m. We may write $2a + mb = 1$ for some integers a and b, and we see that $g^{2a} = g^{2a+mb} = g$ where g is any element having odd order. We conclude that the map $g \longmapsto g^2$ is invertible by the map $g \longmapsto g^a$ and is therefore a bijection as desired. In particular, by the claim in the previous paragraph and since $[\chi, 1_G] = 0$, we have that

$$\sum_{\substack{g \in G \\ o(g) \text{ is odd}}} \chi(g^2) = \sum_{\substack{g \in G \\ o(g) \text{ is odd}}} \chi(g) = -\sum_{\substack{g \in G \\ o(g)=2}} \chi(g) = -\frac{|G|}{q}\chi(x), \tag{4.6}$$

where x is any involution of G. The last equality holds since the involutions of G form a single conjugacy class.

Now, we shall evaluate $\nu_2(\chi)$. We have that

$$\nu_2(\chi) = [\chi^{(2)}, 1_G] = \frac{1}{|G|}\sum_{g\in G}\chi(g^2).$$

By the claim in the second paragraph, every element of G either has odd order or order 2, thus we may split the above sum as

$$\nu_2(\chi) = \frac{1}{|G|}\sum_{\substack{g\in G\\ o(g)=2}}\chi(g^2) + \frac{1}{|G|}\sum_{\substack{g\in G\\ o(g)\text{is odd}}}\chi(g^2).$$

Every term in the first sum equals $\chi(1)$. Thus, by (4.6), we now have

$$\nu_2(\chi) = \frac{\chi(1)}{q} - \frac{\chi(g)}{q} = \frac{1}{q}(\chi(1) - \chi(x)). \tag{4.7}$$

where x is some involution of G.

By the Theorem 4.5(a), $\nu_2(\chi) \in \{-1, 0, 1\}$. However, if $\nu_2(\chi) = 0$, this implies $\chi(1) = \chi(x)$, which implies $\ker\chi \neq 1$ and thereby contradicts the simplicity of G. On the other hand, if $\nu_2(\chi) = -1$, this implies $\chi(g) = \chi(1) + q$, which contradicts the fact that $|\chi(g)| \leqslant \chi(1)$ for all $g \in G$ (Lemma 2.15(c)). Thus, $\nu_2(\chi) = 1$. The fact that $\chi(x) = \chi(1) - q$ is now evident from (4.7), which completes the proof. □

Sublemma. Assuming the notation of Problem 4.11, the set of all involutions in G is a conjugacy class having size $|G|/q$.

Remark: A much stronger statement holds. Given a finite group G and $P \in \mathrm{Syl}_2(G)$ where $P \cap P^g = 1$ for each $g \in G$, then either $P \lhd G$ or the set of involutions in G form a single conjugacy class of G. See Theorem 9.1.4 in *Finite Groups* by Daniel Gorenstein for a proof of this fact.

Proof. In view of the hint, we first prove that any two Sylow subgroups of G intersect trivially. Indeed, if there existed a nonidentity element $y \in S \cap S^g$ for some $g \in G$, we would see that $C_G(y) = S = S^g$. Also, notice that by the Sylow Theorems and the fact that $C_G(x^g) = C_G(x)^g$, we have that $C_G(z)$ is a Sylow 2-subgroup for any involution z of G.

Now, we prove that any two involutions in G are conjugate. Let S_1 and S_2 be distinct Sylow 2-subgroups of G (which exist since G is simple). It suffices by the Sylow Theorems to show that every nonidentity element of $\in S_2$ is conjugate to every nonidentity element of S_1. Indeed, if $y, z \in S_1$ are involutions, then there exist $a, b \in G$ such that $y^a = z^b = x$ for some $x \in S_2$, so $y^{ab^{-1}} = z$.

Now, assume that $g \in S_1$ and $h \in S_2$ are nonconjugate involutions for distinct Sylow 2-subgroups S_1 and S_2. It is a well-known fact in group theory that the subgroup generated by two involutions, say $H = \langle g, h\rangle$, is isomorphic either to the Klein-four

group or to the dihedral group of order $2n$ where $n = o(xy)$.[5] Recall that $|\mathbf{Z}(D_{2n})| = 1$ if n is odd and that $|\mathbf{Z}(D_{2n})| = 2$ if n is even. We claim that n is odd, for otherwise, if n was even, the $\langle g, h\rangle$ is a dihedral group with a central involution, say k. We see that $g, h \in C_G(k) = S_k$ for some Sylow 2-subgroup S_k, which particularly implies $S_k \cap S_1 \neq 1$ and $S_k \cap S_2 \neq 1$. However, we have established above that any two Sylow 2-subgroups of G intersect trivially. Thus, we have $S_1 = S_k = S_2$, contrary to our assumption that $S_1 \neq S_2$. Thus, n is odd.

We write $n = 2m + 1$, and we see that the relation $(gh)^n = 1$ implies $g = (hg)^m h (gh)^m$. Since $(gh)^{-1} = hg$, we see that $h^{(gh)^m} = g$, proving that g and h are conjugates. Since g and h were chosen arbitrarily from S_1 and S_2 respectively, we see that any element of S_1 is conjugate to an element of S_2, and by the second paragraph above, all involutions are conjugate to one another.

Let $\mathscr{K}$ denote the set of all involutions of G. By the Orbit-Stabilizer Theorem and since $\mathscr{K}$ is a conjugacy class, we have that $|G| = |\mathscr{K}||C_G(x)| = |\mathscr{K}|q$, where x is any involution of G. Thus, the number of involutions in G is $|G|/q$, which proves the claim in the hint. □

Problem 4.12. Let $\chi, \psi, \vartheta \in \mathrm{Irr}(G)$. Show that $[\chi\psi, \vartheta] \leqslant \vartheta(1)$.

Proof. This fact is an application of the triangle inequality and the Schwarz Inequality (see the proof of Lemma 4.10). We recall that $|\vartheta(g)| \leqslant \vartheta(1)$ for all $g \in G$, and this gives us that

$$\begin{aligned}[\chi\psi, \vartheta] = |[\chi\psi, \vartheta]| &= \frac{1}{|G|}\left|\sum_{g\in G} \chi(g)\psi(g)\overline{\vartheta(g)}\right| \leqslant \frac{\vartheta(1)}{|G|}\sum_{g\in G}|\chi(g)\psi(g)| \\ &\leqslant \frac{\vartheta(1)}{|G|}\sqrt{\sum_{g\in G}|\chi(g)|^2 \cdot \sum_{g\in G}|\psi(g)|^2} = \frac{\vartheta(1)}{|G|}\sqrt{|G|^2} = \vartheta(1),\end{aligned}$$

which proves that $[\chi\psi, \vartheta] \leqslant \vartheta(1)$. □

[5] This can be reasoned by a fairly routine argument involving generators and relations. For a full proof, see either the link here or Theorem 9.1.1 in *Finite Groups* by Daniel Gorenstein.

5. Induced characters

Problem 5.1. Let $H \leqslant K \leqslant G$. Show that $(\phi^K)^G = \phi^G$.

Proof. This can be proven readily from the definition of character induction, but a cleaner proof arises from using Frobenius Reciprocity. Namely, for any $\chi \in \mathrm{Irr}(G)$, we have

$$[(\phi^K)^G, \chi] = [\phi^K, \chi_K] = [\phi, \chi_H] = [\phi^G, \chi].$$

We conclude that

$$(\phi^K)^G = \sum_{\chi \in \mathrm{Irr}(G)} [(\phi^K)^G, \chi]\chi = \sum_{\chi \in \mathrm{Irr}(G)} [\phi^G, \chi]\chi = \phi^G,$$

as desired. □

Problem 5.2. Let $H, K \leqslant G$ with $HK = G$ and suppose ϕ is a class function of H. Show that $(\phi^G)_K = (\phi_{H\cap K})^K$.

Proof. Since $G = HK$, we may choose a transversal of right cosets of H in G consisting entirely of elements of K: $\{t_i\}_{i=1}^{|G:H|}$. We have that for $k \in K$,

$$\phi^G(k) = \sum_{i=1}^{|G:H|} \phi^\circ(t_i k t_i^{-1}).$$

Observe also that $|G : H| = |K : H \cap K|$ and that $\{t_i\}$ is also a transversal for right cosets of $H \cap K$ in K so that

$$(\phi_{H\cap K})^K(k) = \sum_{i=1}^{|G:H|} \phi^\circ(t_i k t_i^{-1}),$$

proving the equality. □

Problem 5.3. Let $H \leqslant G$ and suppose ϕ is a class function of H and ψ is a class function of G. Show that $(\phi\psi_H)^G = \phi^G\psi$.

Note: Problems (5.1)–(5.3) will be used frequently.

Proof. As in the proof of Problem 5.1, we again use Frobenius Reciprocity to prove this statement. For each $\chi \in \mathrm{Irr}(G)$, we have

$$[(\phi\psi_H)^G, \chi] = [\phi\psi_H, \chi_H] = [\phi, \overline{\psi_H}\chi_H] = [\phi^G, \overline{\psi}\chi] = [\phi^G\psi, \chi].$$

We conclude that

$$(\phi\psi_H)^G = \sum_{\chi \in \mathrm{Irr}(G)} [(\phi\psi_H)^G, \chi]\chi = \sum_{\chi \in \mathrm{Irr}(G)} [\phi^G\psi, \chi]\chi = \phi^G\psi,$$

as desired. □

Problem 5.4. Let $b(G) = \max\{\chi(1) \mid \chi \in \mathrm{Irr}(G)\}$. Show that if $H \leqslant G$ then $b(H) \leqslant b(G) \leqslant |G : H|b(H)$.

Note: Since H is abelian iff $b(H) = 1$, the inequality $b(G) \leqslant |G : H|b(H)$ generalizes Problem 2.9(b).

Proof. Choose $\psi \in \mathrm{Irr}(H)$ such that $\psi(1) = b(H)$. By Corollary 5.4, there is $\chi \in \mathrm{Irr}(G)$ such that ψ is a constituent of χ_H, so $b(H) = \psi(1) \leqslant \chi(1) \leqslant b(G)$.

On the other hand, if we choose $\chi \in \mathrm{Irr}(G)$ such that $\chi(1) = b(G)$ and we choose an irreducible constituent ψ of χ_H, we have $[\psi, \chi_H] \neq 0$, so Frobenius Reciprocity yields that $[\psi^G, \chi] \neq 0$. We have that $\chi(1)$ is a constituent of ψ^G. This means that $\chi(1) \leqslant \psi^G(1) = |G : H|\psi(1) \leqslant |G : H|b(H)$. This verifies both inequalities. □

Problem 5.5. Let F be an arbitrary field and let $H \subseteq G$. Let W be an $F[H]$-module. Show that there exists a unique (up to $F[G]$-isomorphism) $F[G]$-module V such that V has imprimitivity decomposition $\sum \cdot W_i$ where $W_1 \cong W$ and $F[H]$ is the stabilizer of W_1 in this action.

Notes: By Theorem 5.9, $V \cong W^G$. In the following problem, if V is an $F[G]$-module and $H \subseteq G$, we use the notation V_H to denote V viewed as an $F[H]$-module.

Proof. Existence is guaranteed by Theorem 5.9, so we need only prove uniqueness. Let V have imprimitivity decompositions $\sum \cdot W_i$ and $\sum \cdot V_i$. It follows from the Orbit-Stabilizer Theorem that both the number of factors in both direct sums are equal to $|G : H|$. If we take $\{t_i\}_{i=1}^{|G:H|}$ to be a transversal of right cosets, then we may rewrite $\sum \cdot W_i = \sum \cdot W_i t_i$ and $\sum \cdot V_i = \sum \cdot V_i t_i$. If we have an $F[H]$-module isomorphism $\vartheta : W_1 \to V_1$, then we define $\vartheta^* : \sum \cdot W_i t_i \to \sum \cdot V_i t_i$ by setting $\vartheta^*(wt_i) = \vartheta(w)t_i$ for $w \in W_1$ and then extending the map linearly. We claim ϑ^* is an $F[G]$-module isomorphism.

The map ϑ^* is clearly F-linear, so let $g \in G$ and evaluate $\vartheta^*(wt_i g)$. We observe that there is a unique $t_{i,g} \in \{t_i\}$ and $h_{i,g} \in H$ such that $t_i g = h_{i,g} t_{i,g}$. For $w \in W_1$, this yields

$$\vartheta^*(wt_i g) = \vartheta^*(wh_{i,g} t_{i,g}) = \vartheta(w)h_{i,g} t_{i,g}$$

and

$$\vartheta^*(wt_i)g = \vartheta(w)t_i g = \vartheta(w)h_{i,g} t_{i,g},$$

which verifies ϑ^* is an $F[G]$-module isomorphism, so that the module V is the unique $F[G]$-module isomorphism with such an imprimitivity decomposition. □

Problem 5.6. (Mackey) Let $H, K \subseteq G$ and let T be a set of (H, K)-coset representatives so that

$$G = \bigcup_{t \in T} HtK$$

is a disjoint union. Let W be an $F[H]$-module for an arbitrary field F. Show that

$$(W^G)_K = \sum_{t \in T} \cdot \, (W^t_{H^t \cap K})^K$$

Note: Mackey's theorem generalizes Problem 5.2.

Proof. Let $S = \{s_i\}$ be a set of right coset representatives of H in G. We have that $W^G = \sum \cdot W \otimes s_i$. When viewing W^G as an $F[K]$-module, this module decomposes into a direct sum of factors

$$(W^G)_K = \sum_{t \in T} \cdot \big(\sum_j \cdot W \otimes tk_{t,j} \big)$$

where $\{tk_{t,j}\}$ are coset representatives such that $Htk_{t,j} \subseteq HtK$. Each $\sum_j \cdot W \otimes tk_{t,j}$ is an $F[K]$-module in an imprimitivity decomposition. Observe that the stabilizer in K of the summand $W \otimes t$ are the elements of K which also lie in H^t. Thus, each $\sum_j \cdot W \otimes tk_{t,j}$ is induced from the $H^t \cap K$-module $W \otimes t = W^t_{H^t \cap K}$, as desired. □

Problem 5.7. Let $H \subseteq G$ and suppose ψ is a character of H. Let $K \subseteq G$ and assume $(\psi^G)_K \in \mathrm{Irr}(K)$. Show that $HK = G$.

Note: Problem 5.7 is an immediate consequence of Problem 5.6. It can, however, be done without using modules. The key step is to show that $[(\psi^G)_K, (\psi_{H\cap K})^K] \neq 0$ and then conclude that $|G : H| \leqslant |K : K \cap H|$.

Proof. We take the suggestion given as a note to show that $[(\psi^G)_K, (\psi_{H\cap K})^K] \neq 0$. Choose transversals T and S for right cosets of H in G and for $H \cap K$ in H respectively. By Frobenius Reciprocity and Problem 5.1, we have that

$$[(\psi^G)_K, (\psi_{H\cap K})^K] = [\psi^G, (\psi_{H\cap K})^G] = [(\psi^G)_{H\cap K}, \psi_{H\cap K}]$$

Observe that the rightmost expression is nonzero if $[(\psi^G)_H, \psi_H] \neq 0$. However, again, by Frobenius Reciprocity, we have $[(\psi^G)_H, \psi] = [\psi^G, \psi^G]$, which is certainly nonzero. Thus, $(\psi^G)_K$ is an irreducible constituent of $(\psi_{H\cap K})^K$, and in particular, $(\psi^G)_K$ has degree less than or equal to $(\psi_{H\cap K})^K$, but these degrees are $\psi(1)|G : H|$ and $\psi(1)|K : K \cap H|$ respectively. We have that

$$|G : H| \leqslant |K : K \cap H|$$

so

$$|G| = \frac{|H||K|}{|H \cap K|}$$

and $G = HK$. □

Problem 5.8. Let χ be a monomial character of G such that $\chi_K \in \mathrm{Irr}(K)$ for some $K \subseteq G$. Prove that χ_K is monomial as well.

Proof. We may write $\chi = (\lambda)^G$ for some linear $\lambda \in \mathrm{Irr}(H)$ with $H \subseteq G$, so $(\lambda^G)_K \in \mathrm{Irr}(K)$. By Problem 5.7, this implies $HK = G$ so that Problem 5.2 is applicable. We conclude $\chi_K = (\lambda_{H\cap K})^K$. χ_K is therefore monomial. □

Problem 5.9. Let F be an arbitrary subfield of $\mathbb{C}$ and let $H \subseteq G$. Show that if ϑ is a character of H afforded by an F-representation, then so is ϑ^G.

Proof. If W is the corresponding $F[H]$-module to ϑ, then the $W \otimes_{F[H]} F[G]$ is the $F[G]$-module affording ϑ^G by Theorem 5.9. Since $F \subseteq \mathbb{C}$, Theorem 5.8 is applicable, and the character afforded by $W \otimes_{F[H]} F[G]$ is precisely ϑ^G. Now, the corresponding representation to this module is an F-representation affording ϑ^G. □

Problem 5.10. Let $H \subseteq G$ and view $\mathbb{C}[H] \subseteq \mathbb{C}[G]$. Let I be a right ideal of $\mathbb{C}[H]$, and set $J := I\mathbb{C}[G]$. Suppose that I affords ϑ (by viewing I as a $\mathbb{C}[H]$-module). Show that J affords ϑ^G.

Proof. Let T be a transversal of right cosets of H in G. Clearly, $J = \sum_{t\in T} \cdot It$. It is also clear that this is an imprimitivity decomposition of J where the G-stabilizer of I is H, so J affords ϑ^G via Theorem 5.8. □

Problem 5.11. Let $N \trianglelefteq G$ and $\chi \in \mathrm{Irr}(G)$, and suppose that $[\chi_N, 1_N] \neq 0$. Show that $N \subseteq \ker \chi$.

Hint: Use Lemma 5.11.

Remark: It seems that Isaacs intends for the reader to use Lemma 5.11 in order to show that $\ker(1_N)^G = N$. However, this fact is already evident from the formula for character induction.

Proof. By Frobenius reciprocity, the $\chi \in \mathrm{Irr}(G)$ satisfying the above conditions are exactly the irreducible constituents of $(1_N)^G$. Now, since N is normal, we have that $(1_N)^G(x) = |G : N| = (1_N)^G(1)$ for all $x \in N$ by the induction formula. So, $\ker(1_N)^G = N$, and it follows from Lemma 2.21 that each irreducible constituent of this character has kernel containing N. So $N \subseteq \ker \chi$. □

Problem 5.12. Let $\varphi^G = \chi \in \mathrm{Irr}(G)$ for $\varphi \in \mathrm{Irr}(H)$. Show that $Z(\chi) \subseteq H$.

Proof. We prove a result analogous to Lemma 5.11. Namely,

$$Z(\chi) = \bigcap_{x\in G} Z(\varphi)^x \tag{5.1}$$

The proof is entirely analogous to that given in the proof of Lemma 5.12. We have that $g \in Z(\chi)$ if and only if

$$|\chi(g)| = \chi(1),$$

but also,

$$|\chi(g)| = \frac{1}{|H|}\left|\sum_{x\in G} \varphi^\circ(xgx^{-1})\right| \leqslant \frac{1}{|H|}\sum_{x\in G} |\varphi(xgx^{-1})| \leqslant \frac{1}{|H|}\sum_{x\in G} \varphi(1) = |G : H|\varphi(1),$$

so if $g \in Z(\chi)$, then we have equalities above, and $g \in Z(\chi)$ implies that $|\varphi(xgx^{-1})| = \varphi(1)$ for each $x \in G$ which then implies that $g \in Z(\varphi)^x$ for each x. (5.1) now follows, and if $g \in Z(\chi)$, then $g \in H$ since $Z(\chi) = \bigcap_{x\in G} Z(\varphi)^x \subseteq Z(\varphi) \subseteq H$. □

Problem 5.13. (L. Solomon) Show that each row sum in a character table is a nonnegative rational integer.

Hint: Let G act on G by conjugation. Consider the permutation character.

Note: The column sums are also integers. They can be negative.

Remark: The column sums in a character table can indeed be negative although examples seem somewhat difficult to find. For instance, one considers either of the two classes corresponding to an element of order 11 in the Mathieu group of degree 11 (see Character Table 8 in the Appendix). The column sum for such a conjugacy class equals -2.

Proof. Let G act on itself by conjugation and denote its permutation character as χ. Observe that $\chi(g) = |C_G(g)|$ for all $g \in G$. We evaluate $[\chi, \psi]$ where $\psi \in \mathrm{Irr}(G)$. We have that

$$[\chi, \psi] = \frac{1}{|G|} \sum_{g \in G} \chi(g)\psi(g) = \frac{1}{|G|} \sum_{g \in G} |C_G(g)|\psi(g) = \sum_{g \in G} \frac{\psi(g)}{|\mathrm{Cl}(g)|} = \sum_i \psi(g_i),$$

where $\{g_i\}$ denotes a set of conjugacy class representatives. We have expressed a row sum of an arbitrary irreducible character as the inner product of the character with the permutation character χ which implies that the row sum is a nonnegative rational integer.

We now address the claim in the note. Let $E/\mathbb{Q}(\epsilon)$ where ϵ is a primitive $|G|$-th root of unity. Recall that $\mathrm{Gal}(E/\mathbb{Q})$ permutes the conjugacy classes of G. Specifically, if $\epsilon^\sigma = \epsilon^m$ for some m where $\gcd(|G|, m) = 1$, then $\chi^\sigma(g) = \chi(g^m)$. Thus, the Galois group permutes the entries in the column of a character table, and so, the Galois group fixes the column sum, which implies the column sum is a rational integer. □

Problem 5.14. Let G be nonabelian and let $f = \min\{\chi(1) \mid \chi \in \mathrm{Irr}(G),\ \chi(1) > 1\}$. Show

(a). If $|G'| \leqslant f$, then $G' \subseteq Z(G)$.
(b). If $|[G, G']| \leqslant f$, then G' is abelian.
(c). If $H \subseteq G$ and $|G : H| \leqslant f$, then $G' \subseteq H$.

Hint: For (a). If $x \in G$, then $|\mathrm{Cl}(x)| \leqslant |G'|$.

Proof. One sees that since G/G' is abelian, each coset gG' is a union of conjugacy classes in G. Thus, if $x \in G$, we have that $|\mathrm{Cl}(x)| \leqslant |xG'| = |G'|$, and the claim in the hint follows.

For part (a), we have that $\chi_x := (1_{C_G(x)})^G$ is a character of degree $|\mathrm{Cl}(x)|$, which, by the claim in the hint and our assumptions, is less than or equal to f. Notice, that since χ_x is induced a trivial character, we have $[\chi_x, 1_G] \neq 0$ by Frobenius Reciprocity. So, $\chi_x - 1_G$ is a character of degree *strictly* less than f. This implies $\chi_x - 1_G$ (and therefore χ_x itself) is a sum of linear characters of G. The kernel of χ_x is therefore the intersection of some linear characters by Lemma 2.21, so $G' \subseteq \ker \chi_x$. On the other

hand, Lemma 5.11 implies that $\ker \chi_x \subseteq C_G(x)$. However, x was chosen arbitrarily, so G' centralizes every element of G, proving that $G' \leqslant \mathbf{Z}(G)$, as desired.

For part (b), it suffices to show that $G' \subseteq C_G(x)$ for each $x \in G'$. Now, let $x \in G'$. Now, notice that $G'/[G,G'] \subseteq \mathbb{Z}(G/[G,G'])$. Thus, for any $g \in G$, we have $x^g[G,G'] = x[G,G']$. In particular, $\mathrm{Cl}(x) \subseteq x[G,G']$, and we conclude that $|\mathrm{Cl}(x)| \leqslant |[G,G']| \leqslant f$ by our assumptions. Now, set $\psi_x = (1_{C_G(x)})^G$. We have $\psi_x(1) = |\mathrm{Cl}(x)| \leqslant f$, and it follows as in part (a) that ψ_x is a sum of linear characters. The kernels of each of the linear characters contain G', and it follows by Lemma 5.11 that $G' \subseteq \ker \psi_x \subseteq C_G(x)$. Since x was chosen arbitrarily from G', it follows that $G' \subseteq C_G(G')$, so G' is abelian.

For part (c), we consider the trivial character of H induced to G: $(1_H)^G$. As in part (a), by our assumptions, this character is a sum of linear characters, and its kernel is the intersection of the kernels of these linear characters, so $G' \subseteq \ker(1_H)^G$. On the other hand, Lemma 5.11 implies that $\ker(1_H)^G \subseteq H$. So, $G' \subseteq H$. □

Problem 5.15. Let $H \subseteq G$ and suppose that φ is a character of H with $\det \varphi = 1_H$. Let $\chi = \varphi^G$ and show that $(\det \chi)^2 = 1_G$.

Remark: In general, a much stronger statement holds. Assume ψ is a character of $H \leqslant G$. Recall that the set of linear characters of a group is itself a group under pointwise multiplication, so it makes sense to speak of the "order" of a linear character. One can show via a slight tweak to the argument below that $o(\det \psi^G) \mid 2o(\det \psi)$ (see Lemma 2.30 in *Characters of Solvable Groups* by I. Martin Isaacs to see the proof carried out in full.). P. X. Gallagher in (*Abh. Math. Sem. Univ. Hamburg* **28** (1965), 162–167.) gives an explicit formula for the determinant of any induced character in terms of the transfer homomorphism (cf. §10.1 in *A Course in the Theory of Groups* by Derek J. S. Robinson).

Proof. Let W be a $\mathbb{C}[H]$-module affording φ with $\mathbb{C}$-basis $\{w_i\}_{i=1}^n$. By Theorem 5.9, $W^G = \bigoplus_{t\in T}(W \otimes t)$, where T is a right transversal for H in G. The set $\{w_i \otimes t \mid 1 \leqslant i \leqslant n,\, t \in T\}$ forms a basis for W^G. Fix $g \in G$. For each $t \in T$, we have $tg = h_t t_g$ for unique elements $h_t \in H$ and $t_g \in T$. The action of g on the basis is given by

$$(w_i \otimes t) \cdot g = w_i h_t \otimes t_g.$$

Let $\mathfrak{N}(h)$ denote the matrix of $h \in H$ acting on W with respect to $\{w_i\}$. Ordering the basis of W^G by group together the elements of the form $w_i \otimes t$ for some $t \in T$, the matrix $\mathfrak{X}(g)$ given by the action of G on W^G is seen to be a "block-monomial" matrix, which decomposes as a product like so:

$$\mathfrak{X}(g) = D(g) \cdot P(g),$$

where $D(g)$ is the block-diagonal matrix whose diagonal block indexed by $t \in T$ is $\mathfrak{N}(h_t)$ and $P(g)$ is a permutation matrix partitioned into blocks, where the block

corresponding to the entry (t, t_g) is an identity matrix. Since $\det \varphi = 1_H$, we have $\det \mathfrak{N}(h_t) = 1$ for every $t \in T$. Thus,

$$\det D(g) = \prod_{t \in T} \det \mathfrak{N}(h_t) = 1.$$

Furthermore, $\det P(g) = \pm 1$ since the determinant of any permutation matrix is ± 1 depending on whether the permutation is odd or even. It follows that

$$(\det \chi)(g) = \det \mathfrak{X}(g) = \det D(g) \cdot \det P(g) = \pm 1,$$

which gives $((\det \chi)(g))^2 = 1$ for all $g \in G$, and hence $(\det \chi)^2 = 1_G$. □

Problem 5.16. Let H be a maximal subgroup of G, and let $\chi = (1_H)^G$. Let ψ be a nonprincipal irreducible constituent of χ. Show that $\ker \chi = \ker \psi$.

Proof. Since ψ is a constituent of χ, it is clear that $\ker \chi \subseteq \ker \psi$. Observe that $\ker \chi$ is the normal core of H since χ may be viewed as the permutation character of G acting on the right cosets of H in G. Therefore, if we have proper containment of the kernels, we conclude that $\ker \psi \nsubseteq H$. The maximality of H implies then that $H \ker \psi = G$. One sees from a routine computation with the First Orthogonality Relations that ψ restricts to an irreducible character of H. Now, Frobenius Reciprocity yields that $0 \neq [\chi, \psi] = [1_H, \psi_H]$. This implies that $\psi_H = 1_H$. Now, $\ker \psi = H$ or $= G$. If $\ker \psi = G$, then clearly $\psi = 1_G$. Otherwise, if $\ker \psi = H$, then $H \lhd G$, and $\ker \chi = \ker \psi$ consequently. □

Problem 5.17. Let $H \subseteq G$ and let $\chi = (1_H)^G$. Fix a positive integer n. For $g \in G$, let $m(g) = [\chi_{\langle g \rangle}, 1_{\langle g \rangle}]$ and define $\vartheta(g) = n^{m(g)}$. Show that ϑ is a character of G.

Hint: Let Ω be the set of right cosets of H in G. Fix a set with $|S| = n$ and let Λ be the set of all functions from $\Omega \to S$. Then G acts on Λ.

Note: This result provides another necessary condition to add to the list in Theorem 5.18.

Proof. We first make some basic observations. Let T be a transversal of right cosets of H in G, and notice that G acts on T via the map $t \cdot g \longmapsto t_g$ where $Htg = Ht_g$ for $t, t_g \in T$. The character $\chi = (1_H)^G$ is the permutation character of this action. Further, notice that by Corollary 5.15, the integer $m(g)$ is the number of orbits of $\langle g \rangle$ in its action on T.

Now, in view of the hint, let S be a set of size n, and define F to be the set of all functions from T to S. This set has cardinality $|S|^{|T|} = n^{|G:H|}$ by an elementary counting argument. We now define the action of G on F by setting $f^g(t) = f(t_g)$ for each $t \in T$. This is easily verified to be a group action, and we claim that the permutation character of the action of G on F is precisely ϑ.

Now, we let ψ denote the permutation character of G on F so that $\psi(g)$ is the amount of functions in F left fixed by g, i.e.,

$$\psi(g) = |\{f \in F \mid f^g = f\}|.$$

First, assume $f \in F$ is fixed by g, i.e., $f^g = f$. We claim that f is constant on elements of T lying in the same orbit of $\langle g \rangle$. Indeed, the $\langle g \rangle$-orbit to which t belongs is given by the set $\{t_{g^i}\}$ for integers i, and for each $t \in T$, we have that

$$f^g(t_{g^i}) = f(t_{g^{i+1}})$$

for each $t \in T$ and any integer i. Conversely, one sees that the set of all $f \in F$ which are constant on the $\langle g \rangle$-orbits of T are fixed by g. We have deduced that $\psi(g)$ is the amount of functions in F which are constant on the $\langle g \rangle$-orbits of T, of which there are $m(g)$ orbits by our logic in the first paragraph. We now count the number of such functions. The number of values in S that the elements of any given $\langle g \rangle$-orbit may be assigned to is $|S| = n$. Since there are $m(g)$ of these orbits, there are $n^{m(g)}$ such functions. Thus, $\psi(g) = n^{m(g)}$. In particular, $\vartheta = \psi$ is a character since ψ was defined to be a permutation character, which completes the proof. □

Problem 5.18. Let G be a group. Write $a(n) = |\{g \in G \mid o(G) = n\}|$. Then the polynomial

$$f(x) = \frac{1}{|G|} \sum_m a(m) x^{|G|/m}$$

takes on integer values whenever $x \in \mathbb{Z}$.

Hint: Use Problem 5.17.

Proof. The group G acts on itself via right multiplication, i.e., for fixed $g \in G$, we have the map $h \longmapsto hg$. Let χ denote the permutation character of this action. Notice that the orbit of $\langle g \rangle$ under this action has length $o(g)$, and so $\langle g \rangle$ has $|G|/o(g)$ orbits in its action on G. We have established above in Problem 5.17 that the number of orbits of $\langle g \rangle$ equals $m(g)$. In particular, for any positive integer x, the function $x^{|G|/o(g)}$ is a character of G. When x is negative though, $x^{|G|/o(g)}$ may not be a character. However, we are able to establish that either $x^{|G|/o(g)}$ or $-x^{|G|/o(g)}$ is a character of G. Since $(-x)^{|G|/o(g)} = (-1)^{|G|/o(g)} x^{|G|/o(g)}$ and products of characters are characters, it merely suffices to show that $\pm(-1)^{|G|/o(g)}$ is a character to show that $\pm x^{|G|/o(g)}$ is a character for all integers x.

Now, consider $(-1)^{|G|/o(g)}$. If $|G|$ is odd, then $(-1)^{|G|/o(g)} = -1_G$. If G has a noncyclic Sylow 2-subgroup, then computes that $(-1)^{|G|/o(g)} = 1_G$. Now, assume that G has a cyclic Sylow 2-subgroup. It follows from Burnside's normal p-complement theorem that $G = \langle s \rangle N$ where $\langle s \rangle$ is a Sylow 2-subgroup and N is a normal subgroup of odd order. In particular, any element of g may be factorized uniquely as $s^k n$ for an integer k and some $n \in N$. Now, G has a normal subgroup of index 2, say L, and if $g \in L$ then, we see that s^k is not a generator of $\langle s \rangle$. In particular, $|G|/o(g)$ is even

and $(-1)^{|G|/o(g)} = 1$. Otherwise, if $g \in G - L$, then $\langle s^k \rangle = \langle s \rangle$, and so $|G|/o(g)$ is odd and $(-1)^{|G|/o(g)} = -1$. We see now that $(-1)^{|G|/o(g)})$ is identified with the nontrivial character of G/L, proving that $(-1)^{|G|/o(g)}$ is a character. Thus, we have established that either $\pm x^{|G|/o(g)}$ is a character. In particular, $[x^{|G|/o(g)}, \chi] \in \mathbb{Z}$ for any $\chi \in \mathrm{Irr}(G)$.

Now, $[x^{|G|/o(g)}, 1_G] \in \mathbb{Z}$ by the above arguments. The proof will be completed once we establish that $f(x) = [x^{|G|/o(g)}, 1_G]$. We compute

$$f(x) = \frac{1}{|G|} \sum_m a(m) x^{|G|/m} = \frac{1}{|G|} \sum_m \sum_{\substack{g \in G \\ o(g)=m}} x^{|G|/o(g)} = \frac{1}{|G|} \sum_{g \in G} x^{|G|/o(g)} = [x^{|G|/o(g)}, 1_G],$$

as desired. □

Problem 5.19. Let G act doubly transitively on Ω, and let $H \subset G$ with $|G : H| < |\Omega|$. Show that H is transitive on Ω.

Proof. We let $\chi = (1_{G_\alpha})^G = 1 + \psi$ be the permutation character of G on Ω with $\psi \in \mathrm{Irr}(G)$ and $\alpha \in \Omega$. Observe that χ_H is the permutation character of H acting on Ω. By Corollary 5.15, it suffices to show that $[\chi_H, 1_H] = 1$ or equivalently that $[\psi_H, 1_H] = 0$. Were this not the case, $[\psi, (1_H)^G] \neq 0$ and we would have that ψ and 1_G are both irreducible constituents of $(1_H)^G$, and we conclude

$$|G : H| = (1_H)^G \geqslant 1 + \psi(1) = \chi(1) = |G : G_\alpha| = |\Omega|,$$

contrary to our assumption. Thus, $[\psi_H, 1_H] = 0$, as desired. □

Problem 5.20. Assume the notation and hypotheses of Problem 4.11 and show that $|G| = q(q-1)(q+1)$.

Hints: Let $N = N_G(S)$ and show that $|N| = q(q-1)$. Let $\lambda \in \mathrm{Irr}(N)$ with $\lambda \neq 1_N$ and $S \subseteq \ker \lambda$. Let $x \in S$, $x \neq 1$. Show $\lambda^G(x) = \lambda(1)$. Write $\lambda^G = \sum_{\chi \in \mathrm{Irr}(G)} a_\chi \chi$ and conclude

$$\lambda(1) = \sum a_\chi(\chi(1) - q) = |G : N|\lambda(1) - q \sum a_\chi. \tag{5.2}$$

Finish the problem by showing that if $a_\chi \neq 0$, then $\chi(1) \geqslant q+1$ so that $\sum a_\chi \leqslant \lambda(1)$. To show $\chi(1) \geqslant q+1$, compute $[\chi_S, 1_S]$ and use the fact that since $[\chi_N, \lambda] > 0$ and χ is real, then $[\chi_S, 1_S] \geqslant 2$.

Note: Using the result of this problem, it is not very hard to prove that $G = \mathrm{SL}(2, q)$, a theorem originally due to Brauer, Suzuki, and Wall.

Proof. We first demonstrate that $|N| = q(q-1)$. We have established in Problem 4.11 that any two involutions in G are conjugate to one another. Since S is abelian, one has that if $x^g = y$ for $x, y \in S$ and $g \in G$, then $g \in N$. This is seen by recalling that the Sylow 2-subgroups of G intersect trivially with one another. In particular, we see that N (and therefore N/S) acts transitively by conjugation on $S - \{1\}$. We claim that N/S acts regularly. Otherwise, if there existed $g \in N$ for which $x^g = x$

for some $x \in S$, then since $C_G(x) = S$, this implies $g \in S$. Since N/S acts regularly on $S - \{1\}$, we conclude by the Orbit-Stabilizer Theorem that $|N : S| = q - 1$, and therefore $|N| = q(q-1)$.

Now, fix $\lambda \in \mathrm{Irr}(N)$ with $S \subseteq \ker \lambda$. we shall show that $\lambda^G(x) = \lambda(1)$ for all $x \in S$. This follows from the induction formula and the fact that the Sylow 2-subgroups of G intersect trivially. Let T be a transversal for the right cosets of N in G, and observe that $\lambda^\circ(txt^{-1}) = 0$ unless $t \in N$. Thus,

$$\lambda^G(x) = \sum_{t \in T} \lambda^\circ(txt^{-1}) = \lambda(x),$$

as desired.

We now write $\lambda^G = \sum_{\chi \in \mathrm{Irr}(G)} a_\chi \chi$ and evaluate λ^G at a nonidentity element $x \in S$. By the above paragraph, we have that $\lambda^G(x) = \lambda(1)$. On the other hand, by the result in Problem 4, we have that

$$\lambda^G(x) = \sum_{\chi \in \mathrm{Irr}(G)} a_\chi \chi(x) = \sum_{\chi \in \mathrm{Irr}(G)} a_\chi(\chi(1) - q) = \sum_{\chi \in \mathrm{Irr}(G)} a_\chi \chi(1) - q \sum_{\chi \in \mathrm{Irr}(G)} a_\chi$$

$$= \lambda^G(1) - q \sum_{\chi \in \mathrm{Irr}(G)} a_\chi = |G : N|\lambda(1) - q \sum_{\chi \in \mathrm{Irr}(G)} a_\chi,$$

thus verifying (5.2).

Now, observe that by Problem 4.11 $\nu_2(\chi) = 1$ for each $\chi \in \mathrm{Irr}(G)$ and that by Theorem 4.5, all irreducible characters of G take real values. Assuming $a_\chi \neq 0$, by Frobenius Reciprocity, we have that $0 \neq [\chi, \lambda^G] = [\chi_N, \lambda] = [\bar{\lambda}, \chi_N]$. Now, since λ is an irreducible character of the odd-order group N/S, we observe from Problem 3.16 that $\lambda \neq \bar{\lambda}$. Thus, λ and $\bar{\lambda}$ are distinct irreducible constituents of χ_N. We conclude

$$[\chi_S, 1_S] \geqslant [\lambda_S + \bar{\lambda}_S, 1_S] = \lambda(1)[1_S + 1_S, 1_S] = 2\lambda(1) \geqslant 2.$$

We are now able to show that $\chi(1) \geqslant q + 1$. By Problem 4.11, $\chi(x) = \chi(1) - q$ for all $x \in S - \{1\}$. We compute

$$2 \leqslant [\chi_S, 1_S] = \frac{\chi(1)}{q} + \frac{1}{q} \sum_{x \in S - \{1\}} (\chi(1) - q) = \chi(1) - q + 1,$$

from which we obtain $\chi(1) \geqslant q + 1$. By (5.2), we have now that

$$\lambda(1) = \sum a_\chi(\chi(1) - q) \geqslant \sum a_\chi.$$

Now that we have completed the proofs for each statement in the hints,[6] we prove that $|G:N| \leqslant q+1$. Solving for $|G:N|$ in (5.2) and using the fact that $\sum a_\chi \leqslant \lambda(1)$, we obtain that

$$|G:N| = 1 + q\frac{\sum a_\chi}{\lambda(1)} \leqslant 1 + q. \tag{5.3}$$

On the other hand, we also claim that the number n of Sylow 2-subgroups of G satisfies $n \equiv 1 \bmod q$. This is done by considering the action of a fixed Sylow 2-subgroup S on all Sylow 2-subgroups. Let S_1 be a distinct Sylow 2-subgroup from S. For any $x \in S$, one sees that $S_1 \cap S_1^x = 1$ using the fact that distinct Sylow 2-subgroups of G intersect trivially. Thus, we see that the action of S on the Sylow 2-subgroups fixes itself but acts regularly on every remaining orbit, i.e., every S-orbit other than $\{S\}$ has length $|S|$. This implies that the number of Sylow 2-subgroups in G equals $1 + qa$ for some nonnegative integer a. However, since G is simple, $a \neq 0$, and by (5.3), $a < 2$. We conclude $a = 1$, and so,

$$|G| = |G:N||N| = (q+1)q(q-1),$$

which completes the proof. □

Problem 5.21. Let $P(G)$ denote the set of all $\mathbb{Z}$-linear combinations of characters of G of the form $(1_H)^G$ for $H \subseteq G$. Show that $P(G)$ is a ring.

Proof. The fact that $P(G)$ is an abelian group under addition is trivial, as is the validity of the distributive property and the associativity of $P(G)$ under multiplication. Also, $P(G)$ has 1_G as the identity since we may of course write $1_G = (1_G)^G$. It remains therefore to show that $P(G)$ is closed under multiplication.

It suffices to show that $(1_H)^G(1_K)^G \in P(G)$ for $H, K \subseteq G$. By Problem 5.3, we have

$$(1_H)^G(1_K)^G = (1_H((1_K)^G)_H)^G = (((1_K)^G)_H)^G. \tag{5.4}$$

Applying the character-theoretic version of Mackey's formula in Problem 5.6, we choose a set T of double coset representatives for H and K, and we obtain that

$$((1_K)^G)_H = \sum_{t \in T} (1_{K^t \cap H})^H.$$

[6] We remark that at this point in the proof, there is a less tedious way to proceed which however uses some rather nontrivial results. One observes that N is a Frobenius group with an odd-order complement (See Definition 7.1 and Problem 7.1(a)). The Sylow subgroups of a Frobenius complement having odd order are each cyclic by a result of Burnside (See for instance, Theorem 10.3.1(iv) in *Finite Groups* by Daniel Gorenstein.), and as a consequence of Burnside's normal p-complement theorem, we conclude that N/S is solvable. We may then choose $\lambda \in \mathrm{Irr}(N/S)$ to be linear, and the result follows directly from (5.2).

Now, applying this to (5.4), we obtain that

$$(1_H)^G(1_K)^G = \left(\sum_{t\in T}(1_{K^t\cap H})^H\right)^G = \sum_{t\in T}((1_{K^t\cap H})^H)^G = \sum_{t\in T}(1_{K^t\cap H})^G,$$

which proves closure under multiplication and completes the proof. □

Problem 5.22. Let χ be a rational valued character of G and let $n_G(\chi)$ be the least positive integer such that $n_G(\chi)\chi \in P(G)$. Let $N = \ker\chi$ so that χ may be viewed as a character of G/N. Show that

$$n_G(\chi) = n_{G/N}(\chi).$$

Hint: If $n\chi = \sum a_H(1_H)^G$, then $n\chi = \sum a_H(1_{NH})^G$. Show this by writing $(1_H)^G = (1_{NH})^G + \zeta^{(H)}$, where $\zeta^{(H)}$ is a character of G with the property that N is not contained in the kernel of any of its irreducible constituents.

Proof. Notice first that the inequality $n_{G/N}(\chi) \geqslant n_G(\chi)$ is trivial since for any H with $N \leqslant H \leqslant G$, we may view $(1_{H/N})^{G/N}$ as $(1_H)^G$. Thus, it suffices to show $n_G(\chi) \geqslant n_{G/N}(\chi)$, and to show this, it suffices to prove the claim in the hint since we may view $(1_{NH})^G$ as $(1_{HN/N})^{G/N}$.

We first verify that $\zeta^{(H)}$ is an integer for every subgroup H. Notice that by the transitivity and linearity of induction, $\zeta^{(H)} = (1_H)^G - (1_{NH})^G$ is a character if $(1_H)^{NH} - 1_{NH}$ is a character, and $(1_H)^{NH} - 1_{NH}$ is indeed a character since 1_{NH} is a constituent of $(1_H)^{NH}$ by Frobenius Reciprocity.

Now, we show that the kernel of each irreducible constituent of $\zeta^{(H)}$ does not contain N. Let $\psi \in \mathrm{Irr}(G)$ be a constituent of $\zeta^{(H)}$. We have by Frobenius Reciprocity that

$$0 < [\psi, \zeta^{(H)}] = [\psi, (1_H)^G - (1_{NH})^G]$$
$$= [\psi_{NH}, (1_H)^{NH} - 1_{NH}] \leqslant [\psi_N, ((1_H)^{NH})_N - 1_N].$$

Now, Problem 5.2 is applicable, and we conclude $((1_H)^{NH})_N = (1_{H\cap N})^N$, which further gives us

$$0 < [\psi_N, ((1_H)^{NH})_N - 1_N] = [\psi_N, (1_{H\cap N})^N - 1_N].$$

Notice that since $[(1_{H\cap N})^N, 1_N] = 1$, we conclude that $(1_{H\cap N})^N - 1_N$ does not have the principal character as a constituent, and yet ψ_N has a constituent in common with $(1_{H\cap N})^N - 1_N$, which must necessarily be nonprincipal. Hence, ψ_N has a nonprincipal constituent, and so N is not contained in the kernel of ψ. For $\eta \in \mathrm{Irr}(G)$, we conclude $[\eta, \zeta^{(H)}] \neq 0$ only if $N \nsubseteq \ker\eta$.

Setting $n = n_G(\chi)$, for each $\psi \in \mathrm{Irr}(G/N)$, we compute

$$[\psi, n\chi - \sum a_{NH}(1_{NH})^G] = [\psi, \sum a_{NH}\zeta^{(H)}] = 0.$$

On the other hand, each constituent of $n\chi - \sum a_{NH}(1_{NH})^G$ has a kernel containing N, from which it follows that $n\chi - \sum a_{NH}(1_{NH})^G$ is identically zero, and so, we obtain

$$n\chi = \sum a_{NH}(1_{NH})^G,$$

proving that $n_G(\chi) \geqslant n_{G/N}(\chi)$, and by the first paragraph, we conclude $n_G(\chi) = n_{G/N}(\chi)$, as desired. □

Problem 5.23. Let χ be any character of G. Show that

$$|G|\chi = \sum a_\lambda \lambda^G,$$

where λ runs over the set of linear characters of cyclic subgroups and $a_\lambda \in Z$.

Hint: Use Problem 5.3.

Proof. By Corollary 5.23, we have that

$$|G|1_G = \sum a_H (1_H)^G$$

for some cyclic subgroups H. Multiplying both sides by χ and applying Problem 5.3, we obtain

$$|G|\chi = \sum_H a_H (1_H)^G \chi = \sum_H a_H (\chi_H)^G.$$

Since each H is cyclic, χ_H is a sum of linear characters of H, say $\chi_H = \sum \lambda_{H,i}$. By the linearity of induction and the above equation, we have that

$$|G|\chi = \sum_H a_H \sum_i \lambda_{H,i}^G = \sum_{H,i} a_H \lambda_{H,i},$$

as desired.

□

Problem 5.24. Let G be a doubly transitive permutation group on Ω and let $\alpha, \beta \in \Omega$ with $\alpha \neq \beta$. Let $\varphi \in \mathrm{Irr}(G_\alpha)$, and assume that $\varphi_{G_{\alpha\beta}} \in \mathrm{Irr}(G_{\alpha\beta})$. Show that $[\varphi^G, \varphi^G] \leqslant 2$.

Hint: Use Problem 5.6.

Proof. We shall use a not-often stated characterization of doubly transitive groups: G acts doubly transitively on Ω if and only if $G_\alpha \sqcup G_\alpha t G_\alpha$ for $\alpha \in \Omega$ and some $t \in G$.[7] Notice however, by replacing t with an appropriate conjugate, we can assume that $(\alpha)t = \beta$.

Now, by Frobenius Reciprocity, we have that

$$[\varphi^G, \varphi^G] = [\varphi, (\varphi^G)_{G_\alpha}].$$

[7]This fact is proven in an elementary manner, but a proof is given in Theorem 2.7.3 of *Finite Groups* by Daniel Gorenstein. In fact one can show that we can choose t to be an involution, but this will not be necessary for our argument below.

We investigate the character $(\varphi^G)_{G_\alpha}$. By the remarks in the first paragraph, $\{1, t\}$ is a complete set of double coset representatives for (G_α, G_α). By the Mackey formula in Problem 5.6, we have that

$$(\varphi^G)_{G_\alpha} = \varphi + (\varphi^t_{G^t_\alpha \cap G_\alpha})^{G_\alpha}.$$

Now, $G^t_\alpha = G_\beta$, and so $G^t_\alpha \cap G_\alpha = G_{\alpha\beta}$. We now have

$$[\varphi^G, \varphi^G] = [\varphi, (\varphi^G)_{G_\alpha}] = [\varphi, \varphi + (\varphi^t_{G_{\alpha\beta}})^{G_\alpha}] = 1 + [\varphi, (\varphi^t_{G_{\alpha\beta}})^{G_\alpha}].$$

By Frobenius Reciprocity, the rightmost term on the rightmost side equals $[\varphi_{G_{\alpha\beta}}, \varphi^t_{G_{\alpha\beta}}]$. Now, $\varphi_{G_{\alpha\beta}}$ is irreducible by assumption and therefore too is its conjugate by t. Thus, $[\varphi_{G_{\alpha\beta}}, \varphi^t_{G_{\alpha\beta}}]$ is either one or zero. We see now that $[\varphi^G, \varphi^G] \leqslant 2$ by the above equation. □

Problem 5.25. Assume that every Sylow subgroup of G has a faithful irreducible character. Show that G has one also.

Hint: Let S be the product of all of the minimal normal subgroups of G. (This is called the *socle* of G.) Then S is a direct product of simple groups. Find $\vartheta \in \mathrm{Irr}(S)$ such that $\ker \vartheta$ contains no nontrivial normal subgroup of G.

Remark: This problem seems to require a somewhat uncommonly stated result from finite group theory. We shall need this fact not only for this problem but for the next problem as well, so we state it below as a sublemma. The proof is rather elementary, but the reader is nonetheless deferred to Theorem 3.3.12 in *A Course in the Theory of Groups* by Derek Robinson, where the result is proved without assuming the finiteness of order of G and its number of direct factors.

Sublemma. Given any finite group G which is expressible as a direct product of simple groups, say $G = \prod_i S_i$, any normal subgroup N of G has a normal complement, i.e., there exists $M \lhd G$ such that $N \times M = G$. Further, if each S_i is nonabelian, then N is expressible as a direct product of some of the S_i.

Proof of Problem 5.25. By Theorem 2.32, a p-group has a faithful irreducible character iff it has a cyclic center. Recall that a minimal normal subgroup is isomorphic to a direct product of mutually isomorphic simple groups. We claim that for any prime p dividing $|G|$, G has at most one minimal normal subgroup of p-power order. Assume otherwise that there exist minimal normal subgroups N and M having order a power of p, and let P be a Sylow p-subgroup of G. Now, since P a cyclic center, P has a unique central subgroup of order p, say Z. By a result from elementary group theory, we have that $\mathbf{Z}(P) \cap N$ and $\mathbf{Z}(P) \cap M$ are both nontrivial. Therefore, Z is contained in both N and M. However, since N and M are distinct and minimally normal in G, there intersection is trivial, a contradiction.

Now, we may write $S = \prod_p N_p \times \prod_i S_i$ where the N_p are the unique abelian minimal normal subgroups and the S_i are nonabelian simple groups. We define $\vartheta \in \mathrm{Irr}(S)$

by choosing nonprincipal $\lambda_p \in \mathrm{Irr}(N_p)$ for each N_p and nonprincipal $\psi_i \in \mathrm{Irr}(S_i)$ for each S_i, and we set $\vartheta = \prod_p \lambda_p \times \prod_i \psi_i$. We claim $\ker \vartheta$ contains no minimal normal subgroup. On one hand, if $\ker \vartheta$ contained an abelian normal subgroup of G, then this abelian normal subgroup would necessarily contain an N_p for some prime p. However, $\vartheta_{N_p} = \vartheta(1)\lambda_p$, which was chosen to be nontrivial. On the other hand, if $\ker \vartheta$ contained a nonabelian normal subgroup of G, then by the result quoted in the above remark, $\ker \vartheta$ necessarily contains one of the S_i. However, again, ϑ_{S_i} is an integer multiple of ψ_i, which is nonprincipal. Thus, $\ker \vartheta$ does not contain any normal subgroup of G.

Now, we claim that ϑ^G possesses a faithful constituent. Let χ be an irreducible constituent of ϑ^G. We claim that χ is faithful. Assume otherwise that $\ker \chi \neq 1$. In particular, $K := \ker \chi \cap S \neq 1$ since S contains all minimal normal subgroups of G. By Frobenius Reciprocity, we have that $[\chi_S, \vartheta] = [\chi, \vartheta^G] \neq 0$. So, ϑ is an irreducible constituent of χ_S. Thus, $K \subseteq \ker \chi_S \subseteq \ker \vartheta$. However, by our construction, $\ker \vartheta$ does not contain any normal subgroup of G, which forces $K = 1$, contrary to our assumption. □

Problem 5.26. (Passman) Let $X \subseteq G - \{1\}$ be a subset and let $n = |X|$. Assume for all primes $p < n$, that a Sylow p-subgroup of G is cyclic. Show that there exists $\chi \in \mathrm{Irr}(G)$ such the $X \cap \ker \chi = \varnothing$.

Hint: Reduce to the case G is abelian by observing that it suffices to assume $G = \langle X \rangle$ and that if $[x, y] \notin \ker \chi$, then both x and $y \notin \ker \chi$. In the abelian case, let N be maximal such that $N \cap X = \varnothing$. Show that G/N is cyclic.

Note: If $n = 2$, the hypothesis on Sylow subgroups is vacuously satisfied.

Remark: It is not immediately clear how Isaacs' hint is helpful. Specifically, I am unsure of how to reduce to the abelian case. However, a proof is nonetheless available without this reduction. We again refer to Theorem 3.3.12 in Derek Robinson's *A Course in the Theory of Groups* as we have done in Problem 5.25.

Proof. We may assume by induction that G is a group of minimal order possessing a subset X which serves as a counterexample. Further, we may assume that X is chosen to have minimal cardinality in $G - \{1\}$. We first show that $\langle X \rangle = G$. Assume otherwise that $\langle X \rangle = H \subset G$. The condition on Sylow subgroups is clearly inherited by subgroups of G, so by the induction hypothesis, there exists $\psi \in \mathrm{Irr}(H)$ such that $\ker \psi \cap X = \varnothing$. Now, consider an irreducible constituent χ of ψ^G. By Frobenius Reciprocity, ψ is an irreducible constituent of χ_H. From this we conclude that $\ker_H \subseteq \ker \psi$. Thus, $\ker \chi \cap X = \varnothing$. From now on, we will assume that $\langle X \rangle = G$.

Also, every nontrivial normal subgroup N of G contains an element of X. Otherwise, since the condition on Sylow subgroups clearly holds for quotients, there exists $\chi \in \mathrm{Irr}(G/N)$ for which the statement holds by the induction hypothesis.

We now show that G is isomorphic to a direct product of simple groups. Assume for a contradiction that $L := \mathrm{Soc}(G)$ is properly contained in G. By the induction hypothesis, there exists $\psi \in \mathrm{Irr}(L)$ such that $X \cap \ker \psi = \varnothing$. Now, let χ be an irreducible constituent of ψ^G. By Frobenius Reciprocity, ψ is an irreducible constituent of χ_L, and so $\ker \chi_L \subseteq \ker \psi$. In particular, $(\ker \chi \cap L) \cap X = \varnothing$. However, since G is a minimal counterexample, $\ker \chi$ is nontrivial, and as such, there is a minimal normal subgroup N contained in $\ker \chi \cap L$ (since the socle L contains all minimal normal subgroups). By the second paragraph above, there necessarily exists $x \in N$. On the other hand though, $N \subseteq \ker \chi \cap L$ and $\ker \chi \cap L$ contains no elements of X by our choice of χ, a contradiction. Thus, $G = L$, and we may assume from now on that G is a direct product of simple groups.

Write $G = \prod_i S_i$ for simple groups S_i. We claim that for any prime p dividing $|G|$, at most one of the S_i has order p. Otherwise, if both S_i and S_j have order p for $i \neq j$, then by a simple counting argument, $S_i \times S_j$ has $p+1$ distinct subgroups of order p, each of which are normal in G. On the other hand, by the second paragraph above, each normal subgroup of G contains at least one element of X, which implies $p + 1 \leqslant |X| = n$, but our hypothesis on the Sylow subgroups forces the Sylow p-subgroups of G to be cyclic, a contradiction. By grouping together the abelian direct factors of G, we may write $G = C \times N$ where C is cyclic of square-free order and $N = \prod_j T_j$ where each T_j is a nonabelian simple group.

For the final contradiction, we show that G possesses a faithful irreducible character. Write $N = \prod_j T_j$. For each j, choose a nonprincipal $\psi_j \in \mathrm{Irr}(T_j)$. We claim that $\eta := \prod_j \psi_j$ is a faithful irreducible character of N. If η was not faithful, then $\ker \eta$ contains a normal subgroup of N. By the sublemma to Problem 2.25, $\ker \eta$ necessarily contains one of the T_j in its kernel. However, η_{T_j} is an integer multiple of ψ_j, which was chosen to be nontrivial. That η is irreducible follows from Theorem 4.21. Now, let λ be a generator of $\mathrm{Irr}(C)$. Taking $\chi = \lambda \times \eta$, one sees that χ is faithful, a contradiction $\square$

6. Normal subgroups

Problem 6.1. Let $N \lhd G$ with $\vartheta \in \mathrm{Irr}(N)$. Show that $\vartheta^G \in \mathrm{Irr}(G)$ iff $I_G(\vartheta) = N$.

Proof. Assume first that $I_G(\vartheta) = N$. In the notation of Theorem 6.11 (the Clifford Correspondence), $\vartheta \in \mathscr{A}$, and so ϑ^G is irreducible.

Conversely, assume that $\vartheta^G \in \mathrm{Irr}(G)$, and assume $T := I_G(\vartheta) \neq N$. Since ϑ^G is irreducible, by Theorem 6.2, we see that $(\vartheta^G)_N = e\sum_t \vartheta^t$ where the ϑ^t are the $|G:T|$ distinct conjugates of ϑ. Using Frobenius Reciprocity, we have that

$$e = [\vartheta, (\vartheta^G)_N] = [\vartheta^G, \vartheta^G] = 1.$$

On one hand, by the induction formula, $\vartheta^G(1) = |G:N|\vartheta(1)$. On the other hand, by Theorem 6.2, $\vartheta^G(1) = e|G:T|\vartheta(1) = |G:T|\vartheta(1)$. Comparing these formulas, we obtain $|G:N| = |G:T|$, so $N = T$, as desired. □

Problem 6.2. Let $N \lhd G$ and suppose G/N is abelian. Let C be the group of linear character of G/N so that C acts on $\mathrm{Irr}(G)$ by multiplication. (See Problem 2.6) Let $\vartheta \in \mathrm{Irr}(N)$. Show that

$$\vartheta^G = f\sum_{i=1}^{s} \chi_i$$

where f is an integer and the $\chi_i \in \mathrm{Irr}(G)$ constitute an orbit under C.

Hint: Let $\chi \in \mathrm{Irr}(G)$. Then $(\chi_N)^G = \rho\chi$, where ρ is the regular character of $\mathrm{Irr}(G/N)$.

Proof. The claim in the hint follows from the fact that $\rho = (1_N)^G$ and Problem 5.3. Indeed, for any $\chi \in \mathrm{Irr}(G)$,

$$\rho\chi = (1_N)^G\chi = (\chi_N)^G. \tag{6.1}$$

In particular, writing $\rho = \sum_{\lambda\in C}\lambda$, one has

$$(\chi_N)^G = \sum_{\lambda\in C}\lambda\chi. \tag{6.2}$$

Now, if χ is an irreducible constituent of ϑ^G, then by Frobenius Reciprocity, ϑ is an irreducible constituent of χ_N. By the linearity of induction, we have that ϑ^G is a constituent of $(\chi_N)^G$. By (6.2), ϑ^G is a sum of characters of the form $\lambda\chi$ for some $\lambda \in X \subset C$, say

$$\vartheta^G = \sum_{\mu\in X}\mu\chi. \tag{6.3}$$

Also, notice that since ϑ is induced from the normal subgroup N, ϑ^G vanishes on every element of $G - N$ (using the induction formula), and we conclude that C permutes the irreducible constituents of ϑ^G. Thus, the distinct terms of (6.3) together form a C-orbit.

Let χ_i denote the distinct irreducible characters of the form $\mu\chi$ for $\chi \in C$. It only remains to show there is exists f such that $\vartheta^G = f\sum_i \chi_i$. To this end, it merely suffices to show that $[\vartheta^G, \chi_i]$ does not depend on i. Write $\chi_i = \mu_i\chi$ for some $\mu_i \in C$, and notice that $\mu_i\chi_N = \chi_N$. By Frobenius Reciprocity and (6.1), we obtain that

$$[\vartheta^G, \chi_i] = [\vartheta^G, \mu_i\chi] = [\vartheta, \chi],$$

and thus, each irreducible constituent of ϑ^G appears with the same multiplicity as χ does. This completes the proof. □

7. T.I. sets and exceptional characters

8. Brauer's Theorem

9. Changing the field

10. The Schur Index

11. Projective Representations

Problem 11.1. Let $\alpha, \beta \in Z^2(G, F^\times)$, where F is a field. If α and β are equivalent, show that $F^\alpha[G] = F^\beta[G]$.

Proof. Let

$$\alpha(g,h)\beta^{-1}(g,h) = \mu(g)\mu(h)\mu^{-1}(gh) \tag{11.1}$$

for some function $\mu : G \to F^\times$. We write the basis elements of $F^\alpha[G]$ as $\bar{g}$ and $F^\beta[G]$ as $\hat{g}$. We define the isomorphism $\Xi : F^\alpha[G] \to F^\beta[G]$ directly. Given $g \in G$, we define $\Xi(\bar{g}) = \hat{g}\mu(g)$, and we extend this definition linearly. It is clear that this function is bijective. What remains is to show that Ξ preserves products, but by the linearity of Ξ, it suffices to show that $\Xi(\bar{g} \cdot \bar{h}) = \Xi(\bar{g}) \cdot \Xi(\bar{h})$ for each $g, h \in G$. We observe that

$$\Xi(\bar{g} \cdot \bar{h}) = \Xi(\overline{gh})\alpha(g,h) = \widehat{gh}\mu(gh)\alpha(g,h) \tag{11.2}$$

and

$$\Xi(\bar{g}) \cdot \Xi(\bar{h}) = (\hat{g} \cdot \hat{h})\mu(g)\mu(h) = \widehat{gh}\mu(g)\mu(h)\beta(g,h). \tag{11.3}$$

By (11.1), the rightmost sides of (11.2) and (11.3) are equal, proving that Ξ is an isomorphism of F-algebras. □

We remark that the above theorem means that we may speak of a twisted group algebra where the twist is applied by an element of $H^2(G, F^\times)$.

Problem 11.2. For a group G and an abelian group A, let $\alpha \in Z^2(G, A)$ and let $g \in G$. Show that $\alpha(x, y) = \alpha(y, x)$ for $x, y \in \langle g \rangle$.

Proof. Of course, we may assume $x = g^n$ and $y = g^m$ for integers n and m. We first establish that $\alpha(g, 1) = \alpha(1, g) = \alpha(1, 1)$. This is seen by computing

$$\alpha(g \cdot 1, 1)\alpha(g, 1) = \alpha(g, 1 \cdot 1)\alpha(1, 1).$$

Canceling $\alpha(g, 1)$ from both sides yields $\alpha(g, 1) = \alpha(1, 1)$. By symmetry, we obtain

$$\alpha(g,1) = \alpha(1,1) = \alpha(1,g). \tag{11.4}$$

Next, we claim that $\alpha(g, g^i) = \alpha(g^i, g)$ for any integer i. We do induction on i with the base case already being handled by (11.4). Assume $\alpha(g, g^j) = \alpha(g^j, g)$ for all $j < i$. By our induction hypothesis, we have that

$$\alpha(g^i, g) = \alpha(g \cdot g^{i-1}, g) = \alpha(g, g^{i-1} \cdot g) \cdot \frac{\alpha(g^{i-1}, g)}{\alpha(g, g^{i-1})} = \alpha(g, g^i). \tag{11.5}$$

Now, let $n \geqslant 2$ be an integer, and assume inductively (with (11.5) being the base case) that for all integers k, we have that $\alpha(g^{n-1}, g^k) = \alpha(g^k, g^{n-1})$. We compute

$$\alpha(g^n, g^m) = \alpha(g^{n-1} \cdot g, g^m) = \frac{\alpha(g^{n-1}, g \cdot g^m)\alpha(g, g^m)}{\alpha(g^{n-1}, g)} \tag{11.6}$$

Likewise,

$$\alpha(g^m, g^n) = \alpha(g^m, g \cdot g^{n-1}) = \frac{\alpha(g^m \cdot g, g^{n-1})\alpha(g^m, g)}{\alpha(g, g^{n-1})} \tag{11.7}$$

By (11.5) and our induction hypothesis, the rightmost sides of (11.6) and (11.7) are equal. Thus, we obtain

$$\alpha(g^n, g^m) = \alpha(g^m, g^n),$$

as desired. □

Problem 11.3. Let $\alpha \in Z^2(G, A)$ and let $x, y \in G$ commute. Assume that A is divisible. Show that $\alpha(x, y) = \alpha(y, x)$ iff the restriction of α to $\langle x, y\rangle$ is equivalent to the trivial factor set.

Hint: Use Lemma 11.16.

Proof. We first prove the "if" direction. Assume that the restriction of α to $\langle x, y\rangle$ is equivalent to the trivial facts set. In particular, we may assume that $\alpha(g, h) = \mu(g)\mu(h)\mu^{-1}(gh)$ for some function $\mu : \langle x, y\rangle \to \mathbb{C}^\times$ and any $g, h \in \langle x, y\rangle$. However, since we assume x and y commute, we clearly have

$$\alpha(x, y) = \mu(x)\mu(y)\mu^{-1}(xy) = \mu(x)\mu(y)\mu^{-1}(yx) = \alpha(y, x),$$

proving the "if" direction.

For the "only if" direction, we assume from now on that $\alpha(x, y) = \alpha(y, x)$. Construct the central extension (Γ, π) as in 11.16 using the given α where $\langle x, y\rangle \cong \Gamma/\ker\pi$. Each element of Γ is of the form (g, a) for $g \in \langle x, y\rangle$ and $a \in A$. Also, given $g, h \in \langle x, y\rangle$ and $a, b \in A$, we have the relations

$$(g, a)(h, b) = (gh, \alpha(g, h)ab).$$

We claim that it suffices to show that Γ is abelian, for if this were true, then Lemma 11.14 implies that A is complemented in Γ by a group H which is necessarily isomorphic to $\langle x, y\rangle$. In particular, Γ would have a set of coset representatives of A which forms a subgroup of Γ. Using the construction in Lemma 11.9, these coset representatives give the trivial factor set in $Z^2(\langle x, y\rangle, A)$. By Lemma 11.9 and since any two factor sets obtained in this manner from a choice of coset representatives are equivalent, this implies α is equivalent to the trivial factor set.

We now show that Γ is abelian. Let $(x^n y^m, a)$ be an arbitrary element of Γ. We may write

$$(x^n y^m, a) = (x^n, \alpha^{-1}(x^n, y^m))(y^m, a).$$

Now, let $c \in A$. We claim that for any positive integer k, we may factorize $(x^k, c) = \prod_{i=q}^{k}(x, c_i)$ for $c_i \in A$. Observe first that

$$(x, c)(x^{k-1}, \alpha^{-1}(x, x^{k-1})) = (x^k, c),$$

and notice that the claim follows by induction on k. Likewise, by symmetry, we have that $(y^k, c) = \prod_{i=1}^{\ell}(y, d_i)$ for $d_i \in A$. In particular, we conclude that

$$(x^n y^m, a) = \prod_{i=1}^{n}(x, c_i)\prod_{j=1}^{m}(y, d_i) \tag{11.8}$$

for $c_i, d_i \in A$.

Now, given $c, d \in A$, we have that

$$(x, c)(y, d) = (xy, \alpha(x, y)cd) = (xy, \alpha(y, x)cd) = (y, d)(x, c),$$

where $\alpha(x, y) = \alpha(y, x)$ by assumption. Likewise, we have that

$$(x, c)(x, d) = (x^2, \alpha(x, x)cd) = (x, d)(x, c)$$

and

$$(y, c)(y, d) = (y^2, \alpha(y, y)cd) = (y, d)(y, c).$$

It now follows that any element of the form (x, c) or (y, c) commutes with any element of the form (x, d) or (y, d). By (11.8), it now follows that Γ is abelian, proving the claim. □

Problem 11.4. Let $\alpha \in Z^2(G, A)$. Say that $g \in G$ is α-special if $\alpha(g, c) = \alpha(c, g)$ for every $c \in C_G(g)$. Show that if g is α-special, then so is every conjugate of g in G.

Hint: Let (Γ, π) be as in Lemma 11.16 with respect to G, A, and α. Then $\pi(x)$ is α-special iff $\pi(C_\Gamma(x)) = C_G(\pi(x))$.

Note: If α and β are equivalent, then $g \in G$ is α-special iff it is β-special. Thus, we can speak of $\bar{\alpha}$-special elements for $\bar{\alpha} \in H^2(G, A)$.

Proof. We begin by proving the claim provided in the hint. Let $g \in \pi(C_\Gamma(x))$ so that $g = \pi(y)$ for some $y \in C_\Gamma(x)$. We have that

$$g\pi(x) = \pi(y)\pi(x) = \pi(yx) = \pi(xy) = \pi(x)\pi(y) = \pi(x)g,$$

so $\pi(C_\Gamma(x)) \subseteq C_G(\pi(x))$.

Now, let $g \in C_G(\pi(x))$. By our construction in Lemma 11.16, we have that $x = (\pi(x), a)$ for some $a \in A$ and that

$$(g, 1)(\pi(x), a) = (g\pi(x), \alpha(g, \pi(x))a)$$

and

$$(\pi(x), a)(g, 1) = (\pi(x)g, \alpha(\pi(x), g)a).$$

Thus, $(g, 1)(\pi(x), a) = (\pi(x), a)(g, 1)$ iff $\alpha(g, \pi(x)) = \alpha(\pi(x), g)$. Now, if $\pi(x)$ is α-special, then we obtain that $g = \pi((g, 1)) \in \pi(C_\Gamma(x))$ for any $g \in C_G(\pi(x))$, and it follows from the containment in the first paragraph that $\pi(C_\Gamma(x)) = C_G(\pi(x))$. On the other hand, if $\pi(x)$ is not α-special, then we may choose $g \in C_G(\pi(x))$ for which $\alpha(g, \pi(x)) \neq \alpha(\pi(x), g)$, and it follows that $\pi(C_\Gamma(x)) \neq C_G(\pi(x))$. The claim in the hint now follows.

Now, we prove the statement in the problem. Let $g \in G$ be α-special. We may write $g = \pi(x)$ for some $x \in \Gamma$. By the claim in the hint, we have that $\pi(C_\Gamma(x)) = C_G(g)$. Now, let $h \in G$, and choose $y \in \Gamma$ such that $\pi(y) = h$. We have that

$$C_G(g^h) = (C_G(g))^h = \pi(C_\Gamma(x))^h = \pi(y^{-1})\pi(C_\Gamma(x))\pi(y) = \pi(C_\Gamma(x)^y) = \pi(C_\Gamma(x^y)).$$

Since $\pi(x^y) = g^h$, it follows from the claim in the hint that g^h is α-special. □

Problem 11.5. Let $\alpha \in Z^2(G, A)$, and let $\mathscr{K}$ be a conjugacy class of G. Show that $\mathscr{K}$ consists of α-special elements iff there exists a function $\mu : \mathscr{K} \to A$ such that

$$\mu(g)\alpha(g, h) = \mu(g^h)\alpha(h.g^h)$$

for all $g \in \mathscr{K}$ and all $h \in G$.

Hint: See the hint for Problem 11.4. Consider the conjugacy class of x_g in Γ, where x_g is a coset representative of A in Γ corresponding to G.

Proof. We first establish that no such function μ exists if $g \in \mathscr{K}$ is not α-special. If g is not α-special, then there exists some $c \in C_G(g)$ such that $\alpha(g, c) \neq \alpha(c, g)$. However, if such a function μ existed, then we would obtain

$$\mu(g)\alpha(g, c) = \mu(g^c)\alpha(c, g^c) = \mu(g)\alpha(c, g),$$

contrary to our assumption.

Now, assume that $g \in \mathscr{K}$ is α-special. We first show that if such a function μ exists for α, then such a function exists for any equivalent $\beta \in Z^2(G, A)$. Let $\beta(g, h) = \nu(g)\nu(h)\nu^{-1}(gh)\alpha(g, h)$ for all $g, h \in G$. We have that

$$\frac{\mu(g)}{\nu(g)}\beta(g, h) = \frac{\mu(g)\nu(g)\nu(h)}{\nu(g)\nu(gh)}\alpha(g, h) = \frac{\mu(g^h)\nu(h)}{\nu(gh)}\alpha(h, g^h)$$

$$= \frac{\mu(g^h)\nu(g^h)\nu(h)}{\nu(g^h)\nu(gh)}\alpha(h, g^h) = \frac{\mu(g^h)}{\nu(g)}\beta(h, g^h).$$

In particular, if μ is a suitable function for α, then μ/ν is a suitable function for β.

Now, let (Γ, π) be a central extension with coset representatives $\{x_g \mid g \in G\}$ with $x_g x_h = x_{gh}\alpha(g, h)$ (whose existence is guaranteed by Lemma 11.16). By Lemma 11.9, any other choice of coset representatives gives rise to an element of $Z^2(G, A)$ equivalent to α, and so, by the above paragraph, it suffices to verify the existence of a suitable function μ for this new element of $Z^2(G, A)$. We take advantage of the hint to construct this cocycle.

Given $g \in \mathscr{K}$, we claim that no two distinct conjugates of x_g lie in the same coset of A in Γ. Otherwise, there would exist some $c \in C_G(g)$ such that $(x_g)^{x_c} \neq x_g$. In particular, we would have that $x_g x_c \neq x_c x_g$ from which, we obtain $\alpha(g, c) \neq \alpha(c, g)$, contradicting the assumption that g lies in an α-special class. Thus, for fixed $g \in \mathscr{K}$ and for each distinct conjugate g^h of g, we can replace x_{g^h} with the new coset

representative $(x_g)^{x_h}$. This new choice of coset representatives is associated to a new cocycle β, which is equivalent to α.

Now, for any $h \in H$, our choice of coset representatives implies that $(x_g)^{x_h} = x_{g^h}$. Left multiplying by x_h on both sides yields

$$x_h(x_h)^{-1}x_g x_h = x_h x_{g^h},$$

which gives

$$x_{gh}\beta(g,h) = x_g x_h = x_h x_{g^h} = x_{gh}\beta(h,g^h).$$

We conclude that $\beta(g,h) = \beta(h,g^h)$ for all $h \in H$. The problem is now proved by taking $\mu : G \to \mathbb{C}^\times$ to be the function mapping all elements of g to 1. □

Problem 11.6. Let (Γ, π) be a finite central extension of G with $\ker \pi = A$. Choose a set of coset representatives for A in Γ and let $\alpha \in Z^2(G, A)$ be as in Lemma 11.9. Let $\lambda \in \hat{A}$. For each $\chi \in \mathrm{Irr}(\Gamma|\lambda)$, choose a representation $\mathfrak{N}$ affording χ and construct the projective $\mathbb{C}$-representation $\mathfrak{X}$ as in Lemma 11.10. Show that this defines a bijection of $\mathrm{Irr}(\Gamma|\lambda)$ onto the set of similarity classes of irreducible projective $\mathbb{C}$-representations of G with factor set $\lambda(\alpha)$.

Proof. By Lemma 11.10, this is a well-defined, injective map. Given any projective representation $\mathfrak{X}$ of G with factor set $\lambda(\alpha)$, observe that $\lambda(\alpha)$ lies in the image of the standard map, and so the surjectivity of the map follows from Theorem 11.13. Indeed, by following along with the proof of Theorem 11.13, one sees that we may lift $\mathfrak{X}$ to a representation $\mathfrak{N}$ of Γ where the restriction of $\mathfrak{N}$ to A is given by λI. □

Problem 11.7. Let $\alpha \in Z^2(G, \mathbb{C}^\times)$. Show that $\mathbb{C}^\alpha[G]$ is semisimple.

Hint: Show that $\sum(\dim M)^2 = |G|$ where M runs over a representative set of irreducible $\mathbb{C}^\alpha[G]$-modules. Use Problem 11.6.

Remark: We notice that there is an easier proof than what Isaacs seems to suggest from the hint. Our proof is based upon Schur's result (Theorem 11.17).

Proof. For notational convenience, we write $R = \mathbb{C}^\alpha[G]^\circ$ to denote the right regular module of $\mathbb{C}^\alpha[G]$. Upon fixing a basis of R, we obtain a projective representation for R, say $\mathfrak{X}$. Now, let (Γ, π) be a Schur covering of G (whose existence is guaranteed by Theorem 11.17), which is a central extension with $A = \ker \pi$ where $G \cong \Gamma/A$. By Theorem 11.17, the projective representation $\mathfrak{X}$ lifts to an ordinary representation $\mathfrak{N}$ where $\mathfrak{N}(x) = \mathfrak{X}(\pi(x))\mu(x)$ for some function $\mu : \Gamma \to \mathbb{C}^\times$. We conclude that R may be viewed also as an $\mathbb{C}[\Gamma]$-module, and this module is semisimple by Maschke's Theorem. Since $\mathfrak{N}(x)$ and $\mathfrak{X}(\pi(x))$ differ by a scalar factor, one easily verifies that any $\mathbb{C}^\alpha[G]$-submodule of R is also a $\mathbb{C}[\Gamma]$-module, and vice versa.

We show directly that R is completely reducible as a $\mathbb{C}^\alpha[G]$-module. Now, R is completely reducible as a $\mathbb{C}[\Gamma]$-module by Maschke's Theorem. Thus, any $\mathbb{C}[\Gamma]$-submodule V is complemented in R by a $\mathbb{C}[\Gamma]$-submodule W. Since both V and

W are also complementary as $\mathbb{C}^\alpha[G]$-submodules, this proves that R is completely reducible as a $\mathbb{C}^\alpha[G]$-module, and it follows that $\mathbb{C}^\alpha[G]$ is semisimple. □

Problem 11.8. Let $\alpha \in Z^2(G, F^\times)$. Show that $\dim(\mathbf{Z}(F^\alpha[G]))$ is equal to the number of α-special classes of G.

Hint: Use Problem 11.5.

Proof. Let $\mathscr{K}$ be an α-special conjugacy class of G, and let μ be as in Problem 11.5. Define $K_\alpha = \sum_{g\in\mathscr{K}} \mu(g)\bar{g}$. We claim that K_α is central in this F-algebra. Clearly, it suffices to show that K_α commutes with each $\bar{h}$ for $h \in G$. We calculate

$$K_\alpha \bar{h} = \sum_{g\in\mathscr{K}} \mu(g)\bar{g}\bar{h} = \sum_{g\in\mathscr{K}} \mu(g)\alpha(g,h)\overline{gh} = \sum_{g\in\mathscr{K}} \mu(g^h)\alpha(h,g^h)\overline{hg^h}$$

$$= \sum_{g\in\mathscr{K}} \mu(g^h)\bar{h}\overline{g^h} = \bar{h}\sum_{g\in\mathscr{K}} \mu(g)\bar{g} = \bar{h}K_\alpha.$$

Thus, each α-special class $\mathscr{K}$ gives rise to an element $K_\alpha \in \mathbf{Z}(F^\alpha[G])$, and it is clear that the set of each such K_α is linearly independent. We conclude that the number of α-special classes is less than or equal to the the dimension of the center of the F-algebra.

Now, let $L \in \mathbf{Z}(F^\alpha[G])$ where $L = \sum_{g\in G} a_g\bar{g}$ for some $a_g \in F$. Notice that the basis elements $\bar{h}$ permute the terms of L under conjugation, i.e., $\bar{h}^{-1}a_g\bar{g}\bar{h} = a_{g^h}\overline{g^h}$. Rearranging, we obtain $a_{g^h}\bar{h}\overline{g^h} = a_g\bar{g}\bar{h}$. We compute

$$a_{g^h}\alpha(h,g^h)\overline{gh}a_{g^h}\alpha(h,g^h)\overline{hg^h} = a_{g^h}\bar{h}\overline{g^h} = a_g\bar{g}\bar{h} = a_g\alpha(g,h)\bar{g}\bar{h}.$$

We conclude that

$$a_{g^h}\alpha(h,g^h) = a_g\alpha(g,h) \tag{11.9}$$

for each $g, h \in G$. We also conclude that $a_g \neq 0$ iff $a_{g^h} \neq 0$. If $a_g \neq 0$ for some $g \in G$, Problem 11.5 is applicable by viewing $\mu(g) := a_g$ as a function from $\mathrm{Cl}(g)$ into $A = F^\times$, and the existence of suitable coefficients a_{g^h} is possible if and only if g lies in an α-special class. Otherwise, if g does not belong to an α-special class then no such μ exists, implying that $a_{g^h} = 0$ for each conjugate of g. We see that $a_g \neq 0$ only if g lies in an α-special class and that L is a linear combination of the K_α defined above. Thus, $\mathbf{Z}(F^\alpha[G])$ is spanned by the K_α, and the statement now follows. □

Problem 11.9. Let (G, N, ϑ) be a character triple. If $\bar{g} \in G/N$, say that $\bar{g}$ is ϑ-special if ϑ is extendible to $\langle N, g, c\rangle$ for all $c \in G$ with $[g, c] \in N$. Check that this is well defined. Let $\bar{\beta} \in H(G/N, \mathbb{C}^\times)$ be as in Theorem 11.7. Show that $\bar{g} \in G/N$ is ϑ-special iff it is $\bar{\beta}$-special. (See the note following Problem 11.14.)

Note: It is clear without special appeal to Problem 11.4 that conjugates of ϑ-special elements in G/N are ϑ-special.

Proof. We first verify that the definition of ϑ-specialty is well defined. This amounts to showing that the choice of a coset representative for $\bar{g} = gN$ is irrelevant. Assume that ϑ extends to $\langle G, g, c\rangle$ for each $c \in G$ with $[g, c] \in N$. Let $n \in N$. It is clear that $\langle G, gn, c\rangle = \langle G, g, c\rangle$ for any $c \in G$. Now, we claim that $[g, c] \in N$ iff $[gn, c] \in N$. Indeed, if $[g, c] \in N$, then observe that $[gn, c] = n^{-1} \cdot [g, c] \cdot n^c \in N$, and the converse holds by symmetry. It now follows that if ϑ extends to $\langle G, g, c\rangle$ for each $c \in G$ with $[g, c] \in N$, then ϑ extends to $\langle N, gn, c\rangle$ for all $c \in G$ with $[gn, c] \in N$. This proves that the definition is well-defined.

First, assume that $\bar{g}$ is ϑ-special so that ϑ extends to every group of the form $H_c = \langle N, g, c\rangle$ for appropriate c. By Theorem 11.7, the restriction of β to H_c/N is equivalent to the trivial factor set, say, $\beta_{H_c/N}(aN, bN) = \mu(aN)\mu(bN)\mu^{-1}(abN)$ for some function $\mu : G/N \to \mathbb{C}^\times$. Now, the condition that $[g, c] \in N$ is equivalent to $gcN = cgN$. So, we have that

$$\beta(gN, cN) = \mu(gN)\mu(cN)\mu^{-1}(gcN) = \mu(cN)\mu(gN)\mu^{-1}(cgN) = \beta(cN, gN).$$

From this it follows that $\bar{g} = gN$ is $\bar{\beta}$-special.

Conversely, if $\bar{g}$ is $\bar{\beta}$-special, this implies that $\beta(gN, cN) = \beta(cN, gN)$ for each $cN \in C_{G/N}(gN)$. Now, notice that $\mathbb{C}^\times$ is a divisible group (since $\mathbb{C}$ is algebraically closed), so Problem 11.3 is applicable. We conclude that the restriction of β to $\langle gN, cN\rangle$ is equivalent to the trivial factor set. By Theorem 11.7 and since $\langle gN, cN\rangle = \langle N, g, c\rangle/N$, we conclude that ϑ extends to $\langle N, g, c\rangle$ for any $c \in G$ with $cgN = gcN$ (or equivalently $[g, c] \in N$). This proves that $\bar{g}$ is ϑ-special, as desired. □

Problem 11.10. (Gallagher) Let (G, N, ϑ) be a character triple. Show that $|\operatorname{Irr}(G|\lambda)|$ is the number of ϑ-special conjugacy classes of G/N.

Hint: Several of the previous problems are relevant.

Proof. Let $\bar{\alpha} \in H(G/N, \mathbb{C}^\times)$ be as in Theorem 11.7 for some $\alpha \in Z^2(G/N, \mathbb{C}^\times)$. By Problem 11.9, the number of ϑ-special classes of G/N equals the number of $\bar{\alpha}$-special classes. Notice also that by the definition of α-specialty given in Problem 11.4, $g \in G$ is α-special iff g is α^{-1}-special. Set $\beta = \alpha^{-1}$. By Problem 11.8 (and the note to Problem 11.4), the number of $\bar{\beta}$-special classes is equal to the $\mathbb{C}$-dimension of $\mathbf{Z}(\mathbb{C}^\beta[G])$. Now, by Problem 11.7, $\mathbb{C}^\beta[G]$ is semisimple, so Corollary 1.17 is applicable. Namely, by part (e) of this corollary, the dimension of the center of $\mathbf{Z}(\mathbb{C}^\beta[G])$ is equal to $|\mathscr{M}(\mathbb{C}^\beta[G])|$, i.e, the number of isomorphism classes of irreducible $\mathbb{C}^\beta[G]$-modules. Now, let Γ be a Schur cover of G/N as in Theorem 11.17 so that $\beta = \lambda(\gamma)$ for some $\gamma \in Z^2(G, A)$ and $\lambda \in \hat{A}$. By Problem 11.5, $|\mathscr{M}(\mathbb{C}^\beta[G])| = |\operatorname{Irr}(\Gamma|\lambda)|$. We have established that the number of ϑ-special classes equals $|\operatorname{Irr}(\Gamma|\lambda)|$.

Now, by Theorem 11.28 (and its proof), we may define an isomorphism of character triples from (G, N, ϑ) to (Γ, A, λ) since $\bar{\alpha} = \bar{\beta}^{-1} = \eta(\lambda)^{-1}$. By the definition of character-triple isomorphisms, we have that $|\operatorname{Irr}(\Gamma|\lambda)| = |\operatorname{Irr}(G|\vartheta)|$, which we have

established to be equal to the number of ϑ-special classes of G/N. This completes the proof. □

Problem 11.11. The character triple (G, N, θ) is *fully ramified* if there exists $\chi \in \mathrm{Irr}(G|\theta)$ such that $(\chi(1)/\theta(1))^2 = |G : N|$. Suppose (G, N, θ) is fully ramified and $G > N$. Show that no cyclic subgroup of G/N can be its own centralizer.

Hint: Use Problem 11.10

Proof. Notice that our notion of a character triple being fully ramified is the same as the notion in Problem 6.3 of χ and θ being fully ramified in G/N. We conclude from Problem 6.3 that χ is the unique irreducible constituent of θ^G, and by Frobenius Reciprocity, we see that $\mathrm{Irr}(G|\theta) = \{\chi\}$. Problem 11.10 is therefore applicable, and we conclude that G/N has exactly one θ-special class.

We claim that the identity in G/N is θ-special. Indeed, referring to the definition given in Problem 11.9, this is seen by checking that θ extends to $\langle N, c\rangle$ for every $c \in G$. Observe that $\langle N, c\rangle/N$ is cyclic, so by Corollary 11.22, θ in fact extends to each such subgroup.

Now, assume for a contradiction that G/N has a self-centralizing cyclic subgroup, say C/N. First, notice that θ extends to C again by Corollary 11.22. Let gN be a generator of C/N, and observe that $[g, c] \in N$ iff $cN \in C/N$ by the centralizer condition. Also, observe that since C/N. Thus, for each $c \in G$ with $[g, c] \in N$, we have that $\langle N, g, c\rangle = \langle N, g\rangle$ to which θ extends. We conclude that gN is θ-special. However, we have also established that $1N$ is θ-special and that G/N has only one θ-special class. This is a contradiction. □

Problem 11.12. Let (G, N, θ) be a character triple. For $x, y \in G$ with $[x, y] \in N$, define the complex number $\langle\langle x, y\rangle\rangle$ as follows: Let ψ be an extension of θ to $H = \langle N, y\rangle$. Write $\psi^x = \psi\lambda$ for $\lambda \in \mathrm{Irr}(H/N)$ and put $\langle\langle x, y\rangle\rangle = \lambda(y)$. Show that $\langle\langle , \rangle\rangle$ is uniquely defined and that

(a). $\langle\langle x, y\rangle\rangle = \langle\langle xn, ym\rangle\rangle$ for $n, m \in N$;
(b). $\langle\langle x_1x_2, y\rangle\rangle = \langle\langle x_1, y\rangle\rangle\langle\langle x_2, y\rangle\rangle$ whenever $[x_1, y], [x_2, y] \in N$;
(c). $\langle\langle x, y\rangle\rangle = \theta([x, y])$ whenever θ is linear.

Note: By (a) we can view $\langle\langle , \rangle\rangle$ as being defined on commuting pairs of elements of G/N.

Proof. We first demonstrate the $\langle\langle , \rangle\rangle$ is uniquely defined. First, observe that since $[x, y] \in N$, $x \in N_G(H)$, and so $\psi^x \in \mathrm{Irr}(H)$. Further, since $\psi_N = \theta$ and θ is invariant in G, we have that $\psi^x \in \mathrm{Irr}(G|\theta)$. By the Gallagher Correspondence (Corollary 6.17), $\mathrm{Irr}(G|\theta) = \{\psi\beta \mid \beta \in \mathrm{Irr}(H/N)\}$, and each $\psi\beta$ is distinct from one another. Consequently, there exists $\lambda \in \mathrm{Irr}(H/N)$ such that $\psi^x = \psi\lambda$ and that this λ is uniquely defined for the given ψ. Now, we must show that the λ obtained is independent of the choice of extension ψ. Again, by the Gallagher correspondent, any other extension

of θ to H is of the form $\psi\beta$ for $\beta \in \mathrm{Irr}(H/N)$. Observe again that since $[x,y] \in N$, we have $y^{x^{-1}}N = yN$, so $\beta^x(y) = \beta(y)$. Now,

$$(\psi\beta)^x(y) = \psi^x(y)\beta^x(y) = \psi(y)\beta^x(y)\lambda(y) = \psi(y)\beta(y)\lambda(y).$$

This shows that the λ obtained is independent of the extension ψ, and this verifies that $\langle\!\langle , \rangle\!\rangle$ is uniquely defined.

Now, we prove part (a). For any $n \in N$, we have that $\psi^{xn}(g) = \psi^x(g^{n^{-1}}) = \psi^x(g)$ since $n \in N \subseteq H$ and ψ is a class function. Thus, $\langle\!\langle xn, ym \rangle\!\rangle = \langle\!\langle x, ym \rangle\!\rangle$. Also, notice that $H = \langle N, y\rangle = \langle N, ym\rangle$, so $\langle\!\langle x, ym\rangle\!\rangle = \lambda(ym)$, and $\lambda(ym) = \lambda(y)$ since $m \in N \subseteq \ker\lambda$. Thus,

$$\langle\!\langle xn, ym\rangle\!\rangle = \langle\!\langle xym\rangle\!\rangle = \langle\!\langle x, y\rangle\!\rangle,$$

as desired.

For part (b), we have that set $\psi^{x_1} = \psi\lambda_1$ and $\psi^{x_2}\lambda_2$. We have that

$$\psi^{x_1x_2} = \psi^{x_2}\lambda_1^{x_2} = \psi\lambda_1^{x_2}\lambda_2.$$

Notice that since $[x_2, y] \in N$, we have that $y^{x_2^{-1}}N = yN$, so $\lambda_1^{x_2}(y) = \lambda_1(y)$. It follows that $\psi^{x_1x_2}\psi^{x_1x_2} = \psi\lambda_1\lambda_2$. Consequently, $\langle\!\langle x_1x_2, y\rangle\!\rangle = \lambda_1(y)\lambda_2(y) = \langle\!\langle x_1, y\rangle\!\rangle\langle\!\langle x_2, y\rangle\!\rangle$.

For part (c), assume θ (and therefore also ψ) is linear. One has that $\psi^x(y)\overline{\psi(y)} = \lambda(y) = \langle\!\langle x, y\rangle\!\rangle$. Notice that since ψ is linear, we obtain $\langle\!\langle x, y\rangle\!\rangle = \psi([x^{-1}, y^{-1}])$. Observe now that $[x^{-1}, y^{-1}]^{yx} = [x, y]$, and since $[x, y] \in N$ and θ is invariant, we obtain that $\langle\!\langle x, y\rangle\!\rangle = \theta([x, y])$, as desired. □

Problem 11.13. Let (G, N, θ) be a character triple. If $N \subseteq H \subseteq G$, $\psi \in \mathrm{Ch}(H|\theta))$ and $\bar{g} = gN \in G/N$ we define $\psi^{\bar{g}} \in \mathrm{Ch}(H^g|\theta)$ by $\psi^{\bar{g}}(h^g) = \psi(h)$. Check that this is well defined. We say that the isomorphism

$$(\tau, \sigma) : (G, N, \theta) \to (\Gamma, M, \psi)$$

is *strong* provided $(\sigma_H(\psi))^{\tau(\bar{g})} = \sigma_{H^g}(\psi^{\bar{g}})$ for all $\bar{g} \in G/N$, all H with $N \subseteq H \subseteq G$ and all $\psi \in \mathrm{Ch}(G|\theta)$. Show that the isomorphism constructed in Theorem 11.28 is strong.

Proof. We first demonstrate $\psi^{\bar{g}}$ is well defined, i.e., its definition is independent of a choice of a coset representative. This amounts to showing that $\psi^g = \psi^{gn}$ for all $n \in N$. Indeed, notice that since $N \subseteq H^g$, we have $H^g = H^{gn}$ for all $n \in N$. For any $k \in H^g$, we have $\psi^{gn}(k) = \psi((k^{g^{-1}})^{n^{-1}})$. Since ψ is a class function on H and $n \in N \subseteq H$, conjugating $k^{g^{-1}}$ by n^{-1} does not affect the value taken by ψ. It follows that $\psi^g = \psi^{gn}$, and so, $\psi^{\bar{g}}$ is well defined.

One shows that strong isomorphism is an equivalence relation on character triples. Let $(\tau, \sigma) : (G, N, \theta) \to (\Gamma, M, \varphi)$ be a strong isomorphism. The fact that this relation is reflexive is seen by taking both τ and σ to be the identity maps. Further, the fact the composition of strong isomorphisms is itself strong is a routinely verified, and so we will assume that this is a transitive relation.

We now demonstrate that the relation is symmetric. As was demonstrated in the proof to Corollary 11.25, (τ^{-1}, σ^{-1}) is the inverse of (τ, σ) and is a character triple isomorphism. Let K be a subgroup of Γ containing M, $\bar{h} \in \Gamma/M$, and $\xi \in \mathrm{Ch}(K|\varphi)$. Let $H \subseteq G$ such that $K = H^\tau$ and $\bar{g} = \tau^{-1}(\bar{h})$. Defining $\psi = \sigma_H^{-1}(\xi) \in \mathrm{Ch}(H|\theta)$, the strongness of (τ, σ) gives

$$\xi^{\bar{h}} = (\sigma_H(\psi))^{\tau(\bar{g})} = \sigma_{H^g}(\psi^{\bar{g}}).$$

Applying $\sigma_{H^g}^{-1}$ to both sides yields

$$\sigma_{H^{\bar{g}}}^{-1}(\xi^{\bar{h}}) = \psi^{\bar{g}} = (\sigma_H^{-1}(\xi))^{\tau^{-1}(\bar{h})},$$

which proves the inverse is strong.

Now, the character triple isomorphism in Theorem 11.28 is constructed as a composition of the type given in Lemma 11.26 with one of the type given in Lemma 11.27 with the inverse of the type in Lemma 11.26. By the symmetry and transitivity of strong character triple isomorphisms established above, it suffices to show that the isomorphisms described in Lemmas 11.26 and 11.27 are both strong. In the case of Lemma 11.26, this is more or less trivial. In the case of Lemma 11.27 (keeping the same notation), one sees that $\sigma_H(\psi)^{\bar{g}} = (\psi\eta_H)^g = \psi^g \eta_H^g = \psi^g \eta_H = \sigma(\psi^{\bar{g}})$, where $\eta_H^g = \eta_H$ since η is a class function of G. □

Problem 11.14. In the notation of Problem 11.12, show that $\langle\!\langle x, y \rangle\!\rangle = \langle\!\langle y, x \rangle\!\rangle^{-1}$ and $\langle\!\langle x, y_1 y_2 \rangle\!\rangle = \langle\!\langle x, y_1 \rangle\!\rangle \langle\!\langle x, y_2 \rangle\!\rangle$ if $[x, y_1], [x, y_2] \in N$.

Hint: Viewing $\langle\!\langle , \rangle\!\rangle$ as defined on commuting pairs of elements of G/N, it is invariant under strong isomorphisms of character triples (See Problem 11.13).

Proof. Notice first that if $\langle\!\langle x, y \rangle\!\rangle = \langle\!\langle y, x \rangle\!\rangle^{-1}$ holds for all $x, y \in G$ such that $[x, y] \in N$, then so will the identity $\langle\!\langle x, y_1 y_2 \rangle\!\rangle = \langle\!\langle x, y_1 \rangle\!\rangle \langle\!\langle x, y_2 \rangle\!\rangle$. Indeed, by Problem 11.12(b), we have

$$\langle\!\langle x, y_1 y_2 \rangle\!\rangle = \overline{\langle\!\langle y_1 y_2, x \rangle\!\rangle} = \overline{\langle\!\langle y_1, x \rangle\!\rangle \langle\!\langle y_2, x \rangle\!\rangle} = \langle\!\langle x, y_1 \rangle\!\rangle \langle\!\langle x, y_2 \rangle\!\rangle.$$

Thus, we need only prove $\langle\!\langle x, y \rangle\!\rangle = \langle\!\langle y, x \rangle\!\rangle^{-1}$.

We now make concrete the claim in the hint. Given any strong character triple isomorphism (τ, σ) from (G, N, θ) to (G, M, δ), we claim that

$$\langle\!\langle x, y \rangle\!\rangle = \langle\!\langle \tau(x), \tau(y) \rangle\!\rangle.$$

Keeping the notation of Problem 11.12, it follows from Lemma 11.24 that $\sigma_H(\psi)$ extends to $H^\tau = \langle M, \tau(y) \rangle$ since ψ extends to H, and so,

$$\sigma_H(\psi)^{\tau(\bar{x})} = \sigma_H(\psi)\lambda_1 = \sigma_H(\psi)\langle\!\langle \tau(x), \cdot \rangle\!\rangle$$

for some $\lambda_1 \in \mathrm{Irr}(H^\tau/M)$. On the other hand, recall that our assumptions imply $H^x = H$. Since the isomorphism is strong, we have that

$$\sigma_H(\psi)^{\tau(\bar{x})} = \sigma_H(\psi^x) = \sigma_H(\psi\lambda) = \sigma_H(\psi)\lambda^\tau$$

for some $\lambda \in \mathrm{Irr}(H/N)$. Comparing the above two equations yields $\lambda_1 = \lambda^\tau$. Now, we compute

$$\langle\!\langle \tau(x), \tau(y) \rangle\!\rangle = \lambda_1(\tau(y)) = \lambda^\tau(\tau(y)) = \lambda(y) = \langle\!\langle x, y \rangle\!\rangle,$$

which verifies the desired equality.

Now, that the hint has been handled, we may now invoke Problem 11.13 to produce a strongly isomorphic character triple (Γ, A, γ) where $A \subseteq \mathbf{Z}(\Gamma)$. It suffices to prove that $\langle\!\langle x, y \rangle\!\rangle = \langle\!\langle y, x \rangle\!\rangle^{-1}$ with respect to this character triple. We obtain by Problem 11.12(c) that $\langle\!\langle x, y \rangle\!\rangle = \gamma([x, y])$ and that $\langle\!\langle y, x \rangle\!\rangle = \gamma([y, x])$. Recall the commutator identity $[x, y]^{-1} = [y, x]$, and it follows that $\langle\!\langle x, y \rangle\!\rangle = \langle\!\langle y, x \rangle\!\rangle^{-1}$. □

Problem 11.15. Let (G, N, θ) be a character triple.

(a). If $g \in G$ and $\langle\!\langle g, x \rangle\!\rangle \neq 1$ for some $x \in G$, show that $\chi(g) = 0$ for all $\chi \in \mathrm{Irr}(G|\theta)$.

(b). If $\bar{g} \in G/N$, show that $\bar{g}$ is θ-special iff $\langle\!\langle g, x \rangle\!\rangle = 1$ for all x such that $\langle\!\langle g, x \rangle\!\rangle$ is defined.

Note: See Problems 11.12 and 11.9 for definitions.

Proof. For part (a), we begin by recognizing that it is of no loss to assume $G = \langle N, x, g \rangle$ so that G/N is abelian. This is seen by setting $K = \langle N, x, g \rangle$ and observing that if $\chi \in \mathrm{Ch}(G|\theta)$, then $\chi_K \in \mathrm{Ch}(K|\theta)$. Further, it is of course no loss to assume that χ is irreducible.

Notice first that for any $\chi \in \mathrm{Irr}(G|\theta)$, the restriction of χ to $H = \langle N, x \rangle$ has constituents that are extensions of θ to H since the elements of $\mathrm{Irr}(H|\theta)$ are each extensions of θ by the Gallagher Correspondence (Corollary 6.17) and Theorem 11.22. Since the definition of $\langle\!\langle g, x \rangle\!\rangle$ is independent of a choice of extension of θ to $\psi \in \mathrm{Irr}(H)$, we may assume without loss that ψ is a constituent of χ_H. Assuming $\langle\!\langle g, x \rangle\!\rangle \neq 1$, we have that $\psi^g = \psi\lambda$ for nontrivial $\lambda \in \mathrm{Irr}(H/N)$. This implies that $g \in G - I_G(\psi)$, and in particular, $I_G(\psi) \neq G$. Notice also that since G/N is abelian, $I_G(\psi) \lhd G$. Thus, χ is of the form ϕ^G for some $\phi \in \mathrm{Irr}(I_G(\psi)|\psi)$ by Clifford's Theorem 6.11. Since ϕ here is induced from a normal subgroup of G not containing g, we see that $\chi(g) = \phi^G(g) = 0$, as desired.

We now prove part (b). First, assume $\langle\!\langle g, x \rangle\!\rangle \neq 1$ for some $g, x \in G$ with $[g, x] \in N$, and assume for a contradiction that θ extends to $H = \langle N, g, x \rangle$ as η. Notice now that for fixed g, by Problem 11.14, we may view $\lambda(y) = \langle\!\langle g, y \rangle\!\rangle$ as a nontrivial linear irreducible character of H/N. We claim that $\eta(y) = 0$ if $y \notin \ker\lambda$. Indeed, by Problem 11.14, $\langle\!\langle g, y \rangle\!\rangle \neq 1$ implies $\langle\!\langle y, g \rangle\!\rangle \neq 1$, and again by part (a), this implies $\eta(y) = 0$, as desired. Now, for any $y \notin \ker\lambda$, we have $\eta(y) = \eta(y)\lambda(y) = 0$, and we conclude $\eta\lambda = \eta$. However, by the Gallagher Correspondence, the extendability of θ to η implies that $\eta\lambda \neq \eta$, and this is a contradiction. Thus, θ does not extend to H, and it follows from the definition of θ-specialty in Problem 11.9 that $\bar{g}$ is not θ-special.

Conversely, for fixed $g \in G$, assume that $\langle\langle g, x \rangle\rangle = 1$ for all $x \in G$ with $[g, x] \in N$. We let ψ be an extension of θ to $H = \langle N, x \rangle$. Notice that our assumption implies that $\psi^g = \psi$. In particular, ψ is invariant in $\langle N, x, g \rangle$. However, notice that $\langle N, x, g \rangle / \langle N, x \rangle$ is cyclic, and thus, ψ is extendable to $\langle N, x, g \rangle$ by Corollary 11.22. Since x was chosen arbitrarily, it follows from the definition of θ-specialty that $\bar{g}$ is θ-special. □

Problem 11.16. Let E be an elementary abelian p-group of order p^n. Show that $M(E)$ is elementary abelian of order $p^{n(n-1)/2}$.

Hint: Consider a Schur representation group E to get $|M(E)| \leqslant p^{n(n-1)/2}$. For equality, let $\{x_i \mid 1 \leqslant i \leqslant n\}$ be a generating set for E. Define $\alpha_{ij} : E \times E \to \mathbb{C}$ by $\alpha_{ij}(\prod x_\mu^{a_\mu}, \prod x_\mu^{b_\mu}) = \epsilon^{a_i b_j}$, where ϵ is a primitive pth root of unity.

Proof. Before proceeding, we state some relevant commutator identities for a group G with $G' \subseteq \mathbf{Z}(G)$. Consider the well-known and easily proven identity: $[xy, z] = [x, z]^y [y, z]$. Since G' is central, we have that $[xy, z] = [x, z][y, z]$. Analogously, one also has $[x, yz] = [x, y][x, z]$. In particular, we see inductively that

$$\left[\prod_i x_i, \prod_j y_j \right] = \prod_{i,j} [x_i, y_j] \tag{11.10}$$

for any $x_i, y_j \in G$. In particular, we have that

$$[x, y]^m = [x^m, y] = [x, y^m] \tag{11.11}$$

for any $x, y \in G$ and any integer m.

Let Γ be a Schur cover of E with $\mathbf{Z}(\Gamma) \cong M(E)$. We may assume by Corollary 11.20 that $\Gamma' = \mathbf{Z}(\Gamma)$. Thus, $\mathbf{Z}(\Gamma)$ is generated by commutators. We first show that $\mathbf{Z}(\Gamma)$ is an elementary abelian p-group. To this end, it merely suffices to show that every element of $\mathbf{Z}(\Gamma)$ has order p, and in particular, it suffices to show that every commutator in Γ has order p. Choose $x, y \in \Gamma$. By (11.11), one has that $[x, y]^p = [x^p, y]$, and $[x^p, y] = 1$ since $x^p \in \mathbf{Z}(\Gamma)$. It now follows that $M(E)$ is an elementary abelian p-group.

We now show that $M(E)$ has rank at most $n(n-1)/2$. Let $\{g_i \mid 1 \leqslant i \leqslant n\}$ be a preimage of a generating set for E in Γ. It follows from (11.10) that any commutator in Γ may be expressed as a product of commutators of the form $[g_i, g_j]$ for $i < j$. (Recall that $[g_i, g_j] = [g_j, g_i]^{-1}$ and that $[g_i, g_i] = 1$.) It follows that $M(E) \cong \Gamma'$ has a generating set of size $n(n-1)/2$, and we have that $|M(E)| \leqslant p^{n(n-1)/2}$.

Now, taking α_{ij} as defined in the hint, we show that for $i \neq j$, the α_{ij} do not lie in $B^2(G, \mathbb{C}^\times)$. By Problem 11.3, it suffices to show there exist $x, y \in E$ such that $\alpha_{ij}(x, y) \neq \alpha_{ij}(y, x)$. (This requires us to know that $\mathbb{C}^\times$ is divisible, which is a consequence of the fact that $\mathbb{C}$ is algebraically closed.) Keeping the notation in the hint, we evaluate $\alpha_{ij}(x_i, x_j) = \epsilon$ and $\alpha_{ij}(x_j, x_i) = \epsilon^0 = 1$ (assuming that $i \neq j$). This implies that each $\bar{\alpha}_{ij}$ is a nonidentity element of $M(E)$. Since $M(E)$ is an

elementary abelian p-group of rank at most $n(n-1)/2$, it merely suffices to show that $\{\bar{\alpha}_{ij} \mid 1 \leqslant i < j \leqslant n\}$ is linearly independent in $M(E)$. Consider an arbitrary product of these elements

$$\gamma(x,y) = \prod_{1 \leqslant i < j \leqslant n} \alpha_{ij}^{m_{ij}}(x,y)$$

for some integers m_{ij}, and assume that $\bar{\gamma}$ is the identity in $H^2(G, \mathbb{C}^\times)$. We claim that each m_{ij} is divisible by p. If this were not the case, then let $m_{k\ell}$ be indivisible by p for some integers $k < \ell$.

Now, by Problem 11.3, $\gamma(y,z)/\gamma(z,y) = 1$ for any $x, y \in E$. Writing $y = \prod_\mu x_\mu^{a_\mu}$ and $z = \prod_\mu x_\mu^{b_\mu}$, we see that $\gamma(y,z)/\gamma(z,y)$ is ϵ raised to the power of

$$\sum_{1 \leqslant i < j \leqslant n} m_{ij}(a_i b_j - b_i a_j).$$

For a contradiction, it suffices to show that $\gamma(x_k, x_\ell)/\gamma(x_\ell, x_k) \neq 1$. However, one sees from the above equation that $\gamma(x_k, x_\ell)/\gamma(x_\ell, x_k) = \epsilon^{m_{k\ell}}$, which does not equal one since $p \nmid m_{k\ell}$ by assumption. This implies that $\{\bar{\alpha}_{ij} \mid 1 \leqslant i < j \leqslant n\}$ is a linearly independent set in $M(E)$, and we conclude that $M(E)$ has rank exactly $n(n-1)/2$, as desired. □

Problem 11.17. Let $G = A_5$. Show that $|M(G)| = 2$.

Note: In fact, $|M(A_n)| = 2$ for all $n \geqslant 4$ except for $n = 6, 7$, where the Schur multipliers have order 6.

Remark: This solution was adapted from the discussion in this Math Stack Exchange thread.

Proof. First, notice that $|M(A_5)| \geqslant 2$. This is seen by recognizing that $\mathrm{SL}(2,5)$ is a perfect central extension of A_5 by a cyclic group of order 2.

It now suffices to show that $|M(A_5)| \leqslant 2$. First, notice that since the Sylow 3- and Sylow 5-subgroups of G are cyclic, the Schur multiplier of A_5 must be a 2-group by Corollary 11.21.

Let Γ be a perfect Schur cover of A_5 with $\mathbf{Z}(\Gamma) \cong M(A_5)$, and let P be a Sylow 2-subgroup of Γ. Notice that $P/\mathbf{Z}(\Gamma)$ is isomorphic to a Sylow 2-subgroup of A_5, which is a Klein four-group. We claim that $P' = \mathbf{Z}(\Gamma)$. Indeed, $P' \subseteq \mathbf{Z}(\Gamma)$ so that $P' \lhd \Gamma$. To see that equality holds, one sees that Γ/P' has abelian an Sylow 2-subgroup P/P', and it follows from Theorem 5.6 that Γ/P' has a center whose order is indivisible by 2. Thus, $P' = \mathbf{Z}(\Gamma)$, as desired.

Now, we have that $P' \cap \mathbf{Z}(P) = \mathbf{Z}(\Gamma)$, and it follows from Corollary 11.20 that $\mathbf{Z}(\Gamma)$ is isomorphic to a subgroup of $M(P/P')$. Now, by Problem 11.16, $M(P/P')$ has order 2, since P/P' is a Klein four-group. This shows that $M(A_5) \leqslant 2$, as desired. □

Problem 11.18. Let $G = CH$, where H and C are cyclic, $C \lhd G$,and $H \cap C = \mathbf{Z}(G)$. Show that $M(G) = 1$.

Hint: If Γ is a Schur representation group for G, show that $|G'| = |\Gamma'|$.

Remark: Notice that this result implies that generalized quaternion groups have trivial Schur multipliers. Simply take C to be a cyclic subgroup of index 2 and $H = \langle h \rangle$ where h is any element lying outside of C.

Proof. Letting (Γ, π) be a Schur cover of G with $A := \ker \pi \cong M(G)$, one has by Corollary 11.20 that $A \subseteq \Gamma'$. One then trivially has $\pi(\Gamma') = G'$. Therefore, if we verify that $|\Gamma'| = |G'|$ as suggested in the hint, then the result will follow. We approach this by using the sublemma below. Clearly, G satisfies the conditions of the sublemma, and so,

$$|G'| = |C : \mathbf{Z}(G)|.$$

We claim that Γ also satisfies the necessary conditions for the lemma as well. Letting $\bar{H}$ and $\bar{C}$ be the preimages of H and C respectively in Γ, we trivially have that $\Gamma = \bar{C}\bar{H}$, $\bar{C} \trianglelefteq \Gamma$, and $\Gamma/\bar{C}$ is cyclic. The fact that $\bar{C}$ and $\bar{H}$ are abelian follows that $\bar{C}/\mathbf{Z}(\Gamma)$ and $\bar{C}/\mathbf{Z}(\Gamma)$ are cyclic. To see that $\bar{C} \cap \bar{H} = \mathbf{Z}(\Gamma)$, it suffices to show that $\pi(\mathbf{Z}(\Gamma)) = \mathbf{Z}(G)$. Letting $\bar{Z}$ denote the preimage of $\mathbf{Z}(G)$ in Γ, we clearly have $\mathbf{Z}(\Gamma) \subseteq \bar{Z}$. The fact that equality holds follows from the fact that $\bar{H}/\bar{Z}$ and $\bar{C}/\bar{Z}$ are both cyclic and that $\bar{C}\bar{H} = \Gamma$. Thus, $|\Gamma'| = |\bar{C} : \mathbf{Z}(\Gamma)|$ by the sublemma. Also, since $\pi(\mathbf{Z}(\Gamma)) = \mathbf{Z}(G)$, we have

$$|\Gamma'| = |\bar{C} : \mathbf{Z}(\Gamma)| = |C : \mathbf{Z}(G)| = |G'|,$$

which verifies the claim in the hint and therefore proves the result. □

Sublemma. Let $G = CH$ such that the following hold:

(i). $C \trianglelefteq G$.
(ii). $C \cap H = \mathbf{Z}(G)$.
(iii). C and H are abelian.
(iv). G/C is cyclic.

Then, $|G'| = |C : \mathbf{Z}(G)|$.

Proof. We may choose $t \in H$ such that tC generates G/C. We define a homomorphism $\phi : C \to C$ where $\phi(k) = [k, t]$. To see that this is a homomorphism, one sees that $[k_1k_2, t] = [k_1, t]^{k_2}[k_2, t] = [k_1, t]^{k_2}[k_2, t]$, where $[k_1, t]^{k_2} = [k_1, t]$ since C is abelian and normal in G. One computes that the kernel of ϕ is exactly $C_C(t) = \mathbf{Z}(G)$. Letting L denote the image of ϕ, we have that $L = |C : \mathbf{Z}(G)|$ by the First Isomorphism Theorem. The lemma will follow if we show that $L = G'$.

We first observe that L is normal in G since C is normal. This follows from observing that $[k, t]^t = [k^t, t^t] = [k^t, t]$ for any $k \in C$. Now, clearly, $L \subseteq G'$ since L consists of commutators. On the other hand, since $[k, t] \in L$ for any $k \in C$ and since $G = \langle C, t \rangle$, it follows that G/L is abelian. Thus, $L = G'$ as desired. □

Problem 11.19. Let (Γ_1, π_1) and (Γ_2, π_2) be Schur representation groups for G. Define $\Gamma \subseteq \Gamma_1 \times \Gamma_2$ by $\Gamma = \{(x, y) \mid \pi_1(x) = \pi_2(y)\}$, and show that $\Gamma_1' \cong \Gamma' \cong \Gamma_2'$.

Hint: Let $A_i = \ker \pi_i$ and define $\pi : \Gamma \to G$ by $\pi(x, y) = \pi_1(x)$ so that $\ker \pi = A_1 \times A_2$ and (Γ, π) is a central extension of G. Use Theorem 11.19 to show that $|\Gamma'| \leqslant |\Gamma_1'|$. Note that $|\Gamma' : \Gamma' \cap (A_1 \times A_2)| = |G'|$.

Note: One cannot conclude that $\Gamma_1 \cong \Gamma_2$ as the two nonabelian groups of order p^3 show. If $G = G'$, then $\Gamma_i = \Gamma_i'$ and $\Gamma_1 \cong \Gamma_2$ in this case.

Remark: In this proof, we shall frequently use the fact that any surjective group homomorphism $\phi : H \to K$ restricts to a surjective homomorphism of the derived subgroups, i.e., $\phi_{H'} : H' \to K'$ is surjective.

Proof. Notice first that $\pi_1(\Gamma_1') = G'$. Since Γ_1 is a Schur cover of G, we have that $A_1 = \ker \pi_1 \subseteq \Gamma_1'$, so $|\Gamma_1'| = |G'||M(G)|$ by the First Isomorphism Theorem. Notice also π as defined in the hint is the composition of the coordinate projection ρ_1 sending (x, y) to x into the map $\pi_1(x)$. As the restriction $\pi_{\Gamma'} : \Gamma' \to G'$ is a composition of surjective maps, it follows that

$$|\Gamma'| = |\ker \rho_1||M(G)||G'| \geqslant |M(G)||G'| \tag{11.12}$$

with equality holding iff the coordinate projection of Γ' onto Γ_1' is an isomorphism.

Now, one sees that Γ is a central extension of G with kernel $A_1 \times A_2$. Let π_3 denote the projection of Γ onto G. We have that $\pi_3(\Gamma') = G'$ and the restriction of π_3 to Γ' has kernel given by $\Gamma' \cap (A_1 \times A_2)$ Now, it follows from Theorem 11.19 that $|\Gamma' \cap (A_1 \times A_2)| \leqslant |M(G)|$. It now follows from the First Isomorphism Theorem that

$$|\Gamma'| \leqslant |M(G)||G'| \tag{11.13}$$

Together, (11.12) and (11.13) imply that the projection map $\rho_1 : \Gamma \to \Gamma_1$ is an isomorphism. By symmetry, $\Gamma \cong \Gamma_2$ as well. This proves the result. □

12. Character degrees

13. Character correspondence

14. Linear groups

15. Changing the characteristic

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